

A Testable Theory of Atomic Features

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Abstract

We develop and test a theory of language model representations in which there exist *atomic features*. Our main theoretical insight is that in such a model, sparse dictionaries (e.g., SAEs) of increasing size recover an increasing prefix of the most prevalent atoms in the training data. This “recovery principle” yields three testable predictions: many features in small SAEs are shared by all larger SAEs, SAEs trained on different data share features prevalent in both, and sufficiently large SAEs recover both parent and child features. In contrast to conventional wisdom that SAE features are unstable and “split” as size increases, we find that these predictions hold on SAEs of sizes ranging from 512 to 131,072 trained on two large embedding models. From a theoretical perspective, our results suggest the promise of a scientific theory of representations based on atomic features. Practically, our results suggest the promise of scaling SAEs.

1 Introduction

Language models represent text inputs x as vectors $f(x) \in \mathbb{R}^d$. A wide-ranging set of work has sought to better understand these representations. Here, our approach is to treat the linear representation hypothesis [1–14] as a foundation for a scientific theory. We formalize the hypothesis, mathematically derive implications, and then test these predictions on real language model representations.

The linear representation hypothesis (LRH) roughly says that $f(x)$ can be written as $z_1(x)\vec{a}_1 + z_2(x)\vec{a}_2 + \dots + z_M(x)\vec{a}_M$, where we think of $z_1(x), z_2(x), \dots, z_M(x)$ as describing individual features of x —does x mention cats, does x involve algebraic geometry, and so on. Stated succinctly,

$$f(x) = Az(x). \tag{1}$$

We now state a striking first result. Under a natural formalization of the hypothesis [14], where the features given by $z(x)$ are accessible from $f(x)$, this decomposition is unique (Theorem 1). That is, there exists a set of *atomic* features from which language model representations are built. A scientific theory based on the linear representation hypothesis might therefore be called the *atomic theory*.

Of course, Theorem 1 does not prove the existence of atomic features. On the contrary, one might take the strength of Theorem 1’s conclusion as evidence *against* the linear representation hypothesis (LRH). Indeed, there is much reason to be skeptical of the existence of atomic features. Sparse autoencoders (SAEs)—which are designed to recover the features posited by the LRH [15–19]—learn different features depending on the setting. Even SAEs in settings that vary only by random seed learn different features [20, 21]. Coarse features in small SAEs can “split” into granular features in larger SAEs [22–24]. How can we square variation in learned SAE features with the atomic theory?

Theorem 2, which we call the recovery principle, presents a solution. It says that under the atomic theory, SAEs of increasing size trained with appropriate sparsity on a distribution $\mathcal{D}$ recover

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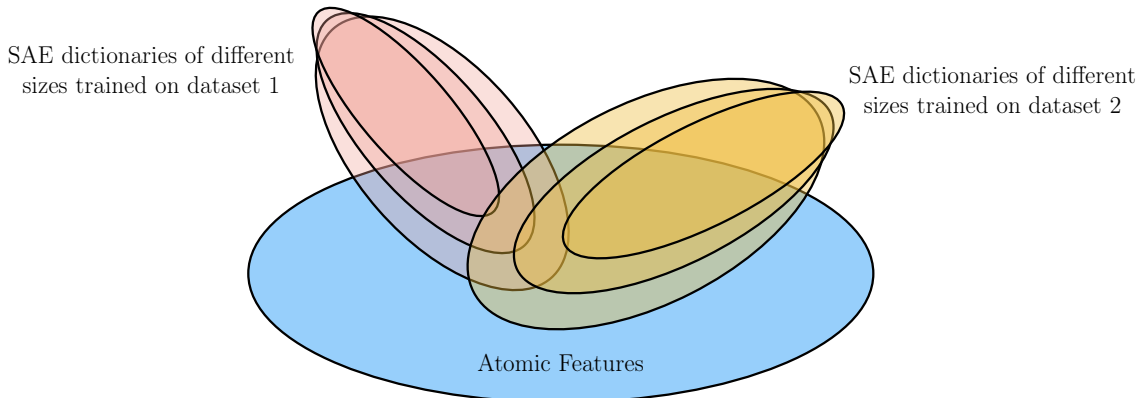


Figure 1: The recovery principle says that SAEs of increasing size recover a growing prefix of atomic features. This is consistent with observed variation in SAEs, while giving testable predictions that counter conventional wisdom: SAEs of different size and trained on different data share features.

a growing prefix of the most prevalent atoms in $\mathcal{D}$. Theorem 3 further shows that the recovery principle holds even when features are hierarchically structured. The recovery principle is consistent with observed variation in SAE features, while also establishing a substantive connection between these SAEs and the hypothesized atoms. Figure 1 illustrates this connection.

Beyond reconciling existing empirical observations with the atomic theory, the recovery principle plays another crucial role in the development of the theory: it makes the theory empirically testable. It yields the following three testable predictions.

- (P1) Many features in smaller SAEs have matches in all larger SAEs trained on the same data (corresponding to the most prevalent atomic features in the distribution).
- (P2) SAEs trained on different data have many matching features (corresponding to the atomic features that are prevalent in *both* distributions).
- (P3) There exist hierarchies where sufficiently large SAEs recover both parent *and* child features.

These predictions—while, strictly speaking, consistent with existing results—run against the conventional wisdom that SAE features are unstable and become strictly more granular as SAE size increases. P3 suggests a particularly surprising phenomenon: parent features lost as SAE size increases (as in feature splitting [22–24]) should eventually return in even larger SAEs. We call this *mesoscale* feature splitting. (In Section 5, we give a “molecular” explanation that also clarifies the importance of correctly choosing sparsity [25].)

We test P1, P2, and P3 by training SAEs of sizes ranging in powers of two from 512 to 131,072 on two large embedding models: gemini-embedding-2 and llama-embed-nemotron-8b.

- Testing P1, we find that about 70% of features recovered by size-4K SAEs simultaneously have matches in SAEs of size 8K, 16K, 32K, 65K, and 131K.
- Testing P2, we find that 26-59% of features are shared across pairs of SAEs trained on five different distributions, compared to mostly 0% and at most 15% for a PCA baseline.
- Testing P3, we find that on three hierarchies (prefixes, geography, and species), recovery at every level is maximized at size 65K or 131K. In fact, we find instances of the predicted mesoscale splitting, where parent recovery initially decreases, but increases again at larger widths.

These results suggest that the atomic theory is a promising scientific theory of language model representations—one that is worth further studying and testing. A practical implication is that by scaling SAEs with appropriate sparsity, we should expect to recover more and more stable atomic features, including both coarse features and granular features. Stepping back, the theory seems to point to a “rigidity principle”: efficient sparse representations of information seem to be pressured towards shared concepts.

2 Overview

We begin by presenting the conceptual ideas of the atomic theory. We start by motivating the formalization of the linear representation hypothesis that serves as the foundation for our work. We then discuss how this formalization leads to atomic features, the recovery principle, and testable predictions. We also give intuition by analogizing to the construction of vocabularies in natural language—noting the apparent pressure towards certain shared concepts.

Formalizing the linear representation hypothesis. A language model representation is a function $f : X \rightarrow \mathbb{R}^d$, where X is the input space (e.g., texts) and $f(x)$ is a representation of the input (e.g., neuron activations or an embedding). The main idea of the linear representation hypothesis, we recall, is to write

$$f(x) = \sum_{i=1}^M z_i(x) a_i, \quad (2)$$

where we call $z_1, z_2, \dots, z_M : X \rightarrow \mathbb{R}$ features (which we think of as measuring the presence of individual concepts in the input) and $a_1, a_2, \dots, a_M \in \mathbb{R}^d$ directions (directions in the representation space that correspond to each concept). Concretely, we can think of $z_1(x)$ as measuring the extent to which x discusses cats, and a_1 to be the direction in which this information is represented within $f(x)$. We call $z_i(x)$ the activation of feature i on input x . Taking $a_1, a_2, \dots, a_M$ to be the columns of $A \in \mathbb{R}^{d \times M}$ and defining the “feature representation” $z : x \mapsto (z_1(x), z_2(x), \dots, z_M(x))$, we can simply write $f(x) = Az(x)$.

We should also expect features to be *accessible* from a representation [14]. If $f(x)$ is a sufficiently expressive representation of the input x , then $z_1(x), z_2(x), \dots, z_M(x)$ should be accessible from $f(x) = Az(x)$ alone. For example, given $x = \text{“Djokovic Overwhelms Nadal for the Wimbledon Title”}$, we should be able to learn a number of properties from $f(x) = Az(x)$: that x discusses tennis, that x mentions Novak Djokovic, that x mentions Rafael Nadal, that x discusses a championship match, that x describes an event in England, that x describes a decisive victory, and so on. Formally, we expect there to exist a function $g : \mathbb{R}^d \rightarrow \mathbb{R}^M$ such that

$$g(Az(x)) = z(x), \quad (3)$$

i.e., mapping representations to features. However, M greatly exceeds d (embeddings typically have d in the thousands, and there are many more concepts), so $A \in \mathbb{R}^{d \times M}$ is not invertible in general. However, results in compressed sensing [14, 26–28] say that it is possible when $z(x)$ is sparse and A is well-behaved (which we make precise in Section 3).

This motivates our formalization of the linear representation hypothesis: that there exists a *well-behaved* matrix $A \in \mathbb{R}^{d \times M}$ and a *K-sparse* function $z : X \rightarrow \mathbb{R}_+^M$ such that for all $x \in X$, $f(x) = Az(x)$. We formalize this as Postulate 1.

Analogizing to dictionaries. Sparsity of $z(x)$ means that while the total number of concepts may be enormous, the number of concepts needed to describe a particular input is small (at most K). This motivates an analogy to natural language, where we call A, z a dictionary in which the directions a_i are the words.¹ Given a “situation” represented by $f(x)$, the idea is to write it as the combination of a small number of words (say, at most K). Then z_i specifies in what situations to use the direction a_i (it can be viewed, therefore, as providing a “definition”). In the setting here, z_i also specifies “how much” the word should be used.

We call M the *dictionary size* (the total number of features) and K the *description length* (the maximum number of features needed to describe an input). Existing work has focused on understanding when dictionaries can be recovered from data for a *fixed* dictionary size and description length [30–34]. In contrast, the results we introduce here are each investigations of the broader space of potential dictionaries. Are dictionaries unique when size and description length can vary? What does an optimal dictionary look like when you restrict its size? What about when tailored to a particular distribution of situations?

Deriving atomicity and a rigidity property. From the postulate, we can derive the atomic property. Theorem 1 considers a “competing” dictionary (B, u) : a well-behaved matrix $B \in \mathbb{R}^{d \times M'}$ and K' -sparse function $u : X \rightarrow \mathbb{R}_+^{M'}$ satisfying

$$f(x) = Bu(x) \tag{4}$$

for all $x \in X$. Then the result shows that if z is sufficiently rich (exploring each combination of features), then if B, u is Pareto-efficient in dictionary size and description length, then either it is equal to A, z or $K' \geq 2K$ (i.e., it sometimes uses *twice* as many features to describe an input). Theorem 1 shows that if a sparse system of representation exists, and the space being described is sufficiently rich, then this sparse system is “rigid.”

To get an intuitive sense of this statement, we draw on the analogy to natural language dictionaries, from which we can state the following informal analog of Theorem 1. Consider a dictionary of M words and a set of situations that can each be exactly described by K words. Suppose further that there exist situations spanning each combination of words. Then any dictionary using one fewer word—and that can still perfectly describe each situation—must for some situation use *twice* as many words. (It can be informative to try to play this “vocabulary construction game” yourself. We return to this game in a moment.)

From theory to testable predictions. Up to this point, we have established a pretty but (thus far) hypothetical picture of the underlying structure of language model representations. How can we know if this picture is a useful description of reality? A natural approach is to try and find the hypothesized atoms. To do this, we can try to apply sparse dictionary learning [15, 29, 32, 35] to find a matrix $\hat{A}$ and sparse function $\hat{z}$ such that

$$f(x) \approx \hat{A}\hat{z}(x). \tag{5}$$

But major challenges arise in practice. First, the true number of atoms M is unknown (and likely very large). Second, the learned dictionary is shaped by the specific training data. Finally, given a learned dictionary, how are we to know that we have recovered the true atoms?

¹Of course, we are drawing from a long line of work on dictionary learning here and throughout [29].

The recovery principle. There is a powerful mathematical consequence of Postulate 1 that helps resolve these issues. Theorem 2 shows that even sparse dictionaries of size $m < M$ that are trained on idiosyncratic (but “well-behaved”) distributions $\mathcal{D}$ approximately recover a subset of atoms. In particular, as we increase the size of a sparse dictionary, we recover a growing prefix of the most prevalent atoms in the distribution. Note that the sparsity K must be chosen appropriately to get this guarantee.²

Notice how the recovery principle implies testable implications. First, by training larger SAEs on a fixed distribution, we should see a growing set of “persistent features”—i.e., features in smaller SAEs that appear in every larger SAE. Second, even if we train SAEs on different data distributions, we should expect to see a set of shared features—the atomic features that are most prevalent in both distributions. This can be contrasted with PCA or SVD, in which learned directions are highly dependent on the data distribution. (SVD had been suggested in the past as a “solution to Plato’s problem” [36], but a primary objection was the sensitivity to distributions; SAE features, under the atomic theory, overcome this objection.) The recovery principle and its implications are further illustrations of the rigidity of sparse systems of representation.

Intuition. Returning to the vocabulary construction game, consider the goal of constructing a dictionary of limited size to best describe situations arising in your daily life using at most K words per situation. (Perhaps, you were confined to tweeting.) You are allowed to invent new words (e.g., referring to combinations of concepts that appear especially frequently for you). The recovery principle says that you would not pick entirely new, invented words; rather you would adopt many existing words—and in particular, the existing words that are most relevant for your own daily life. These words are retained even if you are allowed a larger dictionary.

To give a concrete example, suppose that each day you looked outside your window and saw a tree by a sidewalk. Perhaps, in extremely limited vocabularies, you would define a word “sidewalktree.” This would save you budget in dictionary size and in description length. But you would be unable to describe a “tree” or “sidewalk” on their own. If you wanted to describe an isolated tree, you would need a word that means “nosidewalk” and call it a “nosidewalk” “sidewalktree,” in which case you have now used just as many words and more description length than if you had words for “tree” and “sidewalk.” In this way, there is pressure towards including the atoms “tree” and “sidewalk” in a dictionary. (To actually prove the recovery principle requires some more sophisticated reasoning.)

In a sense, of course, humanity has been playing the vocabulary game for millenia. Different societies have come up with languages of different lengths to describe different distributions of situations. Quite strikingly, these languages have an enormous amount of overlap in the concepts given words. In linguistics, a “Swadesh list” gives concepts that appear in essentially *all* languages, including concepts like I, you, five, short, knee, to drink, to fly, to count, river, sky, red, green, year, and round (among many others). The recovery principle offers some reasoning for this—and hints at a formal notion of “universal concepts.”

Recovery of hierarchical features. The recovery principle shown in Theorem 2 makes an assumption that the training distribution is well-behaved: that for every pair of features z_i, z_j , there exist inputs for which z_i is active but z_j is not (and vice-versa). In Theorem 3, we show that the recovery principle continues to hold (with an adjustment) for distributions where activations are hierarchically correlated (where a parent feature is active whenever a child feature is active). The

²To mix metaphors, the recovery principle suggests that training larger SAEs is something like building larger particle accelerators: by scaling properly, we expect to be able to recover more and more fundamental particles (here, atoms).

result implies that in sufficiently large SAEs, we expect to simultaneously recover parent features and child features. This prediction is especially interesting when compared to existing empirical observations in which a broad parent feature that is recovered in a smaller SAE “splits” as SAE size increases. Under our theory, the expected behavior is that as the size of the SAE increases even further (with sparsity chosen appropriately), we should expect the parent feature to once again be recovered.

3 Theoretical Results

We now mathematically present the atomic theory starting from a single postulate and deriving the testable predictions P1-P3 as consequences. It will be useful to first establish a formal definition of a dictionary, which is the primary object we study throughout.

Definition 1. An (m, k, γ) -dictionary is a pair $B \in \mathbb{R}^{d \times m}, u : X \rightarrow \mathbb{R}_+^m$ where u is k -sparse and $B \in \mathbb{R}^{d \times m}$ has unit-norm columns and satisfies the $(\gamma, 2k)$ -lower restricted isometry property: that is, for all $2k$ -sparse $v \in \mathbb{R}^m$, $\|Bv\|_2 \geq \gamma\|v\|_2$.³

Recall that m is the dictionary’s *size* and k the dictionary’s *description length*. The lower restricted isometry property establishes the key property that $u(x)$ can be recovered from $Bu(x)$. Indeed, if $Bu(x) = Bu'$ for some k -sparse vector u' , then $\|u(x) - u'\| \leq \frac{1}{\gamma}\|Bu(x) - Bu'\|_2 = 0$, so $u(x) = u'$. The dictionaries we study are *overcomplete* ($m > d$) and *sparse* ($k < d$). In what follows, we take $\gamma \in (0, 1)$ to be a fixed constant and refer to an (m, k, γ) -dictionary as simply an (m, k) -dictionary.

We can then succinctly state our foundational postulate about f .

Postulate 1 (The Linear Representation Hypothesis). *There exists an (M, K) -dictionary A, z such that*

$$f(x) = Az(x) \tag{6}$$

for all $x \in X$.

We refer to A, z as the atomic dictionary, with atomic directions $a_1, a_2, \dots, a_M$ and atomic features $z_1, z_2, \dots, z_M$. Our first task is to justify calling these features atomic.

Past work has shown that if z is sufficiently “rich,” then A, z is the unique dictionary with dictionary size and description length exactly equal to M and K [30, 31, 34]. Here, we will establish a notion of uniqueness that is quite a bit broader. We say that an (m, k) -dictionary is *efficient* if, compared to the atomic dictionary, any *one* of the following three conditions holds:

- (a) it has the same dictionary size and description length (i.e., $m = M, k = K$);
- (b) it has smaller description length (i.e., $k < K$);
- (c) it has smaller dictionary size and less than twice the description length (i.e., $m < M$ and $k < 2K$).

Now define richness as follows. For $S \subseteq [M]$ with $|S| = K$, let $E_S = \text{span}\{e_i : i \in S\}$ and $Z_S = z(X) \cap E_S$ (the set of feature activations with support S realized by X). We call z K -rich if Z_S is not contained in any union of a finite number of proper subspaces of E_S .

³For $d = \Omega_{k, \gamma}(\log m)$, lower RIP holds with high probability for random Gaussians [27]

Theorem 1 (Unique Efficient Dictionary). *Suppose z is K -rich. If an efficient dictionary B, u satisfies*

$$f(x) = Bu(x) \quad (7)$$

for all $x \in X$, then $B, u = A, z$ up to a permutation of indices.

Therefore, no dictionary that exactly describes f can have smaller description length than A, z , and any smaller dictionary that exactly describes f must have *double* the description length. One way to view this result is that if a set of inputs can be described by a sparse dictionary, then this sparse dictionary is *rigid*.⁴

The main challenge of the proof is to show that $k < 2K \implies m \geq M$. The key idea is to construct a bipartite graph between directions in the two dictionaries, where a_i and b_j share an edge if b_j is used in the reconstruction of a_i (we will first have to show that a_i has a k -sparse representation in B). Then, we can show that if $k < 2K$, Hall's condition is satisfied for each subset of A directions of size at most $2K$; moreover, no subset of at most K A directions has a neighborhood of size at least $2K$. We then show that these two conditions imply Hall's condition to hold globally. (Interestingly, this appears to be a novel sufficient condition.) Therefore, there exists a one-to-one matching including all A directions, showing that $m \geq M$.

The recovery principle. We now establish the recovery principle: that even undersized SAEs trained to approximate idiosyncratic distributions of inputs recover atoms. This gives testable predictions P1 (features in small SAEs persist in larger SAEs) and P2 (SAEs trained on different data learn overlapping features).

The recovery principle concerns dictionaries that (1) are undersized (i.e., $m < M$) and (2) only approximately describe f on some distribution $\mathcal{D}$ over X . The recovery principle shows conditions under which such dictionaries still recover atoms. Define the dictionary learning objective

$$\mathcal{L}_{\mathcal{D}}(B, u) := \mathbb{E}_{x \sim \mathcal{D}} \|f(x) - Bu(x)\|_2^2 \quad (8)$$

over the reference class of (m, K) -dictionaries. Notice that dictionaries in this reference class have description length matching the atomic dictionary; we show in simulations in Section G that choosing appropriate k is important, but that there is some flexibility. Let $\mathcal{L}_{\mathcal{D}}^*(m)$ be the optimal loss over this class. Taking a fixed constant $C_0 > 1$, let $\hat{A}^{(\mathcal{D}, m)}, \hat{z}^{(\mathcal{D}, m)}$ be an (m, K) -dictionary satisfying

$$\mathcal{L}_{\mathcal{D}}(\hat{A}^{(\mathcal{D}, m)}, \hat{z}^{(\mathcal{D}, m)}) \leq C_0 \mathcal{L}_{\mathcal{D}}^*(m). \quad (9)$$

Conceptually, this is the kind of dictionary that a sparse autoencoder can learn in practice, obtaining loss within a constant factor of optimal. We are interested in if and when $\hat{A}^{(\mathcal{D}, m)}, \hat{z}^{(\mathcal{D}, m)}$ recovers atomic features and directions. Formally, we define recovery as follows:

- Direction recovery: $\hat{A}$ η -recovers a_i if there exists $j \in [m]$ such that $\langle \hat{a}_j, a_i \rangle > \eta$.
- Activation recovery: $\hat{z}$ $(\eta, \mathcal{D})$ -recovers z_i if there exists $j \in [m]$ and a threshold τ such that $\mathbb{E}_{\mathcal{D}}(\mathbf{1}\{z_i > 0\}, \mathbf{1}\{\hat{z}_j > \tau\}) > \eta$.

We show that recovery can happen as long as the distribution $\mathcal{D}$ is well-behaved. As an example of why some kind of assumption on $\mathcal{D}$ is necessary, notice that if two atomic features z_i, z_j were exactly co-occurring on a distribution, then it would not be possible to disentangle them.

⁴A result by Sitharam et al. [37] establishes a related rigidity result: that one cannot achieve a constant fraction of dictionary size without description length blowing up to be on the order of embedding dimension [37]. We state this result as Theorem 4 and give an alternate proof using Sard's theorem.

This motivates the following definition. A distribution $\mathcal{D}$ over X is (ρ, c_-, c_+) -regular if the following conditions hold for $x \sim \mathcal{D}$. First, as a simplification, let $\text{supp}(z(x))$ have size K for all inputs. Let $h_S : (0, 1)^K \rightarrow \mathbb{R}_+$ be the density of $z_S(x)$ conditional on the event $\text{supp}(z(x)) = S$. Then the two conditions are

- $\Pr[j \notin \text{supp}(z(x)) \mid i \in \text{supp}(z(x))] \geq \rho$ for all pairs $i, j \in [M]$ with $i \neq j$.
- $c_- \leq h_S(w) \leq c_+$ for all $w \in (0, 1)^K$.

The idea of the recovery principle, then, is that if $\mathcal{D}$ is regular, then undersized sparse dictionaries can recover the most prevalent atoms in $\mathcal{D}$.

Given $\mathcal{D}$, set $p_i := \Pr[i \in \text{supp}(z(x))]$, the probability that an atom i is active for a given input. Choose $\sigma : [M] \rightarrow [M]$ such that $p_{\sigma(1)} \geq p_{\sigma(2)} \geq \dots \geq p_{\sigma(M)}$, determining the order of prevalence. Intuitively, we may expect these probabilities to follow a power-law distribution $p_{\sigma(i)} \propto i^{-\alpha}$. Define $T(m) = \sum_{j=m+1}^M p_{\sigma(j)}$, which we can call the tail mass beyond m . Then Theorem 2 below says that an atom is recovered in a dictionary of size m whenever its prevalence exceeds the tail mass $T(m)$ by a constant factor.

Theorem 2 (The Recovery Principle). *Let $\mathcal{D}$ be (ρ, c_-, c_+) -regular. Then for every $\eta < 1$, there exists a constant C not depending on m or M such that for all $i \in [M]$ satisfying*

$$p_i \geq CT(m), \tag{10}$$

$\hat{A}^{\mathcal{D}, m}$ η -recovers a_i and $\hat{z}^{\mathcal{D}, m}$ $(\eta, \mathcal{D})$ -recovers z_i

The result guarantees recovery of atoms that appear more frequently than the tail mass beyond the m -th most frequent atom. For example, if atom prevalences follow a power-law distribution with $\alpha > 1$, the result guarantees that at least the $\Omega(m^{1-1/\alpha})$ most prevalent atoms are recovered. In Section G, we show that in simulations, a much larger fraction of atoms can be recovered—suggesting that the bound shown here is not tight (one reason, perhaps, is that the bound here applies for any learned dictionary with low enough loss, and does not take advantage of the structure of *optimal* learned dictionaries). It would be exciting to prove stronger versions of the recovery principle.

To provide some intuition for the result, we sketch the proof. The main lemma shows that if there exists an atomic support $S \subseteq [M]$ such that the learned dictionary approximates representations that lie in $\text{span}(A_S)$, then there must exist a learned support $T_S \subseteq [m]$ such that $\text{span}(\hat{A}_{T_S}) \approx \text{span}(A_S)$. This is non-obvious, since in principle, points in $\text{span}(A_S)$ may have different optimal K -sparse supports in $\hat{A}$. An initial observation is that if we can show that a single learned support is being used to re-construct a large enough subset of $\text{span}(A_S)$, then the span of this support must be aligned with $\text{span}(A_S)$. This approach does not work immediately, since there are $\binom{m}{K}$ possible supports, so each learned support may only be used for a vanishing fraction of the span as m increases. A key insight is that the lower restricted isometry property implies that optimal supports do not change too much for nearby points, allowing us to iteratively identify a large enough ball in which a single learned support is used. From this lemma, we can then intersect recovered supports to show the recovery of individual atoms. Finally, any sufficiently optimal dictionaries—when the tail mass is small—must recover atomic supports spanned by atoms in the head of the distribution.

Predictions P1 and P2 follow as corollaries. Both concern the existence of matching features across dictionaries. For matrices U and V , we define $\mu_t(U, V)$ to be the size of the largest one-to-one matching between columns of U and V such that matches have cosine similarity at least t . Let $\mu_t(U; V_1, V_2, \dots, V_s)$ be the maximum number of columns of U that can be simultaneously matched

to every $V_1, V_2, \dots, V_s$. Then for $t < 1$, there exists a constant C_t such that for $R_{\mathcal{D},t}(m) := \max\{r \in [M] : p_{\sigma(r)} \geq C_t \sum_{j=m+1}^M p_{\sigma(j)}\}$ the following results hold.

Corollary 1 (P1: Stability Across Size). *For a regular distribution $\mathcal{D}$ and dictionary sizes $m_1, \dots, m_s \geq m$,*

$$\mu_t\left(\hat{A}^{(\mathcal{D},m)}; \hat{A}^{(\mathcal{D},m_1)}, \dots, \hat{A}^{(\mathcal{D},m_s)}\right) \geq R_{\mathcal{D},t}(m).$$

That is, there exists a persistent set of features across all dictionaries of size at least m .

Corollary 2 (P2: Stability Across Data). *For any regular distributions $\mathcal{D}$ and $\mathcal{D}'$ on the same atomic representation,*

$$\mu_t\left(\hat{A}^{(\mathcal{D},m)}; \hat{A}^{(\mathcal{D}',m)}\right) \geq |\sigma([R_{\mathcal{D},t}(m)]) \cap \sigma'([R_{\mathcal{D}',t}(m)])|.$$

Here, $\sigma([R_{\mathcal{D},t}(m)]) \cap \sigma'([R_{\mathcal{D}',t}(m)])$ is the set of atoms in both distributions' guaranteed prefixes. So dictionaries trained on different distributions have shared features—the atoms prevalent in both distributions.

The hierarchical recovery principle. To establish P3 (that sufficiently large SAEs recover both parent features and child features), we show that the recovery principle holds under distributions where atomic feature activations have hierarchical correlation: i.e., when the activation of a child atom implies the activation of a parent atom. This violates the regularity condition on $\mathcal{D}$ used in Theorem 2.

We consider a setting with N families, each of which has one parent atom and two child atoms. It is possible to generalize to more than two child atoms. We let $M = 3N$ and $K = 2L$, and re-index the atoms as $i, i1, i2$ for $i \in [N]$. Here, the parent feature z_i is always active when the child features z_{i1} or z_{i2} are active. We then establish a definition of *hierarchical regular* distributions. Under this definition, parent features satisfy the standard regularity assumptions, exactly one child feature activates at a time, and the space of parent-child activation pairs is well-explored. The full definition is given in Section B.

We then define analogs of quantities used in the regular recovery principle. For a given $\mathcal{D}$ and a family $i \in [N]$, let $p_i = \Pr[i \in \text{supp}(z(x))]$ and $\sigma : [N] \rightarrow [N]$ sort the families in monotonically decreasing order of prevalence according to occurrence of the parent feature (so $\sigma(1)$ is the most prevalent family). Define $T(n) = \sum_{j=n+1}^N p_{\sigma(j)}$, the tail mass beyond the n -th family.

Then Theorem 3 below shows that an analog of Theorem 2 holds in hierarchical settings for *activation recovery*. In other words, we should expect to be able to recover features that activate in the presence of the parent, and also features that activate in the presence of each child, yielding P3.

Theorem 3 (The Hierarchical Recovery Principle). *Let $\mathcal{D}$ be hierarchical regular. Then for every $\eta < 1$, there exists a constant C not depending on m or M such that $\hat{z}^{\mathcal{D},m}(\eta, \mathcal{D})$ -recovers the family of features z_i, z_{i1}, z_{i2} for all $i \in [N]$ satisfying*

$$p_i \geq CT \left(\left\lfloor \frac{m}{3} \right\rfloor \right). \quad (11)$$

The following identity provides useful intuition. If $z_j(x) > 0 \implies z_i(x) > 0$, then for all $\lambda \geq 0$ we can re-write

$$z_i(x)a_i + z_j(x)a_j = \hat{z}_i(x)\hat{a}_i + \hat{z}_j(x)\hat{a}_j, \quad (12)$$

where $\hat{z}_i(x) = z_i(x) + \lambda z_j(x)$, $\hat{a}_i = a_i$, $\hat{z}_j(x) = z_j(x)$, and $\hat{a}_j = a_j - \lambda a_i$.⁵ This preserves sparsity and non-negativity of u . More importantly, both $\hat{z}_i$ and $\hat{z}_j$ activate on exactly the same examples as z_i and z_j . Therefore, even though atomic directions and features are not strictly unique under hierarchical correlation, alternative parameterizations are still equivalent in terms of *when* each feature is active.

Mesoscale feature splitting. The prediction of simultaneous recovery of both parent and child features at large SAE sizes seems in tension with empirical results that have demonstrated feature splitting and absorption [22–24]. P3 suggests that as SAE size continues to increase, we should expect lost parent features to return.

We give a “molecular” model of this phenomenon in Section 5, in which small SAEs learn a family molecule, medium SAEs learn individual parent-child molecules (this is the regime where a learned feature matching the parent’s activation is lost), and large SAEs learn an individual parent atom and individual child atoms. This molecular model can explain further phenomena, such as polysemanticity and semantic “clusters” of SAE directions.

4 Experiments: Testing P1, P2, and P3

We now test the predictions P1, P2, and P3 derived from Postulate 1. We study two large embedding models, gemini-embedding-2 [38] and llama-embed-nemotron-8b [39], which give representations $f(x)$ of dimension $d = 3072$ and $d = 4096$, respectively. We train k -sparse autoencoders [35] on embeddings derived from 90M text inputs, comprising 13 datasets used to fine-tune llama-embed-nemotron-8b. An SAE of size m and trained with sparsity k produces a dictionary $\hat{A} \in \mathbb{R}^{d \times m}$, $\hat{z} : X \rightarrow \mathbb{R}^m$, where $\hat{A} = W_{\text{dec}}$ is the SAE decoder and $\hat{z}(x) = \text{TopK}_k(\text{ReLU}(W_{\text{enc}}(f(x) - b_{\text{dec}})))$ are the hidden layer activations. We train SAEs of sizes m in powers of two ranging from 512 to 131,072, with $k = 32, 32, 32, 32, 64, 64, 64, 128, 128$ respectively. (Section D reproduces results with $k = 128$ fixed.)

Figure 2 displays our main results. Details of our experimental setup are given in Section C. Extended analyses for P1, P2, and P3 are collected in Sections D to F, respectively.

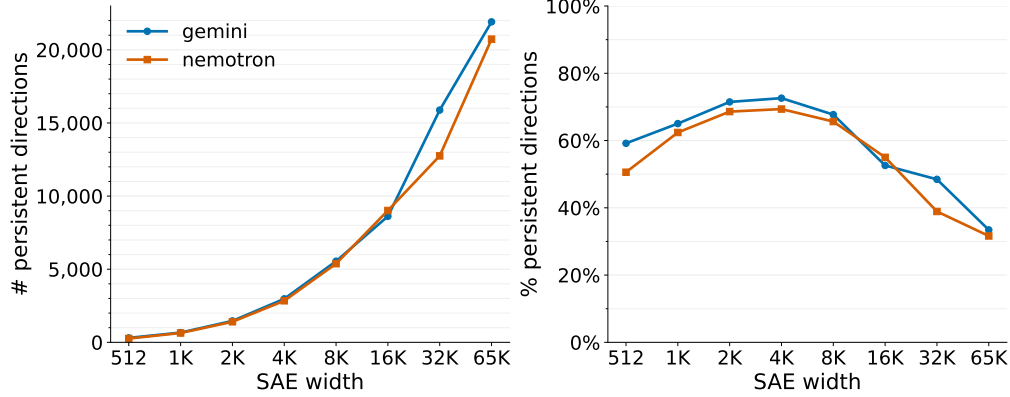
Computing matching features. To test P1 and P2, we need to compute matches across dictionaries. We use the following metrics. Given a bipartite graph, a *maximum matching* is a one-to-one matching with the largest number of matches. The *exclusive matching* is the one-to-one matching consisting of all edges whose endpoints have only one neighbor. Given dictionaries $\hat{A}, \hat{z}$ and $\hat{A}', \hat{z}'$, we draw edges between directions that have cosine similarity at least t and edges between features that have Pearson correlation at least t over a specified distribution. We then compute the maximum and exclusive matching. Given a matching criterion (maximum or exclusive), we compute a matching independently between the source dictionary and each larger dictionary. A direction or feature is *persistent* if it is matched in every comparison. We use a threshold of $t = 0.7$ as our default, following Paulo and Belrose [20].

P1: Persistent features across size. We start by testing P1, which says that many features in smaller SAEs have matches in all larger SAEs. Many directions and features have matches in *all* larger SAEs. Under independent maximum matching at threshold $t = 0.7$, in SAEs of size 4,096, 72.6% of Gemini directions and 69.4% of Nemotron directions are persistent (Figure 2a).

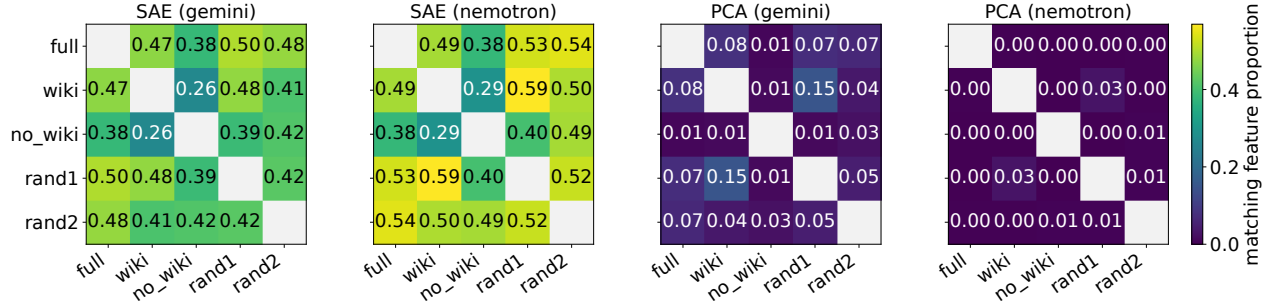
⁵This identity is studied by Chanin et al. [22] as a potential source of feature absorption; an important distinction in our assumptions is that we allow child feature activations to be larger than the parent, meaning that $\hat{z}_i = z_i(x) + \lambda z_j(x) \geq 0 \implies \lambda \geq 0$.

Figure 2: Testing P1, P2, and P3—the predictions of the atomic theory.

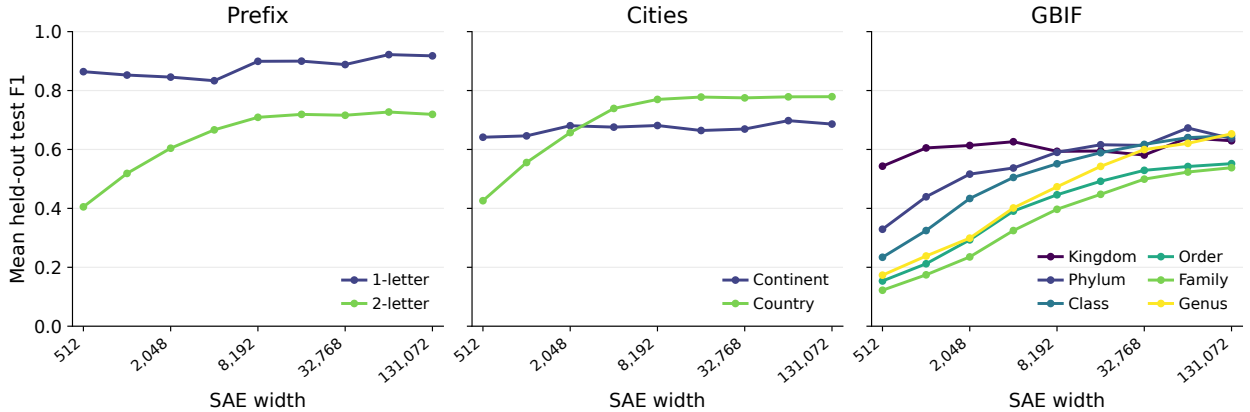
(a) **P1: Persistent features across size.** Number and proportion of directions with matches in every larger SAE. Similarity threshold $t = 0.7$. About 70% of 4K SAE directions are retained in size-8K, 16K, 32K, 65K, and 131K SAEs.



(b) **P2: Stability over data distributions.** Proportion of directions shared between SAE and PCA dictionaries trained on different distributions. Similarity threshold $t = 0.7$. We use absolute cosine similarity for PCA directions. 26%-59% of SAE feature directions have matches, compared to mostly 0% and at most 15% for PCA.



(c) **P3: Hierarchical recovery.** Mean held-out test F1 by hierarchy level and SAE width for Prefix, Cities, and GBIF. Each curve is an unweighted mean over category-model pairs, pooling Gemini and Nemotron. Average recovery is highest across all levels at sizes 65K and 131K.



By activation similarity, 59.7% of Gemini features and 66.4% of Nemotron features are persistent (Figure 7). Section D shows results under exclusive matching and under different thresholds.

We also compare feature prevalence across SAE sizes. Persistence suggests that larger SAEs do not learn strictly more granular features. We find that larger SAEs learn a similar number of coarse features as smaller SAEs while learning a longer tail of granular features; in contrast, KMeans clusters become strictly smaller as the number of clusters increases (Figure 10). Consistent with recovering prevalent atomic features, more prevalent features are more likely to be persistent (Figure 12).

P2: Shared features across training data. We test P2, which says that SAEs trained on different data learn overlapping features. We use four additional data splits: `wiki`, `no_wiki`, `rand1`, and `rand2`. The pair (`wiki`, `no_wiki`) contain separate datasets (`wiki` contains datasets that are built using Wikipedia data, and `no_wiki` contains the datasets that are not). The pair (`rand1`, `rand2`) contain separate datasets as well, but the partition is chosen randomly rather than by content type. None of the four data splits contain the multilingual dataset MIRACL. We train SAEs with $m = 16384$ and $k = 64$ on each distribution. We compare to PCA baseline dictionaries, consisting of the d principal components fitted to each distribution. To compute matchings between PCA directions, we use absolute cosine similarity, since the signs of principal components are arbitrary.

Figure 2b shows the proportion of features shared across pairs of different dictionaries trained on different datasets for $t = 0.7$. SAE dictionaries have many cross-dataset matches (between 26% and 59%), while PCA dictionaries have substantially fewer (at most 15%, median 1%). The lower matching rates in the PCA dictionaries suggest that the high amount of overlap in SAE dictionaries is not explained only by similarity in the underlying data distributions. Figure 14 also shows that SAEs have many cross-distribution matches based on activation similarity (22-61%). Under exclusive matching, SAE dictionaries continue to have many matching directions (25-57%) whereas the PCA match proportion remains at most 15% (Figure 13).

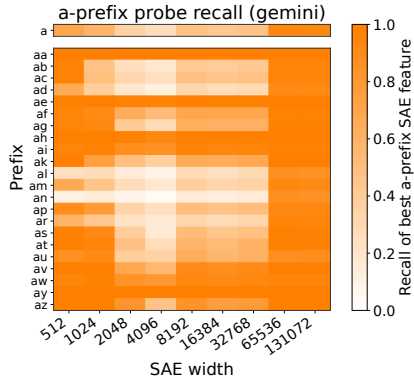
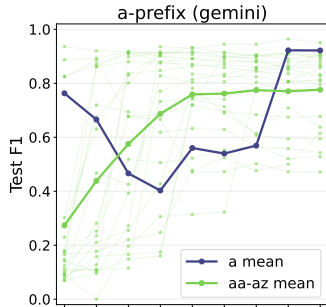
Figure 15 shows that features with greater prevalence in both distributions are more likely to be selected in direction matching, consistent with the recovery of overlapping prevalent atoms.

P3: Hierarchical feature recovery. We test P3 by studying how well SAEs of different sizes recover a ground-truth set of hierarchical features defined by three datasets: `prefix`, `cities`, `gbif`.

For each ground-truth binary feature $z_\ell^* : X \rightarrow \{0, 1\}$ defined over inputs from each dataset, and for each SAE size from 512 to 131K trained on `full`, we report the test F1 score $F1_{\mathcal{E}_{\text{test}}}(z_\ell^*, \mathbf{1}\{\hat{z}_j \geq \tau\})$, where $j \in [m]$ and $\tau > 0$ are selected to maximize the same score under $\mathcal{E}_{\text{train}}$. Here $\mathcal{E}_{\text{train}}$ and $\mathcal{E}_{\text{test}}$ are the empirical distributions on the train and held-out test splits of the labeled dataset. Dataset and experiment details are given in Section C.5. Figure 2c shows that across levels of the hierarchy, mean test F1 generally improves with size and is highest at width 65K or 131K. Figure 18 confirms that this corresponds to increasing *simultaneous* recovery of parent and child features.

We also find examples of mesoscale feature splitting. Figure 3a shows that F1 recovery of the parent ‘a’ prefix feature decreases as SAE size increases initially (splitting), but increases again at even larger sizes. (Figure 19 shows all 25 Gemini letter prefixes with eligible two-letter children, including ‘a’ and excluding ‘x’.) However, this effect disappears when we increase the sparsity of smaller widths to match the largest widths, suggesting that feature splitting may primarily be the result of over-constrained sparsity. See Section 5 for more discussion.

In addition to testing how well SAEs recover ground-truth hierarchies, we also extract hierarchical structure directly by examining correlation in feature activations over the corpus `full` (Section F.4). We find 114,762 parent-child pairs in the 131K gemini SAE. Figure 3b shows examples of extracted three-level hierarchies, with top-level features “Food” and “Music.” Example lineages include “Food” $\rightarrow$ “Potatoes” $\rightarrow$ “Hash browns / rosti” and “Music” $\rightarrow$ “Jazz” $\rightarrow$ “John Coltrane.”



(a) Mesoscale feature splitting.

Level 1	Level 2	Level 3
Food	Cheese	Cheddar; Mozzarella / related cheeses; Blue cheeses; Gruyère / Emmental; Brie / Camembert
	Potatoes	Potato cultivars; Mashed potatoes; Baked potatoes; Hash browns / rösti; Tater tots
	Fast food	Burger King; Wendy’s; Jack in the Box; Hardee’s; McDonald’s burgers
	Rice dishes	Biryani; Nasi goreng; Fried rice
	Sausages	Chorizo; Salami; German sausages; Hot dogs
	Nutritional information	Fast-food calorie questions; Calorie-breakdown template; Percent Daily Value text; Vitamins / minerals
Music	Jazz	Charlie Parker / bebop; John Coltrane; Miles Davis; Mahavishnu / John McLaughlin; Marsalis family
	Hip-hop / rap	N.W.A; Eminem; Snoop Dogg; Tupac Shakur; Lil Wayne
	Country music	George Jones; Rascal Flatts; Patty Loveless / Vince Gill; Hank Williams
	Rock / metal	Hard-rock musicians / bands; Nu / alternative metal; Power / symphonic metal; Doom / stoner metal
	Electronic / dance	Detroit techno cluster; Trance mixes / compilations; Basement Jaxx; deadmau5
	The Beatles	Please Please Me / early Beatles; Hey Jude; Let It Be / Get Back; Abbey Road / late Beatles

(b) Learned feature hierarchies.

Figure 3: **Hierarchical feature recovery.** (a) Top: parent and child F1 for a-prefixes. Parent recovery follows a U-shape. Bottom: recall of the selected a-prefix feature, broken down by prefix. (b) Extracted Food and Music hierarchies in the Gemini 131K SAE. Activating examples for each feature are given in Tables 8 and 9.

5 A Molecular Model

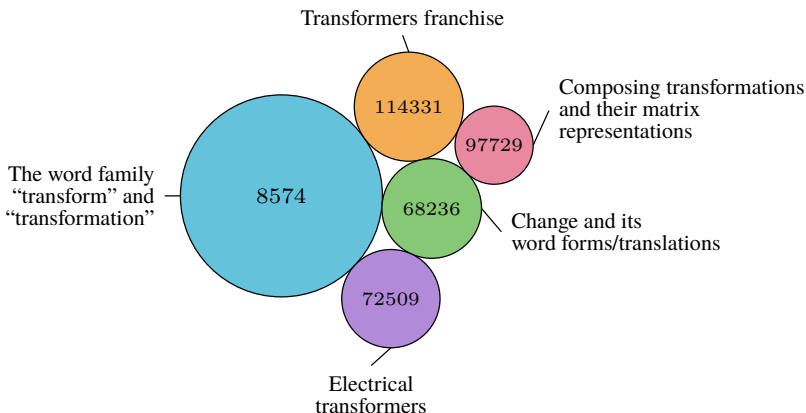
We now sketch a molecular model that helps explain several additional empirical phenomena: polysemanticity, feature splitting, and semantic clusters of directions [24]. In the model, “molecules” are learned SAE directions that are linear combinations of atomic directions. The recovery principle says that molecules in small SAEs eventually split into stable atoms in larger SAEs. A prediction, then, is that since the above empirical phenomena stem from molecules, they disappear at sufficiently large widths. The results and explanations here are intended to be illustrative—we hope that future work formalizes and systematically tests the molecular extension of the atomic theory.

The intuition for molecules can be conveyed with a minimal example. If two atomic features $z_i(x)$ and $z_j(x)$ are heavily correlated over $\mathcal{D}$, then it can be efficient for $\hat{A}$ (trained to minimize $\mathcal{L}_{\mathcal{D}}$) to mix the atomic directions a_i and a_j into a single learned direction, saving budget in terms of dictionary size (and also description length). The direction can be visualized as follows:

$$\textcircled{i} \textcircled{j} \quad (13)$$

In the empirical results that follow, we use learned directions and features in the size-131K gemini SAE as proxy atoms. We then decompose learned directions in smaller SAEs as linear combinations of proxy atomic directions, giving proxy molecules. Details are given in Section H.

Table 1: A polysemantic “transform”-related feature in the 4K gemini SAE, represented as a molecule of 131K features. It activates on mentions of “transform” as well as strings like “are autobots stronger than decepticons” (referencing the Transformers franchise). Examining activating examples in the table, the 131K features (treated as proxy atoms) seem comparatively monosemantic (e.g., representing the Transformers franchise). More examples for each atom are given in Table 11.



z_{3290}^{4K}	Text	z_{8574}^{131K}	z_{68236}^{131K}	z_{114331}^{131K}	z_{72509}^{131K}	z_{97729}^{131K}
0.387	This is a list of transforms in mathematics.	0.252	0.087	0.000	0.067	0.014
0.330	what is the importance of data transformation?	0.255	0.102	0.000	0.009	0.000
0.318	A transformation group is a group G that acts on a set X by bijections. In other words, for each element g in G , there ...	0.250	0.047	0.000	0.000	0.000
0.285	A non-linear transformation is a mathematical function that doesn't follow the rules of linearity. In simple terms, a linear ...	0.211	0.068	0.000	0.000	0.014
0.257	To transform the given sum, utilize the decomposition $\frac{1}{k(k+1)(k+2)!} = \frac{1}{k!} - \frac{1}{(k+1)!} + \frac{1}{(k+2)!}$...	0.217	0.124	0.000	0.000	0.000
0.163	The Metamorphosis (German: Die Verwandlung) is a novella written by Franz Kafka which was first published in 1915.	0.085	0.078	0.000	0.000	0.000
0.148	what is ctrl t in photoshop?	0.079	0.043	0.000	0.000	0.000
0.204	... voice actors Peter Cullen and Hugo Weaving voice Optimus Prime and Megatron respectively.	0.030	0.000	0.210	0.000	0.000
0.169	are autobots stronger than decepticons?	0.000	0.000	0.242	0.032	0.000
0.338	A transformer is a device that transfers electrical energy from one circuit to another by magnetic coupling.	0.165	0.055	0.036	0.299	0.000
0.186	why crgo core is used?	0.055	0.022	0.000	0.177	0.000
0.168	Translation: 15 units left, 3 units down; Reflection: across x-axis; Scale: along y-axis by a factor of 2.	0.075	0.061	0.000	0.000	0.063
0.152	1. Reflect about the x-axis: $-A(1, 5) \rightarrow A'(1, -5)$ $-B(0, 1) \rightarrow B'(0, -1)$ 2. Translate vertically by 3: $-A'(1, -5) \rightarrow A''(1, -2)$ $-B'(0, -1) \rightarrow B''(0, 2)$ 3. Rotate about the origin...	0.037	0.075	0.000	0.000	0.097
<i>Selective examples for individual atoms</i>						
0.159	At transform plate boundaries plates grind past each other side by side. This type of boundary separates the North American plate from the Pacific plate along the San Andreas fault...	0.095	0.022	0.000	0.000	0.000
0.000	how do i change my azure subscription?	0.000	0.239	0.000	0.000	0.000
0.202	who does optimus prime fight in transformers autobots	0.000	0.000	0.270	0.000	0.000
0.172	How should I wire a center-tapped transformer with input voltages of 0V and 230V and output voltages of 0V, 20V, 0V, and 20V? Can I safely connect the two middle outputs together...	0.049	0.000	0.000	0.199	0.000
0.060	To represent the combined rotation of a 3D point (x, y, z) , you can multiply the rotation matrices corresponding to each axis in the order of rotation: first Z-axis, then X-axis...	0.000	0.000	0.000	0.000	0.102

dict. size	i	$\hat{a}_i$ decoder molecule	recall on ah	recall on ab	recall on ap	recall on as
512	108	$0.91 a_a^* + 0.11 a_{as}^* + 0.08 a_{ab}^* + 0.03 a_{ap}^* + 0.02 a_{ah}^*$ 	1.000	0.962	0.877	0.993
2048	1457	$0.92 a_a^* + 0.10 a_{ah}^*$ 	1.000	0.236	0.339	0.388
	1794	$0.85 a_{ab}^* + 0.36 a_a^*$ 	0.000	0.873	0.003	0.000
	413	$0.84 a_{ap}^* + 0.34 a_a^*$ 	0.000	0.003	0.887	0.000
	183	$0.79 a_{as}^* + 0.36 a_a^*$ 	0.029	0.003	0.000	0.783
131K	38618	$1.00 a_a^*$ 	1.000	0.950	0.932	0.980
	121657	$1.00 a_{ah}^*$ 	0.871	0.000	0.000	0.000
	100965	$1.00 a_{ab}^*$ 	0.000	0.749	0.000	0.000
	120576	$1.00 a_{ap}^*$ 	0.000	0.003	0.798	0.000
	32170	$1.00 a_{as}^*$ 	0.029	0.000	0.000	0.765

Table 2: Empirical a-prefix feature splitting viewed in terms of molecules. A family molecule splits into parent-child molecules into atoms. In the middle regime, no molecule corresponds to a feature that activates on the full family (feature splitting). Such a feature exists in the small and large regimes.

Polysemanticity. Some SAE features are polysemantic, mixing loosely related concepts. In the molecular view, this is due to a direction that is a mixture of atomic directions.

Table 1 gives a direction in the 4K gemini SAE written as a molecule, with atoms corresponding to the word transform, as well as the Transformers franchise, electrical transformers, mathematical compositions of transforms, and the concept of change. Intuitively, these are mixed together in the small SAE since they are correlated.

The table shows how texts that activate the polysemantic 4K SAE feature also activate the proxy atomic features. For example, “are autobots stronger than decepticons?” activates the original feature, as well as the Transformers franchise feature in the 131K SAE, but not the other proxy atomic features.

We also find that many inputs that activate the broader “change” feature in 131K do not activate the polysemantic 4K feature. This is an example of asymmetry between the molecular representation of learned *directions* and the activation pattern of the corresponding feature. This plays a central role in the explanation for feature splitting below.

Feature splitting and absorption. Consider a family with one parent atom and two child atoms, and suppose that the parent feature activation is highly correlated with the child feature activation. Concretely, we can consider a pet atom, along with a dog atom and cat atom. Then consider what happens when a different number of features are “budgeted” to this family.

- If only one feature is budgeted, then the learned direction is a mixture of the parent direction

with small contributions from each child direction (this is the first un-normalized principal component).

(14)

This direction is used in the reconstruction of every input that activates the family. Therefore, its corresponding feature has the same activation pattern as the parent atom.

- If two features are budgeted, then the optimal learned directions include one for each parent-child pair; each is given by a sum of the parent direction and child direction.

(15)

Each molecule is used only for inputs that activate the specific child. Therefore, there is no learned feature that activates for all members of the family (giving “feature splitting”). Notice how this solution also minimizes description length (k).

- If three features are budgeted, then the learned directions can match the actual parent and child atoms (up to the re-parameterization given in (12)).

(16)

In particular, as shown in Theorem 3, there is a learned feature with an activation pattern corresponding to that of the parent and each child.

This gives a theoretical explanation of mesoscale feature splitting behavior that is consistent with P3. The example illustrates a second reason for feature splitting: over-restriction of the dictionary’s description length (sparsity).

Table 2 illustrates this behavior in real SAEs, where we have restricted our attention to a subset of proxy atoms that correspond to one- and two-letter “a” prefixes. At small dictionary size, there is a large molecule that is active for ah, ab, ap, and as. At intermediate size, there are instead molecules that mix the broad parent atom with the individual ah, ab, ap, and as child atoms. None are active for the whole family, and so the feature “a” appears to disappear (though its direction is still included in the molecules). Finally, at the largest size, individual atoms are recovered, including a feature that activates on the whole family.

Correlated directions forming semantic clusters. Various papers report seeing clusters of learned SAE directions that are correlated. As a concrete example, in Anthropic’s SAEs trained on Claude 3 Sonnet, Templeton et al. [24] find that “around the Golden Gate Bridge feature 34M/31164353 ... there are features corresponding to particular locations in San Francisco such as Alcatraz and the Presidio” [24]. Our model suggests that these directions are molecules rather than atoms: the directions are correlated since they share a common atom (corresponding to the shared geographical region). Indeed, Templeton et al. [24] note a broader correlated San Francisco direction in a smaller SAE, consistent with the feature splitting view presented above.

In underbudgeted regimes (in either dictionary size or description length) learning these molecules can be efficient. Our results suggest, however, SAEs with enough budget should learn individual atoms, rather than molecules. Then, at large budgets, we should see the observed correlation in directions decrease substantially.

We find evidence consistent with this prediction. Figure 26 compares the decoder directions selected as a-prefix probes in our 32K and 131K gemini SAEs. The median cosine over the 231 unique off-diagonal pairs is 0.144 at width 32K and 0.008 at width 131K. Direction correlation between extremely related features (e.g., the ‘ap’-prefix and the ‘ad’-prefix) can be very small or even negative.

Given the large size of the SAEs trained by Anthropic, our model predicts that these SAEs are underbudgeted in description length ($k < K$).⁶ The theory and observations here are consistent with work by Chanin et al. [22], who show that mis-specified k leads to correlation in learned feature directions, and suggest using orthogonality in learned features as a way to tune sparsity.

A broad implication is that many undesirable aspects of learned SAEs are explained by molecules. So problems can be solved by scaling SAEs, with appropriate sparsity [22], so that molecules split into their constituent atoms.

6 Discussion and Further Results

We developed an atomic theory of language model representations, deriving three testable predictions from the LRH and validating these predictions on SAEs trained on two large embedding models. (A natural next step is to test these predictions on other representations—e.g., residual streams.) Here, we discuss further applications of the atomic theory and connections to related work, alongside additional experiments.

The Recovery Principle. The recovery principle suggests the promise of scaling standard SAEs. In particular, large SAEs should be able to simultaneously recover stable coarse and granular features, reducing the need for incorporating hierarchical structure directly (cf., [23, 40–43]). Two aspects of the recovery principle are worth further investigation. First, Theorem 2 gives fairly weak guarantees in terms of the number of recovered atoms (e.g., a vanishing fraction of SAE size m). In Section G, we find that SAEs trained on simulated data can recover a much larger proportion of atomic features. Second, Theorem 2 requires exact sparsity K . While our results suggest that correct sparsity specification is important, our simulations again suggest some robustness. Practically, it is important to know when sparsity is being chosen correctly [25].

Multi-dimensional, non-linear, and manifold hypotheses. Some work has noted that representations can exhibit higher-dimensional non-linear structure [44–47]. In Section I, we suggest that these observations are consistent with the atomic theory. We show that correlations in feature activations alone (absent any correlation in their associated directions) are enough to produce manifold structure. Figure 27 shows that small sets of features recover manifolds on dates and hex colors. (As a “bonus,” Table 12 shows that these color features generalize across hex codes, text, images.)

A rigidity principle, platonic representations, and human interpretability. A growing literature documents increasing similarity in representations learned between different AI models [48–51]; work studying “algorithmic monoculture” has theoretically and empirically analyzed homogeneity in model outputs and its implications [52–62].

A “rigidity principle” illustrated by Theorems 1 to 3 suggests an explanation: that sparse systems of representation tend to involve the same concepts (recall, again, the vocabulary construction game). If AI models all use sparse representations, then these should be the same. Formally, we might

⁶In the case of Anthropic’s SAEs, which do not use a hard sparsity constraint, the sparsity penalty is too high.

hypothesize that different representations factor as $f = Az$ and $f = A'z$ for different choices of $A \in \mathbb{R}^{d \times M}$ and $A' \in \mathbb{R}^{d' \times M}$, where the atomic feature representation $z : X \rightarrow \mathbb{R}^M$ is shared. In the language of the platonic representation hypothesis [48], one could hypothesize a set of *platonic atoms*. Then, as in the recovery principle, larger and larger models could be learning a growing subset of these platonic atoms, explaining increasing representational similarity. (Empirically, homogeneity seems to be increasing with model size beyond what can be explained by increasing accuracy [59, 60].)

In Section J, we identify matches between features in SAEs trained on gemini and nemotron. We find that size-131K gemini and nemotron SAEs have 11K matching features at Pearson similarity threshold 0.7 and 27K matching features at threshold 0.5. Example shared features (shown in Tables 13 to 24) include “mechanical bearings,” “comments in source code,” “trade tariffs and protectionism,” “trigonometric half-angle identities,” and “astragalus and milkvetch species.”

The fact that SAE features trained on AI models are *human*-interpretable [15, 18, 19, 24] is quite amazing, and not obvious a priori: we could have imagined AI models learning different, inscrutable representations. One hypothesis is that since both humans and AI rely on sparse representations, there is pressure towards shared concepts.

Code

Code to reproduce data preparation, SAE training, and experiments is available in this repository: https://github.com/kennylpeng/atomic_theory.

AI Use

The conceptual ideas in the paper are those of the authors. We used AI to prove the ultimate theorem statements in the paper (after proving restricted versions ourselves) and help draft appendix sections. We reviewed and edited proofs (and rewrote proofs of central claims for clarity). Lean verifications for theoretical results are given in this repository: <https://github.com/nikhgarg/atomic-features-formalization/>, with autoformalization and verification using the AppliedModelingLib project [63].

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Appendix contents

A	Proof of Theorem 1 (Existence of Atomic Features)	26
B	Proofs of Theorems 2 and 3 (the Recovery Principle)	29
C	Experimental Setup	36
C.1	Data and SAE training details	36
C.2	Matching criteria and computation	38
C.3	P1: Persistence across SAE widths	40
C.4	P2: Shared features across training distributions	40
C.5	P3: Hierarchical datasets and feature-recovery protocol	41
D	Extended Experiments for P1: Persistence across Size	43
D.1	Fixed sparsity	43
D.2	Variation by seed	43
D.3	Exclusive matching	44
D.4	Persistence by activation similarity	44
D.5	Sensitivity to the matching threshold	45
D.6	Feature activation rates	46
D.7	Relationship between persistence and prevalence	49
E	Extended Experiments for P2: Stability across Distributions	50
E.1	Cross-distribution stability under exclusive matching	50
E.2	Cross-distribution stability by activation similarity	51
E.3	Feature prevalence and cross-distribution matching	52
F	Extended Experiments for P3: Hierarchical Feature Recovery	53
F.1	Hierarchical recovery at fixed sparsity	53
F.2	Counts and proportions of recovered categories and adjacent pairs	53
F.3	Recovery of letter-prefix hierarchies	53
F.4	Empirical extraction of feature hierarchies	57
G	Simulation experiments	61
G.1	Recovery under width and sparsity variation	61
G.2	Persistent features across simulated SAE widths	64
H	A Molecular Model of SAE Scaling	67
H.1	Computing the empirical molecules	67
I	An Atomic View of Feature Manifolds	71
J	An Atomic View of the Platonic Representation Hypothesis	75

A Proof of Theorem 1 (Existence of Atomic Features)

For a matrix $A \in \mathbb{R}^{d \times M}$, define

$$U_K(A) := \bigcup_{\substack{S \subseteq [M] \\ |S|=K}} \text{span}(A_S), \quad (17)$$

where $A_S := \{a_i\}_{i \in S}$.

Lemma 3. *We have that*

$$U_K(A) \subseteq U_{K'}(B) \quad (18)$$

Proof. Fix $S \subseteq [M]$ with $|S| = K$, and write $E_S := \text{span}\{e_i : i \in S\}$. For each $T \subseteq [M']$ with $|T| = K'$, define the linear subspace

$$L_T := \{w \in E_S : Aw \in \text{span}(B_T)\}. \quad (19)$$

Since $Az(x) = Bz'(x)$ and $z'(x)$ is at most K' -sparse,

$$z(X) \cap E_S \subseteq \bigcup_{\substack{T \subseteq [M'] \\ |T|=K'}} L_T. \quad (20)$$

By K -richness, these finitely many linear subspaces cannot all be proper subspaces of E_S . Thus $L_T = E_S$ for some T , which implies

$$\text{span}(A_S) = A(E_S) \subseteq \text{span}(B_T). \quad (21)$$

Taking the union over all $S \subseteq [M]$ with $|S| = K$ gives $U_K(A) \subseteq U_{K'}(B)$, as desired. $\square$

The inclusion (18) is the primary consequence of richness that we invoke.

We first prove that $K' \geq K$. Invoking Lemma 3, we have that $U_K(A) \subseteq U_{K'}(B)$. Consider $S \subseteq [M]$ with $|S| = K$. Then $\text{span}(A_S) \subseteq U_K(A) \subseteq U_{K'}(B)$. Since a finite-dimensional subspace cannot be covered by finitely many proper linear subspaces, there exists some $T \subseteq [M']$ with $|T| = K'$ such that $\text{span}(A_S) \subseteq \text{span}(B_T)$. Since $|S| = K < \text{spark}(A)$, as implied by the lower RIP, $\text{rank}(A_S) = K$. Therefore, $K' = \text{rank}(B_T) \geq K$.

Next we prove that if $M', K' = M, K$, then $(A, z) \equiv (B, u)$. Suppose that $M' = M$ and $K' = K$. Invoking Lemma 3, we have that $U_K(A) \subseteq U_{K'}(B)$. As in the proof above, for each $S \subseteq [M]$ with $|S| = K$, there must exist some $T \subseteq [M]$ with $|T| = K$ such that $\text{span}(A_S) \subseteq \text{span}(B_T)$. Since both are K -dimensional (since $\text{spark}(A), \text{spark}(B) > 2K$), we must have $\text{span}(A_S) = \text{span}(B_T)$. Define $g : \binom{[M]}{K} \rightarrow \binom{[M]}{K}$, $S \mapsto T$. Since $\text{span}(A_S) \neq \text{span}(A_{S'})$ when $S \neq S'$ (again from the spark condition), g is a bijection.

Moreover,

$$|S \cap S'| = \dim(\text{span}(A_S) \cap \text{span}(A_{S'})) = \dim(\text{span}(B_{g(S)}) \cap \text{span}(B_{g(S')})) = |g(S) \cap g(S')|. \quad (22)$$

Now for each $i \in [M]$, we can choose $S, S' \in \binom{[M]}{K}$ such that $S \cap S' = \{i\}$, so $g(S) \cap g(S') = \{j\}$ for some $j \in [M]$. This implies that

$$\text{span}(a_i) = \text{span}(A_S) \cap \text{span}(A_{S'}) = \text{span}(B_{g(S)}) \cap \text{span}(B_{g(S')}) = \text{span}(b_j). \quad (23)$$

So $b_j = a_i$ since columns are unit norm and coefficients must be non-negative. Since columns of A are pairwise linearly independent, each i maps to a distinct j according to a permutation matrix P .

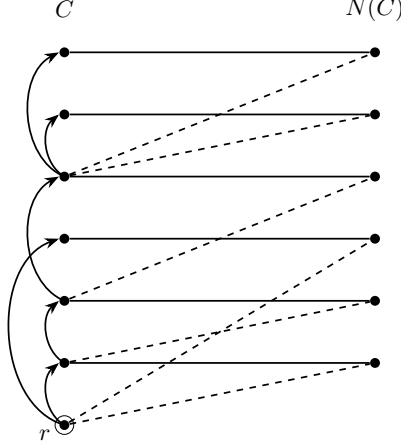


Figure 4: A sketch of how the tree construction completes the proof. Removing one vertex r from C , there exists a matching denoted by the solid lines from $C \setminus \{r\}$ to $N(C)$ (by the “Hall minimality” of C). We can then construct a tree on C where the existence of an edge ij implies that $m(j) \in F_i$ (so $F_i \cap F_j$ is non-empty). Splitting the tree into its two parts, this gives us a subset of C that has a large neighborhood, providing the desired contradiction.

Then for each $x \in X$, $Bz'(x) = Az(x)$. Since $\text{spark}(B) > 2K$, there is a unique K -sparse choice of $z'(x)$. In particular, $z'(x) = P^T z(x)$.

Finally, we show that if $M' < M$, then $K' \geq 2K$.

The proof rests on the following new characterization of when Hall’s condition holds: if Hall’s condition is satisfied for subsets of size at most $2K$ and subsets of size at most K have neighborhoods of size at most $2K - 1$, then Hall’s condition is satisfied globally.

Lemma 4 (Local-to-global Hall lemma). *Let $G = (X, Y, E)$ be a bipartite graph, and let $K \geq 1$. For $S \subseteq X$, let $N(S) := \{y \in Y : (x, y) \in E \text{ for some } x \in S\}$. Suppose that for every $S \subseteq X$,*

- (i) *if $|S| \leq 2K$, then $|N(S)| \geq |S|$;*
- (ii) *if $|S| \leq K$, then $|N(S)| \leq 2K - 1$.*

Then G satisfies Hall’s condition: $|N(S)| \geq |S|$ for every $S \subseteq X$. Consequently, there exists a matching that saturates X , and in particular $|Y| \geq |X|$.

Proof of Lemma 4. Assume for sake of contradiction that (i) and (ii) are satisfied, but that there exists $C \subseteq X$ such that $|N(C)| < |C|$. Take C to be inclusion minimal, in the sense that $|N(C')| \geq |C'|$ for all $C' \subsetneq C$. Then $|N(C)| = |C| - 1$, since otherwise we can freely remove any vertex from C to obtain C' violating Hall’s condition. By (i), we have that $|C| \geq 2K + 1$. We now show that (ii) is contradicted.

Choose any $r \in C$. By the minimality of C , every strict subset of C satisfies Hall’s condition. Therefore, we can construct a perfect matching $m : C \setminus \{r\} \rightarrow N(C)$.

We now iteratively construct a tree $T = (C, E_T)$ alongside a bijection $g : E_T \rightarrow N(C)$. Start with the set of vertices $R = \{r\}$ and the empty set of edges $E_T = \emptyset$. At each step, choose a vertex $j \in C \setminus R$ such that $m(j) \in N(\{i\})$ for some $i \in R$. Then add j to R , the edge ij to E_T , and set $g(ij) := m(j)$. Such a vertex j exists as long as $R \neq C$. Indeed, suppose that there were not such a vertex. This would mean that for every $u \in N(R)$, $m^{-1}(u) \in R \setminus \{r\}$. Since m is a bijection, this implies that $|N(R)| \leq |R| - 1$. Therefore, if $R \neq C$, this would contradict the minimality of C .

A key property of T is that if $g(ij) = m(j)$, then $m(j) \in N(\{i\})$ and $m(j) \in N(\{j\})$.

Now take any subtree of T with $2K + 1$ vertices (equivalently, $2K$ edges). Let the two sides of the tree (which is bipartite) have vertices S_1 and S_2 . Without loss of generality, suppose $|S_1| \leq K$. Every edge ij in the subtree has one endpoint in S_1 . Therefore, for each such edge ij , $g(ij)$ is in $N(S_1)$. Since g is a bijection, this implies that $|N(S_1)| \geq 2K$. This contradicts (ii), as desired.

Finishing the proof. We now complete the proof. Invoking Lemma 3, we have that $U_K(A) \subseteq U_{K'}(B)$. Therefore, for all $i \in [M]$, since $a_i \in U_K(A) \subseteq U_{K'}(B)$, a_i has K' -sparse representation in B . Since $\text{spark}(B) > 2K'$, this representation is unique. Let $F_i \subseteq [M']$ denote the support of this representation.

Now consider a bipartite graph $G = ([M], [M'], E)$ where $E = \{(i, j) : i \in [M], j \in F_i\}$. Then we have $N(S) = \cup_{i \in S} F_i \subseteq [M']$.

Our approach is to show that if $K' \leq 2K - 1$, then G satisfies the two conditions in Lemma 4, implying that $M' \geq M$.

To show (i) holds, observe that since $|S| \leq 2K < \text{spark}(A)$, then $\text{rank}(A_S) = |S|$. Also, $\text{span}(A_S) \subseteq \text{span}(B_{\cup_{i \in S} F_i})$. Now observe that $|N(S)| = |\cup_{i \in S} F_i| \geq \text{rank}(B_{\cup_{i \in S} F_i}) \geq \text{rank}(A_S) = |S|$. (Notice that this property holds regardless of K' .)

To show (ii) holds, observe that if $|S| \leq K$, $\text{span}(A_S) \subseteq U_K(A) \subseteq U_{K'}(B)$. Since a subspace can't be covered by the union of a finite number of strictly lower-dimensional subspaces, there exists $T_S \subseteq [M']$ with $|T_S| = K'$ such that $\text{span}(A_S) \subseteq \text{span}(B_{T_S})$. For each $i \in S$, since $a_i \in \text{span}(A_S)$ and F_i is the support of the unique K' -sparse representation of a_i in B , then $F_i \subseteq T_S$. Therefore, $|N(S)| = |\cup_{i \in S} F_i| \leq |T_S| \leq K' \leq 2K - 1$, as desired.

Define a featurization of f to be a dictionary that exactly represents f .

Theorem 4 (Generic rigidity). *Fix d, M, K with $M > d \geq 2$. For almost every $A = (a_1, \dots, a_M) \in (\mathbb{S}^{d-1})^M$, the following holds. Suppose that (A, z) is a K -rich (M, K) -featurization of $f : X \rightarrow \mathbb{R}^d$. Then every (M', K') -featurization (A', z') of f , for $M' > d$, satisfies*

$$K' \geq 1 + (d - 1) \frac{M - M'}{M}. \quad (24)$$

Since $d + 1 \geq \text{spark}(A') > 2K'$, (24) implies that $M' > \frac{M}{2}$.

Proof of Theorem 4. Suppose that (A, z) is a K -rich (M, K) -featurization of $f : X \rightarrow \mathbb{R}^d$. Then by Lemma 3, we have that $a_i \in U_{K'}(B)$. We show that if

$$K' < 1 + (d - 1) \frac{M - M'}{M}, \quad (25)$$

then for almost every $A = (a_1, a_2, \dots, a_M) \in (\mathbb{S}^{d-1})^M$, there does not exist $B \in \mathbb{R}^{d \times M'}$ such that $a_1, a_2, \dots, a_M$ all have K' -sparse representations in B . (Notice that this argument does not require that $U_{K'}(B)$ include all of $U_K(A)$, but rather just $a_1, a_2, \dots, a_M$, so it seems natural that the bound given here could be further improved.)

The idea is a dimension counting argument on parameter spaces. In particular, we first define a smooth manifold that parameterizes A . We then define a smooth manifold that parameterizes (M', K') -featurizations B, z' that can represent $a_1, a_2, \dots, a_M$. Finally, we define a smooth map between the B -manifold and the A -manifold, with the property that every representable A -featurization must be in the image of this map. Then invoking Sard's theorem, if the dimension of the B -manifold is smaller than the A -manifold, then the image of the map has measure zero. So for a measure one subset of A , no (M, K) -featurization A, z has an (M', K') -featurization.

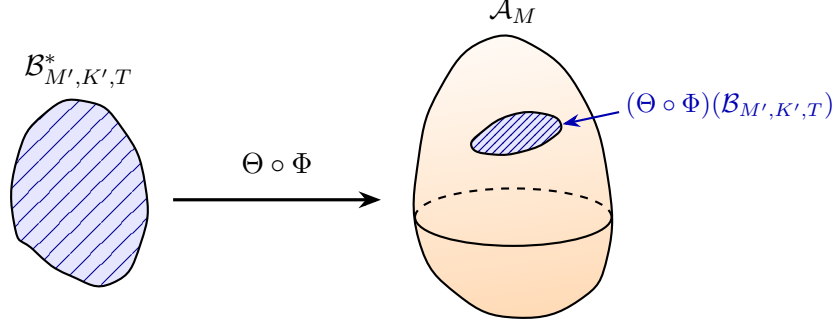


Figure 5: When $\dim(\mathcal{B}_{M',K',S}) < \dim(\mathcal{A}_M)$, the set of (M, K) -factorizations that also have an (M', K') -factorization is measure zero under a generic distribution over (M, K) -factorizations.

Let $\mathcal{A}_M = (\mathbb{S}^{d-1})^M$. Let $\mathcal{B}_{M',K',T} = (\mathbb{S}^{d-1})^{M'} \times \prod_{i=1}^M S^{|T_i|-1}$. Let $T = (T_1, \dots, T_M)$ where $T_i \subseteq [M']$ with $1 \leq |T_i| \leq K'$. Define the smooth function $\Phi : \mathcal{B}_{M',K',T} \rightarrow (\mathbb{R}^d)^M$ such that

$$(B, \gamma_1, \gamma_2, \dots, \gamma_M) \mapsto \left(\sum_{j \in T_1} \gamma_{1j} b_j, \sum_{j \in T_2} \gamma_{2j} b_j, \dots, \sum_{j \in T_M} \gamma_{Mj} b_j \right). \quad (26)$$

Now define $\Theta : (\mathbb{R}^d \setminus \{0\})^M \rightarrow \mathcal{A}_M : (a'_1, a'_2, \dots, a'_M) \mapsto (\frac{a'_1}{\|a'_1\|}, \frac{a'_2}{\|a'_2\|}, \dots, \frac{a'_M}{\|a'_M\|})$, which is also smooth. Let $\mathcal{B}_{M',K',T}^* = \Phi^{-1}((\mathbb{R}^d \setminus \{0\})^M)$. Since $(\mathbb{R}^d \setminus \{0\})^M$ is open, $\mathcal{B}_{M',K',T}^*$ is itself a smooth manifold. $\mathcal{B}_{M',K',T}^*$ has dimension at most $M'(d-1) + M(K'-1)$. $\mathcal{A}_M$ has dimension $M(d-1)$. So if

$$M'(d-1) + M(K'-1) < M(d-1), \quad (27)$$

by Sard's theorem, $(\Theta \circ \Phi)(\mathcal{B}_{M',K',T}^*) \subseteq \mathcal{A}_M$ has measure zero. Now observe that every $(a_1, a_2, \dots, a_M)$ that has a K' -sparse decomposition in $(b_1, b_2, \dots, b_{M'})$ must lie in the image of $(\Theta \circ \Phi)(\mathcal{B}_{M',K',T}^*)$ for some T . Since there are a finite number of supports T , and the union of a finite number of measure zero sets has measure zero, this implies that with probability one, $(a_1, a_2, \dots, a_M)$ does not have a K' -sparse decomposition in any $(b_1, b_2, \dots, b_{M'})$. Similarly, we can repeat the argument for all M', K' satisfying (27).

Finally, observe that since $d+1 \geq \text{spark}(A') > 2K'$, then $K'-1 < \frac{d-1}{2}$, so

$$\frac{d-1}{2} > (d-1) \frac{M-M'}{M} \quad (28)$$

$$\frac{M}{2} > M - M' \quad (29)$$

$$M' > \frac{M}{2}, \quad (30)$$

as desired.

B Proofs of Theorems 2 and 3 (the Recovery Principle)

Let us recall our assumptions. We assume throughout that A and $\hat{A}$ have unit-norm columns and satisfy the $(\gamma, 2K)$ -lower restricted isometry property. To simplify, we simply say that a matrix is $(\gamma, 2K)$ -stable. We also assume that $z(x)$ and $\hat{z}(x)$ are nonnegative, and that $\hat{z}(x)$ is at most K -sparse. The atomic support $S(x) := \text{supp}(z(x))$ has size K almost surely. Conditional on any

positive-probability support S , the coefficients $z_S(x)$ lie in $(0, 1)^K$ and have density h_S satisfying $c_- \leq h_S \leq c_+$. The densities may differ across supports. All probabilities are with respect to $\mathcal{D}$. We abbreviate $\hat{A}^{(\mathcal{D}, m)}, \hat{z}^{(\mathcal{D}, m)}$ by $\hat{A}, \hat{z}$ and write

$$E := L_{\mathcal{D}}(\hat{A}, \hat{z}), \quad E_S := \mathbb{E}_{x \sim \mathcal{D}} \left[\|Az(x) - \hat{A}\hat{z}(x)\|_2^2 \mid S(x) = S \right].$$

For equal-dimensional subspaces $U, V \subseteq \mathbb{R}^d$, let $\text{gap}(U, V) := \sup_{u \in U: \|u\|_2=1} \text{dist}(u, V)$; the same value is obtained after exchanging U, V . We set $\text{gap}(\{0\}, \{0\}) = 0$. The first three lemmas do not require regularity, so we can use them for both the ordinary and hierarchical settings.

The core idea is the following lemma, which shows that if a full atomic support span is accurately reconstructed by the learned dictionary, then there must be a single learned support whose span matches the atomic span (in a way that is independent of m and M). Then, by intersecting these spans, we can show the recovery of individual atoms. A key challenge in showing the result is that the atomic span can, in theory, be “tiled” by many different learned spans (the number of which depends on m).

Lemma 5. *For every $\epsilon > 0$, there exists $\delta = \delta(\epsilon, K, \gamma, c_-) > 0$ such that any positive-probability support S with $E_S \leq \delta$ has a learned support $T \subseteq [m]$ of size K satisfying $\text{gap}(\text{span}(A_S), \text{span}(\hat{A}_T)) \leq \epsilon$.*

Proof. Fix $\epsilon > 0$. Set $r_0 := \gamma/4$ and $r_{j+1} := \gamma r_j / (16K)$ for $j = 0, \dots, K-1$. Write $R := r_K$ and choose $\alpha := R \min\{\epsilon, 1/8\}$. Note that these quantities are independent of m, M .

Define $U_K(\hat{A}) := \bigcup_{T \subseteq [m]: |T| \leq K} \text{span}(\hat{A}_T)$ and $g(z) := \text{dist}(A_S z, U_K(\hat{A}))$. The conditional density lower bound gives $E_S \geq c_- \int_{(0,1)^K} g(z)^2 dz$. Moreover, since distance to a fixed set is 1-Lipschitz and $\|A_S\|_{\text{op}} \leq \sqrt{K}$, we have $|g(z) - g(z')| \leq \sqrt{K} \|z - z'\|_2$.

Suppose $g(z_0) > \alpha$ at some $z_0 \in [1/4, 3/4]^K$. Then $g(z) > \alpha/2$ throughout the coefficient ball of radius $\alpha/(2\sqrt{K})$ centered at z_0 . This ball lies inside $(0, 1)^K$ since unit norms and stability imply $\gamma \leq 1$, and our choice of α makes its radius smaller than $1/4$. Writing ω_K for the volume of the unit ball in $\mathbb{R}^K$, choose

$$\delta := \frac{c_- \omega_K}{2^{K+3} K^{K/2}} \alpha^{K+2}.$$

The contribution of this ball would give $E_S \geq c_- (\alpha^2/4) \omega_K (\alpha/(2\sqrt{K}))^K = 2\delta$, a contradiction. Thus $g(z) \leq \alpha$ throughout $[1/4, 3/4]^K$.

Write $V := \text{span}(A_S)$ and $y_0 := A_S(1/2, \dots, 1/2)$. Since $\sigma_{\min}(A_S) \geq \gamma$, the image $A_S[1/4, 3/4]^K$ contains the ball $\mathcal{B}_0 := B_V(y_0, r_0)$. Indeed, any displacement in V of norm at most $\gamma/4$ equals $A_S w$ for some $\|w\|_2 \leq 1/4$. Here and below, B_V denotes a closed ball taken within V .

For each $y \in \mathcal{B}_0$, choose a best signed K -sparse coefficient vector $v(y)$. Such a vector exists because $U_K(\hat{A})$ is a finite union of subspaces. The preceding bound gives $\|y - \hat{A}v(y)\|_2 \leq \alpha$. Since $v(y) - v(y')$ is at most $2K$ -sparse, stability and the triangle inequality imply that for all $y, y' \in \mathcal{B}_0$,

$$\|v(y) - v(y')\|_2 \leq \frac{\|\hat{A}(v(y) - v(y'))\|_2}{\gamma} \leq \frac{\|y - y'\|_2 + 2\alpha}{\gamma} \quad (31)$$

We now construct nested balls $\mathcal{B}_j := B_V(y_j, r_j)$ and sets $T_j \subseteq [m]$ of size j such that $T_j \subseteq \text{supp}(v(y))$ throughout $\mathcal{B}_j$. Initially, take $T_0 = \emptyset$.

Suppose $\mathcal{B}_j, T_j$ have been constructed, with $j < K$. Since $\dim V = K > j$, choose a unit vector in V orthogonal to $\text{span}(\hat{A}_{T_j})$. At least one of the two points obtained by displacing y_j by $\pm r_j/2$ in

this direction has distance at least $r_j/2$ from $\text{span}(\hat{A}_{T_j})$. Call this point y_{j+1} . Then, since errors are at most α and since columns are unit-norm,

$$\frac{r_j}{2} \leq \text{dist}(y_{j+1}, \text{span}(\hat{A}_{T_j})) \leq \alpha + \sum_{\ell \notin T_j} |v_\ell(y_{j+1})|. \quad (32)$$

Since $\alpha \leq r_j/4$ and $v(y_{j+1})$ has at most K nonzero entries, some $i \notin T_j$ satisfies $|v_i(y_{j+1})| \geq r_j/(4K)$.

Set $T_{j+1} := T_j \cup \{i\}$ and $\mathcal{B}_{j+1} := B_V(y_{j+1}, r_{j+1})$. This ball lies inside $\mathcal{B}_j$, since its center is displaced by $r_j/2$ and $r_{j+1} \leq r_j/2$. For every $y \in \mathcal{B}_{j+1}$, (31) and $2\alpha \leq r_{j+1}$ give $|v_i(y) - v_i(y_{j+1})| \leq 2r_{j+1}/\gamma = r_j/(8K)$. Hence $|v_i(y)| \geq r_j/(8K) > 0$ throughout the smaller ball. The coordinates in T_j remain nonzero by the induction hypothesis, completing the construction.

After K steps, $T := T_K$ has size K , so sparsity forces $\text{supp}(v(y)) = T$ throughout $\mathcal{B}_K$. Consequently, $\text{dist}(y, \text{span}(\hat{A}_T)) \leq \alpha$ for every $y \in B_V(y_K, R)$.

Finally, let P_T be the orthogonal projection onto $\text{span}(\hat{A}_T)$. For any unit vector $w \in V$, both $y_K + Rw$ and $y_K - Rw$ belong to the final ball. Therefore,

$$2R\|(I - P_T)w\|_2 \leq \|(I - P_T)(y_K + Rw)\|_2 + \|(I - P_T)(y_K - Rw)\|_2 \leq 2\alpha.$$

Noting that I is the projection onto $\text{span}(A_S)$ and that both A_S and $\hat{A}_T$ have dimension K , so taking the supremum over unit vectors $w \in V$ bounds their gap by $\alpha/R \leq \epsilon$. $\square$

The next lemma shows that approximate matching of support spans preserves their intersections. Importantly, its bound does not depend on the number of supports being intersected.

Lemma 6. *Suppose a nonempty finite collection of pairs S_ℓ, T_ℓ , with $|S_\ell| = |T_\ell| = K$, satisfies*

$$\text{gap}(\text{span}(A_{S_\ell}), \text{span}(\hat{A}_{T_\ell})) \leq r \quad \text{for every } \ell.$$

Write $I := \bigcap_\ell S_\ell$ and $J := \bigcap_\ell T_\ell$. If $r(1 + 2K/\gamma) < 1$, then

$$|I| = |J|, \quad \text{gap}(\text{span}(A_I), \text{span}(\hat{A}_J)) \leq r(1 + 2K/\gamma).$$

Proof. Take a unit vector $u \in \text{span}(A_I)$. For each ℓ , choose $v_\ell \in \mathbb{R}^m$, supported on T_ℓ , such that $\|u - \hat{A}v_\ell\|_2 \leq r$. Fix one pair, indexed by 1. For every $j \in T_1 \setminus J$, some T_ℓ omits j . Stability then gives $|(v_1)_j| \leq \|v_1 - v_\ell\|_2 \leq 2r/\gamma$. Deleting these at most K coordinates from v_1 , and using the unit norms of the learned columns, gives

$$\text{dist}(u, \text{span}(\hat{A}_J)) \leq \|u - \hat{A}v_1\|_2 + \sum_{j \in T_1 \setminus J} |(v_1)_j| \leq r + 2Kr/\gamma.$$

Exchanging A, S_ℓ with $\hat{A}, T_\ell$ gives the same bound in the reverse direction. If the two intersection spans had different dimensions, the larger would contain a unit vector orthogonal to the smaller, contradicting the bound being less than 1. Their dimensions are $|I|$ and $|J|$ by stability, proving both claims. This argument also covers the case in which an intersection is empty. $\square$

We next connect these geometric matches with the actual learned activations. This step uses both nonnegativity of $\hat{z}$ and the upper density bound. It does not assume that $\hat{z}(x)$ minimizes the reconstruction error for each x .

Lemma 7 (Activation recovery). *For every $\eta \in (0, 1)$, there exist $r, c, \tau > 0$, depending only on η, K, γ, c_+ , with the following property.*

Let $\mathcal{G}$ be a collection of positive-probability atomic supports. Suppose each $S \in \mathcal{G}$ has a learned support T_S of size K satisfying $\text{gap}(\text{span}(A_S), \text{span}(\hat{A}_{T_S})) \leq r$. Suppose also that an atomic coordinate i of prevalence $p_i := \Pr(z_i > 0) > 0$ and a learned coordinate j satisfy $j \in T_S$ if and only if $i \in S$, for every $S \in \mathcal{G}$.

If $E + \Pr(S(x) \notin \mathcal{G}) \leq cp_i$, then

$$\text{F1}_{\mathcal{D}}(\mathbf{1}\{z_i > 0\}, \mathbf{1}\{\hat{z}_j > \tau\}) > \eta.$$

Proof. We first derive a bound for any $0 \leq r \leq 1/2$ and $\tau > 0$ satisfying $2Kr \leq \gamma\tau$. Write $\beta := \Pr(S(x) \notin \mathcal{G})$.

Fix $S \in \mathcal{G}$, and let P_{T_S} denote orthogonal projection onto $\text{span}(\hat{A}_{T_S})$. Let H_S be the $K \times K$ matrix satisfying $\hat{A}_{T_S} H_S = P_{T_S} A_S$. It exists and is unique because $\hat{A}_{T_S}$ has full column rank. For an input with support S , let $v_S(x) \in \mathbb{R}^m$ have coordinates $H_S z_S(x)$ on T_S and zero elsewhere. These auxiliary coefficients may have either sign.

Since the atomic coefficients lie in $(0, 1)^K$ and the columns have unit norm, $\|Az(x)\|_2 \leq K$. The gap bound therefore gives $\|Az(x) - \hat{A}v_S(x)\|_2 \leq Kr$. Writing $e(x) := \|Az(x) - \hat{A}\hat{z}(x)\|_2$, stability and the triangle inequality imply

$$\|\hat{z}(x) - v_S(x)\|_2 \leq \frac{\|\hat{A}(\hat{z}(x) - v_S(x))\|_2}{\gamma} \leq \frac{e(x) + Kr}{\gamma}. \quad (33)$$

Here stability applies because $\hat{z}(x) - v_S(x)$ is at most $2K$ -sparse.

If $j \in T_S$, the coordinate $(v_S(x))_j$ is a linear function of $z_S(x)$ with at least one coefficient of magnitude $\gamma/(2K)$ or larger. To see this, for every $w \in \mathbb{R}^K$,

$$\sqrt{K}\|H_S w\|_2 \geq \|P_{T_S} A_S w\|_2 \geq \sqrt{1 - r^2} \|A_S w\|_2 \geq \frac{\gamma}{2} \|w\|_2.$$

Thus $\sigma_{\min}(H_S) \geq \gamma/(2\sqrt{K})$. Since H_S is square, each row has at least this norm, and hence has an entry of magnitude at least $\gamma/(2K)$.

Holding the other $K - 1$ atomic coefficients fixed, the values of the corresponding coefficient for which $|(v_S(x))_j| \leq 2\tau$ form an interval of length at most $8K\tau/\gamma$. Integrating over the remaining coordinates and using $h_S \leq c_+$ gives

$$\Pr(|(v_S(x))_j| \leq 2\tau \mid S(x) = S) \leq \frac{8Kc_+\tau}{\gamma}. \quad (34)$$

No independence of the coefficients is needed.

By Markov's inequality, outside an event of probability at most $\beta + 4E/(\gamma^2\tau^2)$, we have $S(x) \in \mathcal{G}$ and $e(x) \leq \gamma\tau/2$. On this complement, (33) and $Kr \leq \gamma\tau/2$ give $\|\hat{z}(x) - v_S(x)\|_2 \leq \tau$, where $S = S(x)$.

If $i \notin S$, then $j \notin T_S$, so $(v_S(x))_j = 0$ and $\hat{z}_j(x) \leq \tau$. Thus there is no false positive on this complement. If $i \in S$ but $\hat{z}_j(x) \leq \tau$, nonnegativity of $\hat{z}_j(x)$ implies $-\tau \leq (v_S(x))_j \leq 2\tau$. Summing (34) over supports in $\mathcal{G}$ containing i , the probability of these false negatives is at most $(8Kc_+\tau/\gamma)p_i$.

Let $Y := \mathbf{1}\{z_i > 0\}$ and $\hat{Y} := \mathbf{1}\{\hat{z}_j > \tau\}$. The total disagreement probability is therefore at most $\beta + 4E/(\gamma^2\tau^2) + (8Kc_+\tau/\gamma)p_i$. Since $\Pr(Y = 1) = p_i > 0$, the definition of the F1 score gives

$$\begin{aligned} \text{F1}_{\mathcal{D}}(Y, \hat{Y}) &= 1 - \frac{\Pr_{\mathcal{D}}(Y \neq \hat{Y})}{\Pr_{\mathcal{D}}(Y = 1) + \Pr_{\mathcal{D}}(\hat{Y} = 1)} \\ &\geq 1 - \frac{\beta}{p_i} - \frac{4E}{\gamma^2\tau^2 p_i} - \frac{8Kc_+\tau}{\gamma}. \end{aligned} \quad (35)$$

Finally, choose $\tau > 0$ sufficiently small that $8Kc_+\tau/\gamma < (1 - \eta)/2$, and choose $0 < r \leq \min\{1/2, \gamma\tau/(2K)\}$. Choose $c > 0$ sufficiently small that $c(1 + 4/(\gamma^2\tau^2)) < (1 - \eta)/2$. These choices depend only on η, K, γ, c_+ . If $E + \beta \leq cp_i$, then the sum of the first two error terms in (35) is at most $c(1 + 4/(\gamma^2\tau^2))$, so the F1 score exceeds η . $\square$

The ordinary recovery principle. We now impose regularity of the support law: for a fixed $\rho \in (0, 1]$ and distinct atoms i, k with $p_i > 0$, assume $\Pr(k \notin S(x) \mid i \in S(x)) \geq \rho$. Let σ order the atom prevalences nonincreasingly and let $T(m) := \sum_{j=m+1}^M p_{\sigma(j)}$. Write $L_{\mathcal{D}}^*(m)$ for the infimum of the dictionary learning loss over the unit-norm, $(\gamma, 2K)$ -stable reference class with nonnegative, at most K -sparse codes. We do not need the infimum to be attained.

Proof of Theorem 2. Retain the m most prevalent columns of A , together with their true coefficients. This gives an admissible dictionary. For every input, the squared norm of the omitted sum is at most $K \sum_{k \notin \sigma([m])} z_k(x)^2$, which is at most K times the number of omitted active atoms. Therefore,

$$E \leq C_0 L_{\mathcal{D}}^*(m) \leq C_0 K T(m). \quad (36)$$

We will show that an atom of prevalence $p_i > 0$ is recovered whenever $E \leq cp_i$, for a sufficiently small constant $c > 0$.

It suffices to take $0 < \eta < 1$. Let r_*, c_*, τ be the constants supplied by Lemma 7. Choose $0 < r \leq r_*$ small enough that $r(1 + 2K/\gamma) < \min\{\gamma/(4\sqrt{2}), \sqrt{1 - \eta^2}\}$. Let δ be the threshold supplied by Lemma 5 with $\epsilon = r$, decreasing it if necessary so that $\delta < c_- \gamma^2 (\gamma/(8K))^{K-1}/32$. Finally, choose $c > 0$ such that $c < \rho\delta$ and $c(1 + \delta^{-1}) \leq c_*$. These choices depend only on the parameters in the theorem.

Call a positive-probability support S good if $E_S \leq \delta$, and write $\mathcal{G}$ for the good supports. Their complement has probability $\beta \leq E/\delta$. For each good support, choose T_S as in Lemma 5, so its span has gap at most r from $\text{span}(A_S)$.

Fix an atom i with $p_i > 0$ and $E \leq cp_i$, and let $\mathcal{G}_i := \{S \in \mathcal{G} : i \in S\}$. Since $\beta < \rho p_i \leq p_i$, this collection is nonempty. For every $k \neq i$, regularity gives $\Pr(i \in S(x), k \notin S(x)) \geq \rho p_i > \beta$. Hence some good support contains i and omits k , implying $\bigcap_{S \in \mathcal{G}_i} S = \{i\}$. Lemma 6 now gives a single learned coordinate j such that

$$\bigcap_{S \in \mathcal{G}_i} T_S = \{j\}, \quad \text{gap}(\text{span}(a_i), \text{span}(\hat{a}_j)) \leq r(1 + 2K/\gamma). \quad (37)$$

Thus $j \in T_S$ whenever a good support contains i . If a good support S omits i , adding S to the preceding intersection makes the atomic intersection empty. Lemma 6 makes the learned intersection empty as well, so $j \notin T_S$. We have proved $j \in T_S$ if and only if $i \in S$ on all good supports, not just on those containing i .

The line bound in (37) gives $|\langle a_i, \hat{a}_j \rangle| > \eta$. We must also fix its sign. If the inner product were negative, the same bound and our choice of r would give $\|a_i + \hat{a}_j\|_2 < \gamma/4$. Choose any $S \in \mathcal{G}_i$ and consider coefficients with $z_i \in (1/2, 1)$ and $z_k \in (0, \gamma/(8K))$ for $k \in S \setminus \{i\}$. Since $\hat{z}$ is nonnegative and at most K -sparse, $\hat{z} + z_i e_j$ is at most $2K$ -sparse and has norm at least z_i . Stability and the triangle inequality therefore give, on this coefficient region,

$$\|Az - \hat{A}\hat{z}\|_2 \geq (\gamma - \|a_i + \hat{a}_j\|_2)z_i - \sum_{k \in S \setminus \{i\}} z_k \geq \gamma/4.$$

The density lower bound gives this region conditional probability at least $(c_-/2)(\gamma/(8K))^{K-1}$. Thus $E_S \geq c_- \gamma^2 (\gamma/(8K))^{K-1}/32$, contradicting goodness and our choice of δ . Consequently, $\langle a_i, \hat{a}_j \rangle > \eta$, as required for signed direction recovery.

For activation recovery, observe that $E + \beta \leq E(1 + \delta^{-1}) \leq c_* p_i$. Lemma 7 therefore applies to the same coordinate j and gives an F1 score greater than η at threshold τ . This threshold is shared by all recovered coordinates. Finally, take $C := C_0 K / c$. By (36), the condition $p_i \geq CT(m)$ implies $E \leq cp_i$ and completes the proof. $\square$

The hierarchical recovery principle. We make precise the hierarchical regularity invoked in the main text. Let $M = 3N$ and $K = 2L$, with families i, i_1, i_2 for $i \in [N]$. An input chooses a set $F(x) \subseteq [N]$ of exactly L active families. Each active family contributes its parent and exactly one child, so $S(x) = \bigcup_{i \in F(x)} \{i, i_{b_i(x)}\}$ with $b_i(x) \in \{1, 2\}$. The law of $F(x)$ satisfies $\Pr(k \notin F(x) \mid i \in F(x)) \geq \rho$ for distinct parents i, k whenever $\Pr(i \in F(x)) > 0$. In addition, for a fixed $\kappa \in (0, 1/2]$, we require

$$\Pr(b_i(x) = b \mid F(x) = F) \geq \kappa \quad (i \in F, b \in \{1, 2\}) \quad (38)$$

for every positive-probability family set F . Conditional on each resulting atomic support, all K coefficients have a joint density between c_- and c_+ on $(0, 1)^K$, as above. We call such a distribution hierarchical regular.

The child choices need not be independent, and neither do the active coefficients. Independent uniform child choices give the special case $\kappa = 1/2$. The child-choice bound and the full conditional density bounds are part of this definition; they do not follow from the child-implies-parent condition alone.

As in the main text, now let $p_i := \Pr(i \in F(x))$ be the parent prevalences, let σ sort these prevalences, and let $T(n) := \sum_{j=n+1}^N p_{\sigma(j)}$. For a learned dictionary of width m , we set $n := \lfloor m/3 \rfloor$, with $T(0) := \sum_{i=1}^N p_i = L$.

Proof of Theorem 3. Retain all three atoms in each of the $n = \lfloor m/3 \rfloor$ most prevalent families. If $3n < m$, fill the remaining slots with unused columns of A and set their codes to zero. This gives an admissible dictionary of exactly m unit-norm columns. Each omitted active family contributes two omitted atoms, so the same truncation calculation as before gives

$$E \leq C_0 L_{\mathcal{D}}^*(m) \leq 2C_0 K T(n). \quad (39)$$

We show that a family of prevalence $p_i > 0$ is recovered whenever $E \leq cp_i$, for a sufficiently small constant $c > 0$.

Again it suffices to take $0 < \eta < 1$. Let r_*, c_*, τ be the constants from Lemma 7. Choose $0 < r \leq r_*$ with $r(1 + 2K/\gamma) < 1$, and let δ be the threshold from Lemma 5 with $\epsilon = r$. Choose $c > 0$ so that $c < \kappa\rho\delta$ and $c(1 + \delta^{-1}) \leq \kappa c_*$. As before, let $\mathcal{G}$ contain the positive-probability supports with $E_S \leq \delta$, choose their learned supports T_S , and write $\beta \leq E/\delta$ for the probability of the remaining supports.

Fix a family i with $p_i > 0$ and $E \leq cp_i$. Each child has prevalence at least κp_i . Moreover, for a different family k , (38) and parent regularity give

$$\Pr(i_b \in S(x), k \notin F(x)) \geq \kappa \Pr(i \in F(x), k \notin F(x)) \geq \kappa \rho p_i > \beta.$$

Thus, even after discarding the bad supports, either child can occur with i , and each chosen child can occur without any specified other family. It follows that the collections below are nonempty and

$$\bigcap_{S \in \mathcal{G}: i \in S} S = \{i\}, \quad \bigcap_{S \in \mathcal{G}: i_b \in S} S = \{i, i_b\} \quad (b \in \{1, 2\}). \quad (40)$$

Indeed, the other child is excluded whenever i_b occurs, and the displayed probability bound excludes every atom belonging to any other family.

Applying Lemma 6 to these intersections gives learned sets

$$J_i := \bigcap_{S \in \mathcal{G}: i \in S} T_S = \{j_i\}, \quad J_{i_b} := \bigcap_{S \in \mathcal{G}: i_b \in S} T_S = \{j_i, j_{i_b}\}.$$

Here $J_i \subseteq J_{i_b}$ because every support containing i_b also contains i . The good supports containing i are exactly the union of those containing i_1 and those containing i_2 . Hence $J_{i_1} \cap J_{i_2} = J_i$, so j_i, j_{i_1}, j_{i_2} are distinct.

We claim that each of these learned coordinates has the same membership pattern as its corresponding atom across all good supports. For the parent, this is the same argument as in Theorem 2: adding a good support omitting i to its intersection forces that support's learned set to omit j_i . For a child i_b , every good support containing i_b contains J_{i_b} on the learned side. If a good support omits the family, its atomic intersection with $\{i, i_b\}$ is empty, so its learned intersection with J_{i_b} is empty as well. If it instead contains i and the other child, that atomic intersection is $\{i\}$. Lemma 6 makes the learned intersection a singleton, which must be $\{j_i\}$. In particular it omits j_{i_b} . This proves the claimed membership patterns.

Finally, $E + \beta \leq E(1 + \delta^{-1}) \leq \kappa c_* p_i$, and each of the three target coordinates has prevalence at least κp_i . Lemma 7 therefore gives an F1 score greater than η for each target, using the distinct learned coordinates j_i, j_{i_1}, j_{i_2} . The same threshold τ works for all three and for all recovered families. Take $C := 2C_0 K/c$ and invoke (39). This proves the theorem. $\square$

C Experimental Setup

This section describes the shared data, training setup, and matching metrics, followed by the setup for the main P1, P2, and P3 experiments in Figures 2a to 2c, respectively. Extended experiments (including results and setup) are discussed in Sections D to F.

Code. Code to reproduce data preparation, SAE training, and evaluation are given in the repository at https://github.com/kennylpeng/atomic_theory.

C.1 Data and SAE training details

Training corpus. The full training mix, `full`, combines 13 source datasets used to fine-tune the Nemotron embedding model. Table 3 counts the stored *embedding vectors* available to SAE training: a question and its answer, for example, are embedded separately and contribute two vectors.

Table 3: SAE training datasets.

Source collection	# embeddings
Emotion [64]	19,930
FEVER [65, 66]	11,079,420
GooAQ [67]	6,024,992
HotpotQA [68]	10,564,510
MS MARCO [69]	9,351,785
Natural Questions [70]	200,462
NFCorpus [71]	10,503
PAQ, sampled 10M pairs [72]	20,000,000
SciFact [73]	11,475
SQuAD [74]	175,198
TriviaQA [75]	146,692
WebInstructSub [76]	4,670,440
MIRACL	27,572,151
Full mix (<code>full</code>)	89,827,558

Corpus. Each dataset gives a query and an answer. We embed both separately. We sample 10M question-answer pairs from PAQ. For MIRACL [77], we sample up to 500,000 passages per language. All available passages are used when a language has fewer than 500,000. (Due to an error, our nemotron embeddings are missing 500,000 Thai passages and 100,000 sampled Telugu passages present for Gemini. Thus the full corpora contain 89.8M Gemini vectors and 89.2M Nemotron vectors.) When comparing activations between the two embedding families, we use the 89.2M shared embeddings.

Embeddings. For gemini embeddings, we use `gemini-embedding-2-preview` with the task type “RETRIEVAL_DOCUMENT” and dimension $d = 3072$. The nemotron representations are generated from `nvidia/llama-embed-nemotron-8b` with $d = 4096$. We truncate at 10,000 characters for gemini and 256 tokens for Nemotron.

Additional data splits for P2 experiments. We train a separate $m = 16,384, k = 64$ SAE for each of the following mixtures and each embedding model:

- **rand1**: emotion, fever, gooaq, natural-questions, sampled PAQ, and scifact; 37,336,279 vectors.
- **rand2**: hotpotqa, msmarco, nfcopus, squad, trivia-qa, and WebInstructSub; 24,919,128 vectors.
- **wiki**: fever, hotpotqa, natural-questions, sampled PAQ, squad, and trivia-qa; 42,166,282 vectors.
- **no_wiki**: emotion, gooaq, msmarco, nfcopus, scifact, WebInstructSub; 20,089,125 vectors.

Each mixture contains six complete prepared datasets. The **wiki/no_wiki** and **rand1/rand2** pairs are two partitions of the same 12 non-MIRACL datasets; both MIRACL collections occur only in **full**. **wiki/no_wiki** splits based on if the dataset includes Wikipedia data or not.

SAE architecture and objective. We use the ordinary (non-batch) TopK SAE implementation from <https://github.com/bartbusmann/BatchTopK/tree/main>. For an embedding $x \in \mathbb{R}^d$ the forward pass is

$$h(x) = \text{ReLU}(W_{\text{enc}}(x - b_{\text{dec}})), \quad (41)$$

$$z(x) = \text{TopK}_k(h(x)), \quad (42)$$

$$\hat{x} = W_{\text{dec}}z(x) + b_{\text{dec}}, \quad (43)$$

where $W_{\text{enc}} \in \mathbb{R}^{m \times d}$, $W_{\text{dec}} \in \mathbb{R}^{d \times m}$, and $b_{\text{dec}} \in \mathbb{R}^d$. There is no encoder bias. TopK_k retains the k largest nonnegative activations in each example.

Let D be the set of dead neurons: neurons with no positive post-TopK activation in the last 512 consecutive training batches. For each example, $z^{\text{AuxK}}(x)$ retains the largest $\min\{32, |D|\}$ entries of the *pre-TopK* activation vector $h(x)$ restricted to D , with all other entries set to zero. For a batch of B embeddings, the implemented loss is

$$\mathcal{L} = \frac{1}{Bd} \sum_{i=1}^B \left[\|x_i - \hat{x}_i\|_2^2 + w_{\text{aux}} \|W_{\text{dec}}z^{\text{AuxK}}(x_i) - (x_i - \hat{x}_i)\|_2^2 \right]. \quad (44)$$

The auxiliary term is zero when D is empty. Its residual $x_i - \hat{x}_i$ remains in the gradient computation. We use no ℓ_1 penalty. The auxiliary coefficient is $w_{\text{aux}} = 0.25$ except for the Gemini width-131,072 SAE, which uses $w_{\text{aux}} = 1.0$ (we found that at 0.25 in this setting, the number of dead neurons would remain large).

Optimization and training budget. We train SAEs given in Table 4. The four additional distribution mixtures give 26 SAEs in the main width/split experiments. We use float32 parameters, batch size $B = 2048$, and Adam with learning rate 5×10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.99)$, $\epsilon = 10^{-8}$, and no weight decay or learning-rate schedule. Gradients are clipped to global norm 1. The encoder is initialized with Kaiming-uniform weights, the decoder with the normalized transpose of the encoder, and the bias with zeros. Before each optimizer step, decoder directions are normalized and their radial gradient components are removed.

All of these SAEs train for 1,000,000 optimizer steps except the Gemini width-131,072 SAE, which trains for 2,000,000 steps. These budgets correspond to 2.048 billion and 4.096 billion sampled embeddings, respectively, with repeated sampling from the finite corpora above. A shard is selected with probability proportional to its row count and supplies four batches before the next shard selection. Each batch is a randomly positioned contiguous block with its rows shuffled; shards smaller than one batch are sampled with replacement. Training uses NVIDIA H100, A100 (80GB), or A6000 GPUs. Seeds are fixed separately for each run, rather than paired between embedding models. The synthetic experiments use a separate training recipe described in Section G.

Table 4: Sparsities for the 26 SAEs in the main width and distribution experiments. Subset runs all use $m = 16,384$ and $k = 64$. The Gemini width-131,072 run uses 2M steps and auxiliary coefficient 1.0; every other listed run uses 1M steps and coefficient 0.25.

Width or subset	k
512	32
1,024	32
2,048	32
4,096	32
8,192	64
16,384	64
32,768	64
65,536	128
131,072	128
wiki	64
no_wiki	64
rand1	64
rand2	64

Further sparsity and seed SAE controls. Table 5 lists six additional fixed- $k = 128$ SAEs and three alternate-seed Gemini SAEs.

Table 5: The nine additional SAEs used for the fixed-sparsity and alternate-seed controls. All use 1,000,000 optimizer steps, batch size 2048, and auxiliary coefficient 0.25.

Control	Model	Width	k
Fixed k	Gemini	512	128
Fixed k	Gemini	4,096	128
Fixed k	Gemini	32,768	128
Fixed k	Nemotron	512	128
Fixed k	Nemotron	4,096	128
Fixed k	Nemotron	32,768	128
Alternate seed	Gemini	512	32
Alternate seed	Gemini	4,096	32
Alternate seed	Gemini	32,768	64

C.2 Matching criteria and computation

The P1, P2, and cross-model comparisons combine a pairwise similarity score with a rule for selecting matches. We define the shared metrics here; the experiment-specific sections specify the dictionaries and evaluation corpora. We use a default threshold of $t = 0.7$ [20].

C.2.1 Direction similarity

For nonzero dictionary directions u and v in the same embedding space, we use signed cosine similarity,

$$s_{\text{dir}}(u, v) = \frac{\langle u, v \rangle}{\|u\|_2 \|v\|_2}.$$

For SAEs, u and v are decoder columns. The PCA baseline uses $|s_{\text{dir}}(u, v)|$ for both maximum-cardinality and exclusive matching, because the signs of principal components are arbitrary.

C.2.2 Activation similarity

For two SAE features evaluated on the same examples $x_1, \dots, x_N$, let a_n and b_n be their continuous post-TopK activations. With $\bar{a} = N^{-1} \sum_n a_n$ and $\bar{b} = N^{-1} \sum_n b_n$, their signed Pearson correlation is

$$s_{\text{act}}(a, b) = \frac{\sum_{n=1}^N (a_n - \bar{a})(b_n - \bar{b})}{\sqrt{\sum_{n=1}^N (a_n - \bar{a})^2} \sqrt{\sum_{n=1}^N (b_n - \bar{b})^2}}.$$

Every evaluation example receives equal weight, including examples with zero activation. The metric depends on the choice of evaluation data.

C.2.3 One-to-one matching and persistence

For either similarity score, eligible pairs have similarity at least t and form the edges of a bipartite graph. A maximum-cardinality matching is the largest set of eligible pairs in which each feature occurs at most once. We denote this maximum cardinality by $\mu_t(U, V)$. For PCA directions, we use the absolute-cosine modification above. Reported proportions divide the matching count by the source dictionary width, including inactive and unmatched features in the denominator.

For all maximum-matching persistence results, we compute a maximum-cardinality matching M_ℓ independently on the threshold graph between the full source dictionary U and each larger target dictionary V_ℓ , using retained candidate edges for empirical activation comparisons (Section C.2.4). Let

$$S_\ell = \{u \in U : \text{there exists } v \in V_\ell \text{ with } (u, v) \in M_\ell\}, \quad S_{\text{persistent}} = \bigcap_{\ell=1}^s S_\ell.$$

Maximum-cardinality matchings can be nonunique, so the persistent set and its size can depend on tie-breaking. We retain ascending original source and target feature IDs, canonicalize each graph as a sorted CSR matrix, and use SciPy 1.15.3’s `maximum_bipartite_matching` with `perm_type='column'`. This fixes the matching choices for these results.

C.2.4 Candidate graph construction

Direction comparisons evaluate all pairs. For activation comparisons, we use 512-dimensional CountSketch representations to propose 32 candidates per feature in each search direction, with the same evaluation rows and sketch hashes for both dictionaries. We recompute candidate correlations over the full evaluation corpus and retain the five highest-scoring candidates per feature in each direction. Thresholded matching uses the union of retained pairs with correlation at least t .

The sketch hash is the fixed SplitMix64 transformation of the original global row index, implemented by `row_hash` in `scripts/match_cross_model_sae_activations_131k.py`.

The within-family width sweeps use 89,827,558 Gemini rows and 89,227,558 Nemotron rows, respectively. Same-width cross-model comparisons use the 89,227,558 aligned shared rows, with hashes based on their original Gemini offsets. Split comparisons instead use the row distribution identified in each evaluation manifest. These frozen manifests record ordered shard identities, offsets, model-checkpoint SHA-256 hashes, and producing-code hashes. The cross-model `alignment_audit.json` additionally records SHA-256 hashes of each shard’s ordered source-row IDs after checking dataset/configuration/split/shard identity, text columns, and row order across the two

models. The reproduction workflows preserve these manifests and retained neighbor arrays; the code archive’s `SOURCE_MANIFEST.json` identifies the supplied source revision.

Pairwise maximum matching is exact on these retained graphs, and its cardinality is a lower bound on exhaustive pairwise Pearson matching.

C.2.5 Alternative exclusive matching criterion

The exclusive direction comparisons in Figures 6 and 13 count a pair (u, v) only when its similarity exceeds t and neither endpoint has another above-threshold partner. We use $s(u, v) = s_{\text{dir}}(u, v)$ for SAEs and $s(u, v) = |s_{\text{dir}}(u, v)|$ for PCA, applying the same score to pair admission and competing partners. The requirements are

$$s(u, v) > t, \quad s(u, v') \leq t \text{ for all } v' \neq v, \quad s(u', v) \leq t \text{ for all } u' \neq u. \quad (45)$$

The exclusive pairs themselves form a feasible one-to-one matching, so their count is bounded above by μ_t . Lowering t can reduce the exclusive count by introducing competitors, whereas maximum-cardinality counts cannot decrease.

C.3 P1: Persistence across SAE widths

We apply the independent pairwise matching and source-set intersection procedure in Section C.2.3 separately within Gemini and Nemotron, using each source SAE below width 131,072 and every larger trained SAE as targets. We report counts and source-width-normalized proportions. The largest SAE is a target only, since it has no larger comparison dictionary. Figure 2a uses the signed decoder-cosine metric in Section C.2.1.

C.4 P2: Shared features across training distributions

Figure 2b compares dictionaries trained on `full`, `wiki`, `no_wiki`, `rand1`, and `rand2`, separately for each embedding model and dictionary family. SAE dictionaries contain 16,384 directions; PCA uses all d principal components, with $d = 3072$ for Gemini and $d = 4096$ for Nemotron.

PCA dictionaries. We fit PCA separately for each embedding model and training mixture, using every embedding in that mixture. We stream the data to accumulate the count N , sum $s = \sum_{i=1}^N x_i$, and second-moment matrix $Q = \sum_{i=1}^N x_i x_i^\top$ in float64, then compute

$$\bar{x} = s/N, \quad \Sigma = \frac{Q - N\bar{x}\bar{x}^\top}{N-1}.$$

We diagonalize this centered covariance matrix and order its orthonormal eigenvectors by decreasing eigenvalue. The PCA dictionary consists of these directions in the original embedding space; P2 uses the full d -component basis. PCA fitting uses the stored embeddings without additional length normalization or coordinate standardization.

Dictionary matching. For each pair of training distributions, Figure 2b applies the metric in Section C.2. Exclusive direction matching and distribution-specific activation comparisons are reported in Section E.

C.5 P3: Hierarchical datasets and feature-recovery protocol

For Figure 2c, we evaluate individual SAE features against binary category labels in three domains: word prefixes, geography, and organism taxonomy. The labels define hierarchical relationships, such as a two-letter prefix implying its first letter or a city’s country determining its continent (the organism taxonomy has more levels).

English prefixes (prefix). We use the English frequency list from the [wordfreq package](#). We retain the top 200,000 words in frequency order. Each normalized word is embedded on its own. The first character defines a parent label, and the first two characters define a child label for words of length at least two. We retain categories with at least 100 positive word rows across the full evaluation collection, giving 26 one-letter and 216 two-letter categories per embedding model.

Geography (cities). We use all 69,182 records in the experiment’s [GeoNames cities5000 snapshot](#), with country labels from each record’s ISO country code and continent labels obtained from `countryInfo.txt`. We retain 87 country and six continent categories with at least 100 city rows each. For the reported results, we embed two question variants of every city name: **what country is [city] in** for country prediction and **what continent is [city] in** for continent prediction, following the use of elicitation in Bhalla et al. [45]. (We found that without this elicitation, SAEs at no widths reliably could detect the continent.)

Organism taxonomy (gbif). We use the [GBIF Backbone Taxonomy](#). From its `Taxon.tsv` and `VernacularName.tsv` tables, we select accepted species having at least one nonempty English vernacular name. This yields 123,495 species. We embed one scientific-name row per species and 259,286 deduplicated English common-name strings, leading to 382,781 name rows in total. Binary labels follow the species’ kingdom, phylum, class, order, family, and genus. We retain categories with at least 100 distinct species. This gives 4 kingdoms, 12 phyla, 33 classes, 131 orders, 276 families, and 43 genera. Once labels are constructed, F1 is evaluated over name rows, so species with more names can contribute more evaluation examples.

Selection and test splits. We partition the data into selection and test sets. Word rows are divided equally within each two-letter-prefix stratum, with one-character words forming their own strata. City rows are divided equally within each country, and both question variants inherit this partition. GBIF partitions the underlying species into 61,748 selection species and 61,747 test species, with scientific names inheriting the species assignment. Common-name rows are assigned a single split, even when their species associations span both partitions.

Every probe reads these saved split labels unchanged, and the same assignments are used across both embedding models and all widths. Exact reruns require these row tables and their source snapshots: the provided preparation scripts supply deterministic seed-1729 reconstructions for the word and city splits, but do not reproduce the historical row-by-row assignments. Fresh source downloads can also change the evaluation collections.

Embedding and SAE evaluation. We compute gemini and nemotron embeddings for each input and compute activations for the nine full-mix SAEs for each model, with widths 512 through 131,072 and the corresponding TopK sparsities in Table 4.

Selecting a single-feature probe. For a binary target z_i^* and SAE activations $\hat{z}$, define

$$g_{j,\tau}(x) = \mathbf{1}\{\hat{z}_j(x) \geq \tau\}, \quad \tau > 0.$$

For each coordinate active on selection rows, we sort its positive selection activations and evaluate every distinct observed value as an inclusive threshold. We choose the coordinate and threshold maximizing selection F1, requiring at least one selection true positive.

For $E = \{z_i^* = 1\}$ and $F = \{g_{j,\tau} = 1\}$, we use

$$\text{F1}_{\mathcal{D}}(E, F) = \frac{2 \Pr_{\mathcal{D}}(E \cap F)}{\Pr_{\mathcal{D}}(E) + \Pr_{\mathcal{D}}(F)} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

on the relevant split. This evaluates how well a single thresholded feature identifies a category.

Figure 2c details. Each point in Figure 2c is the unweighted mean of held-out test F1 over all retained categories at that level and both embedding models. The number of category–model pairs is twice the number of categories listed above, and the same pairs contribute at every width. The main-text a-prefix example, empirical Food and Music hierarchies, and fixed-sparsity recovery analysis are described in Section F.

D Extended Experiments for P1: Persistence across Size

We extend the main persistent feature experiment in Figure 2a with controls for sparsity and training seed, activation-based matching, alternative matching criteria and similarity thresholds, and analyses of feature prevalence.

D.1 Fixed sparsity

At widths 512, 4,096, and 32,768, we replace each original source SAE by its $k = 128$ counterpart, retaining every original larger-width target and using signed cosine at threshold 0.7 with the independent pairwise matching and intersection criterion in Section C.2.3.

Table 6: Fixed-sparsity source replacement, using the same full-sweep target sets as the seed control. Only the source changes from its original k to $k = 128$; all original larger-width targets are retained. Same-width counts match the original and replacement source dictionaries. All comparisons use signed decoder cosine similarity ≥ 0.7 ; percentages divide by source width.

Model	Original width/ k	Same-width cross-sparsity matches	Original source persistence	$k = 128$ source persistence
Gemini	512/32	389 (76.0%)	303 (59.2%)	221 (43.2%)
	4,096/32	2,949 (72.0%)	2,974 (72.6%)	2,405 (58.7%)
	32,768/64	16,466 (50.3%)	15,881 (48.5%)	12,893 (39.3%)
Nemotron	512/32	410 (80.1%)	259 (50.6%)	210 (41.0%)
	4,096/32	3,286 (80.2%)	2,841 (69.4%)	2,607 (63.6%)
	32,768/64	18,198 (55.5%)	12,750 (38.9%)	13,609 (41.5%)

Table 6 shows persistence remains substantial, but is somewhat lower in general, especially for smaller widths. This seems to suggest that the conventional wisdom to increase k alongside m is reasonable (and also is somewhat in alignment with our simulations Section G).

D.2 Variation by seed

We repeat the direction comparisons for gemini SAEs 512/32, 4,096/32, and 32,768/64 (width/ k) with different training seeds. Table 7 shows a substantial number of matches across seeds, and that persistent remains similar (suggesting that persistence doesn’t vary heavily between SAEs). The same-width column uses pairwise maximum matching (Section C.2.3); the persistence columns use independent pairwise matching and intersection (Section C.2.3). Consistent with [20], there are a substantial number of different features learned when varying the random seed (especially for larger widths); simultaneously, there are many features that have matches even in all larger dictionaries.

Table 7: Gemini direction stability across training seeds. Persistence uses the independent pairwise matching and intersection criterion.

Width/ k	Same-width cross-seed matches	Original source persistence	Alternate source persistence
512/32	470 (91.8%)	303 (59.2%)	293 (57.2%)
4,096/32	3,564 (87.0%)	2,974 (72.6%)	2,979 (72.7%)
32,768/64	17,856 (54.5%)	15,881 (48.5%)	15,998 (48.8%)

D.3 Exclusive matching

Figure 6 repeats the source–target comparisons in Section C.3 using the exclusive persistence criterion in Section C.2.5.

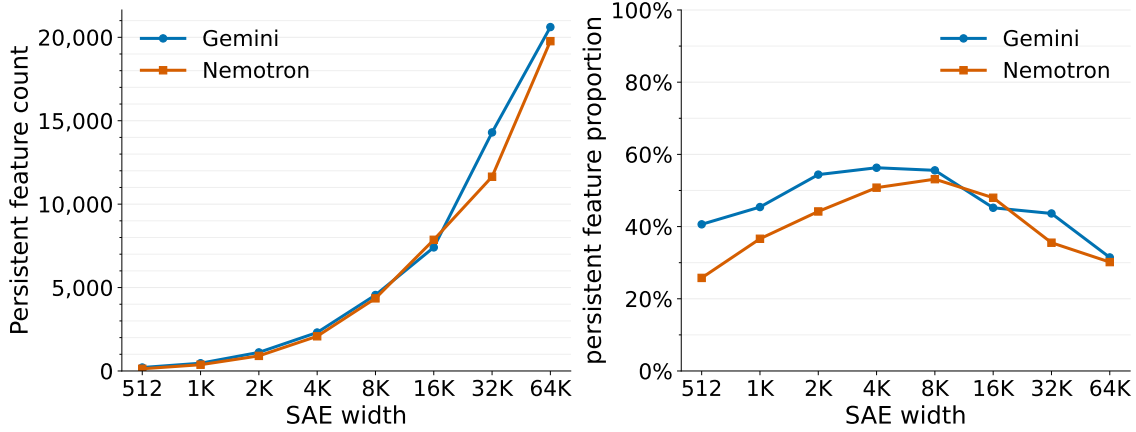


Figure 6: Persistent stability under strict exclusive matching at 0.7. A source direction counts only if it has a signed-cosine match above 0.7 in every larger trained dictionary, with no competing partner above 0.7 at either endpoint in each comparison.

D.4 Persistence by activation similarity

Figure 7 repeats the source–target comparisons in Section C.3 using signed Pearson correlation at threshold 0.7, with the metric and candidate-graph procedure defined in Sections C.2.2 and C.2.4.

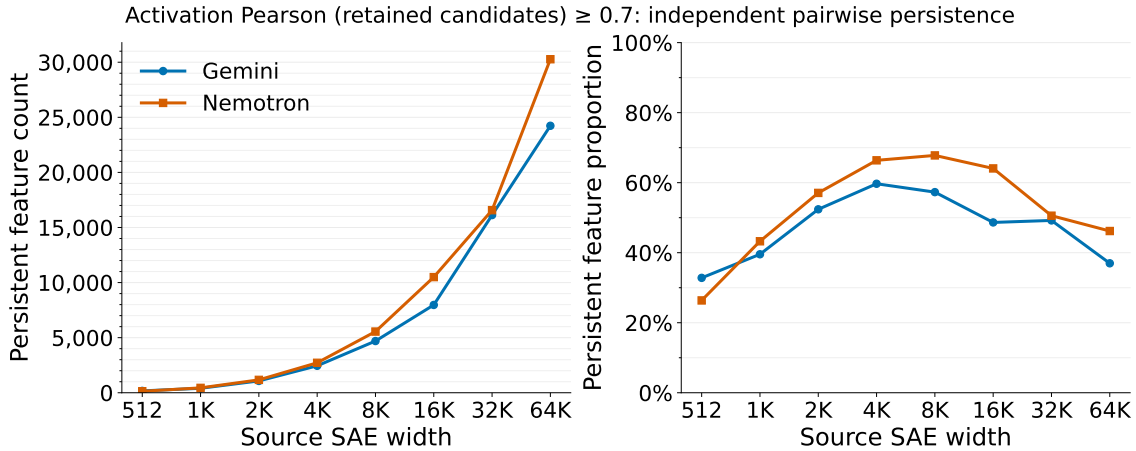


Figure 7: Persistent stability across SAE widths using activation similarity at signed Pearson correlation at least 0.7. Left: the number of source features matched in every independently computed pairwise maximum matching to a larger SAE on the retained candidate graphs. Right: this count divided by source width. Both panels compare Gemini and Nemotron, using the fixed tie-breaking procedure in Section C.2.3.

D.5 Sensitivity to the matching threshold

We vary the matching threshold over $t \in \{0.5, 0.6, 0.7, 0.8\}$ using the independent pairwise matching and intersection criterion in Section C.2.3. Figure 8 uses signed decoder cosine similarity, and Figure 9 uses signed activation Pearson correlation.

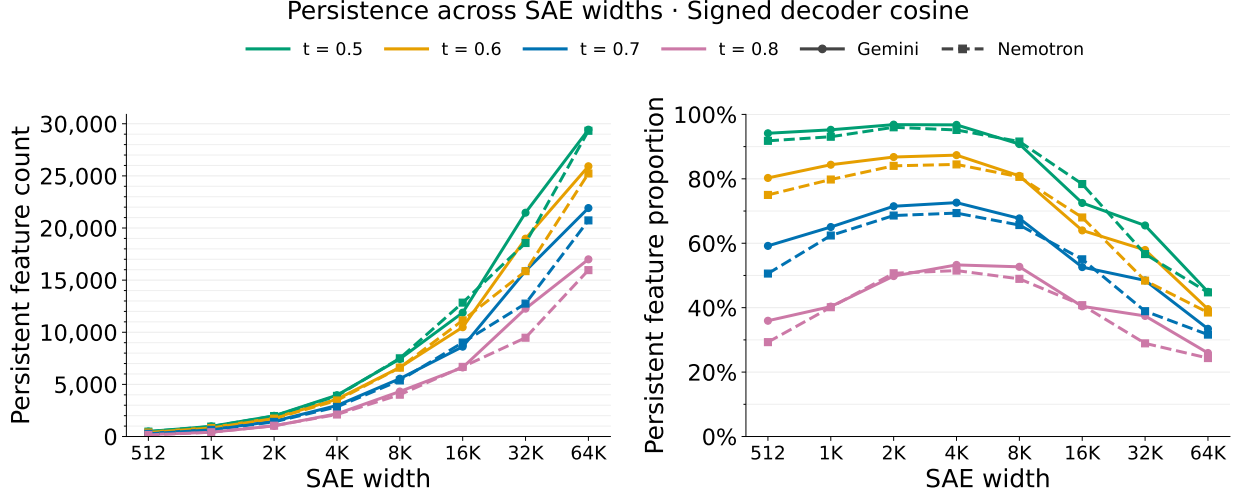


Figure 8: Threshold sensitivity of direction persistence.

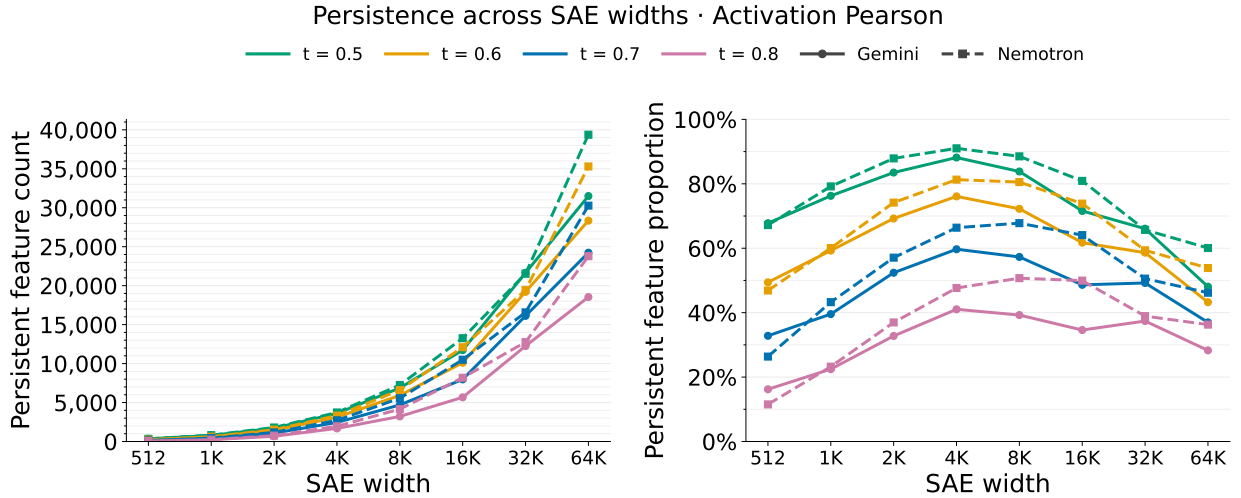


Figure 9: Threshold sensitivity of activation persistence.

D.6 Feature activation rates

Figure 10 compares prevalence across SAE widths and KMeans cluster counts. For each SAE feature, we compute the fraction of the model’s full evaluation corpus on which its continuous post-TopK activation exceeds the SAE’s pooled median positive activation. This is one threshold per SAE, computed over all positive activation events across features and examples.

For KMeans, prevalence is the fraction of examples assigned to each nearest centroid. We use Euclidean KMeans in the full embedding space, trained on every row of the corresponding full corpus for 100 total iterations, at 512, 4,096, 16,384, and 131,072 clusters. Both methods are evaluated on the same full corpus within each embedding family: 89,827,558 Gemini examples and 89,227,558 Nemotron examples.

We plot kernel density estimates of $\log_{10}$ activation rates, using a bandwidth of 0.12 and scaling each curve to integrate to the number of features with positive rates. Zero-rate features are omitted from the logarithmic axis. The vertical axis uses a square-root scale with tick labels in the original density units.

The total mass and average prevalence are partly constrained by dictionary width and sparsity. For N examples and an SAE of width m with TopK parameter k , there are at most Nk positive activation events. Strictly exceeding their pooled median selects at most half of these events, with ties potentially reducing this number. Thus, if p_j denotes feature j ’s above-median prevalence, then $\frac{1}{m} \sum_{j=1}^m p_j \leq k/(2m)$. For KMeans, cluster prevalences sum to one and have mean $1/m$. Consequently, a growing low-prevalence tail and changes in the count-scaled density area partly reflect these constraints. The observed curve shapes provide descriptive evidence consistent with the recovery principle, rather than identifying atoms on their own.

We observe a “nested” behavior, in which larger SAEs tend to include a similar number of coarse features as smaller SAEs, while also learning a longer tail of more granular features, consistent with the recovery principle. The KMeans prevalence distributions instead shift toward smaller clusters as the number of clusters increases.

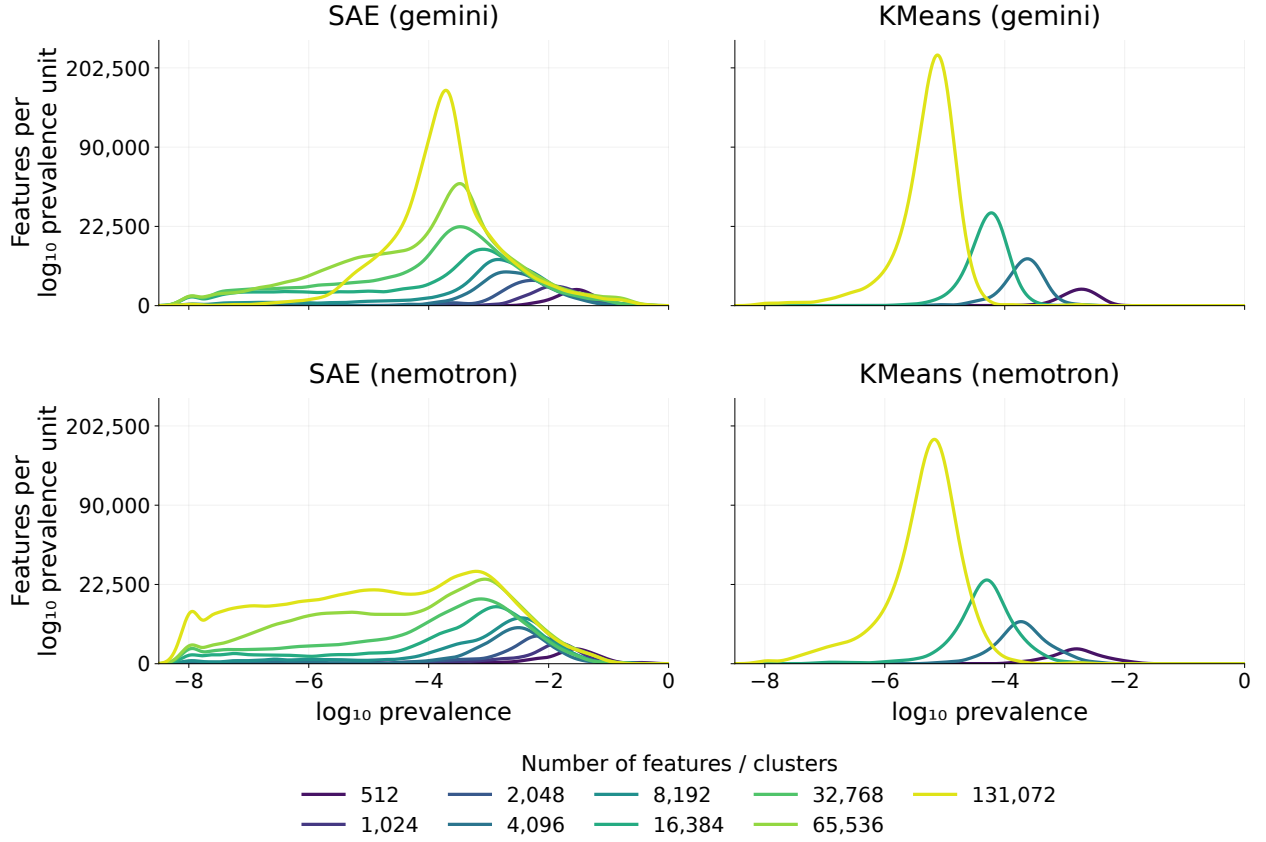


Figure 10: Prevalence across SAE widths (left) and KMeans cluster counts (right), for Gemini (top) and Nemotron (bottom). SAE prevalence uses activation strictly greater than each SAE’s pooled median positive activation; KMeans prevalence uses nearest-centroid assignment. KMeans models are trained on the full corpus for 100 iterations. Colors indicate dictionary size. Curves show feature-count-scaled densities of $\log_{10}$ prevalence, with shared axes and a square-root vertical scale. Zero-rate features are omitted from the logarithmic axis.

To control for varying sparsities, Figure 11 repeats the comparison with TopK sparsity fixed at $k = 128$ for all SAE widths. We use widths 512, 4,096, 32,768, 65,536, and 131,072, with the same prevalence definitions, full evaluation corpora, and KMeans baselines as above.

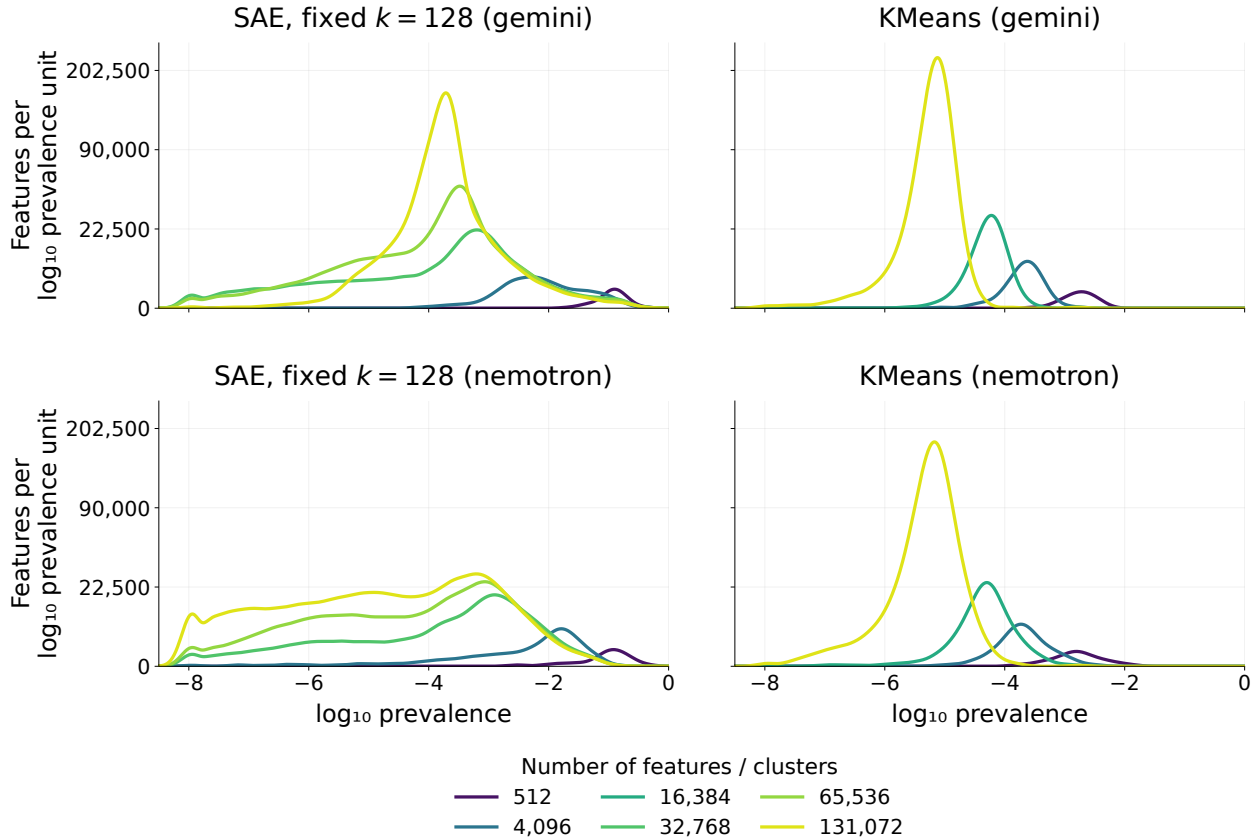


Figure 11: Prevalence for SAEs with fixed TopK sparsity $k = 128$ (left) and KMeans (right), for Gemini (top) and Nemotron (bottom). SAE widths are 512, 4,096, 32,768, 65,536, and 131,072; KMeans cluster counts are 512, 4,096, 16,384, and 131,072. SAE prevalence is the fraction of examples with activation strictly above each SAE’s pooled median positive activation; KMeans prevalence is nearest-centroid assignment frequency after 100 full-corpus training iterations. Colors indicate the number of features or clusters. Curves show feature-count-scaled densities of $\log_{10}$ prevalence with bandwidth 0.12, shared axes, and a square-root vertical scale. Zero-rate features are omitted from the logarithmic axis.

D.7 Relationship between persistence and prevalence

We directly compare feature prevalence with direction persistence using the median activation threshold from Figure 10. For an SAE with continuous post-TopK activations $\hat{z}_i(x)$, let τ_{SAE} be the pooled median of all strictly positive activation events, across features and examples in its embedding model’s full evaluation corpus. We define

$$p_i(D) = \frac{1}{|D|} \sum_{x \in D} \mathbf{1}\{\hat{z}_i(x) > \tau_{\text{SAE}}\}. \quad (46)$$

There is one threshold per SAE, rather than a separate median for each feature; the denominator includes every example. We use the full corpora for these measurements.

For each source width in the original sweep, we compare p_i for persistent features and non-persistent features.

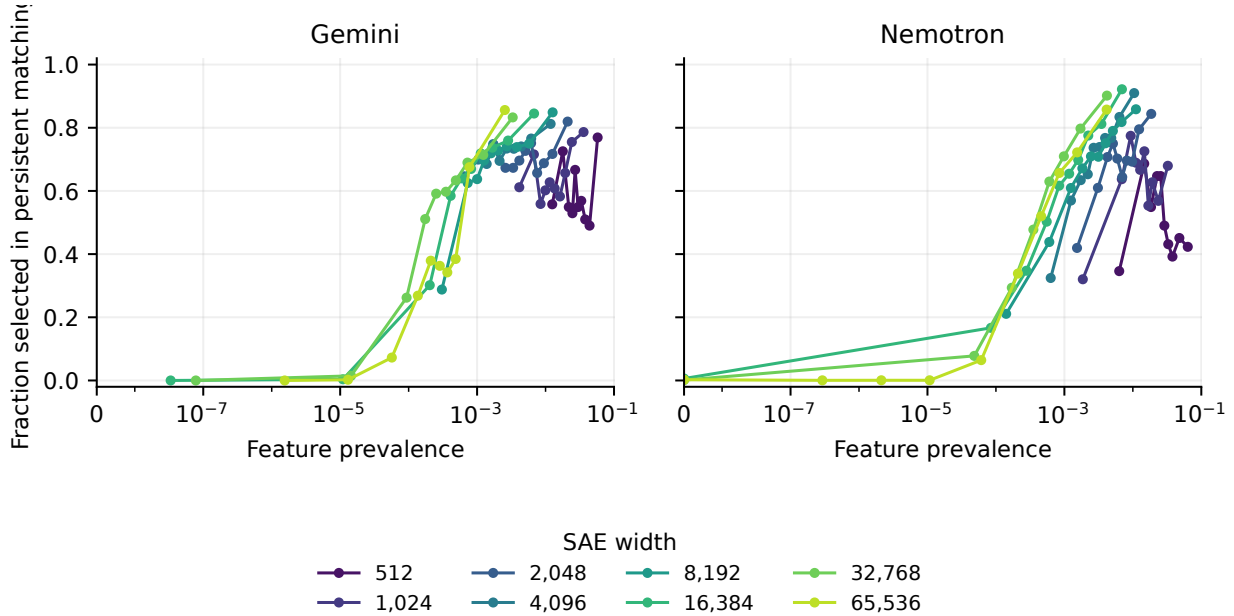


Figure 12: Median-threshold feature prevalence and direction persistence under independent pairwise matching and intersection in Gemini (left) and Nemotron (right). Each curve represents a source width in the original sparsity schedule. Points group source features into up to ten prevalence-quantile bins, keeping ties together, and show the median prevalence and fraction selected in the persistent matching within each bin. All source features are included. Bin sizes can differ; lines connect descriptive bin summaries. The horizontal axis is logarithmic above 10^{-8} with a linear segment retaining zero-prevalence bins.

We measure the AUC of feature prevalence as a predictor of persistence. At widths 16,384, 32,768, and 65,536, AUCs are 0.814, 0.801, and 0.822 for Gemini, and 0.818, 0.905, and 0.917 for Nemotron. Restricting to features with nonzero prevalence gives respective ranges of 0.787–0.821 and 0.792–0.905. The association is weaker at the smallest widths.

Overall, these results are consistent with the recovery principle’s prediction that the most prevalent atomic features are recovered.

E Extended Experiments for P2: Stability across Distributions

We extend the direction-based comparison in Figure 2b using the exclusive matching criterion, activation-based matching, and the relationship between cross-distribution stability and cross-distribution prevalence.

E.1 Cross-distribution stability under exclusive matching

Figure 13 applies the exclusive criterion from Section C.2.5 to SAE and PCA dictionaries trained on different distributions, at $t = 0.7$. This tests whether cross-distribution agreement persists when each match must have an unambiguous partner, allowing either sign for PCA.

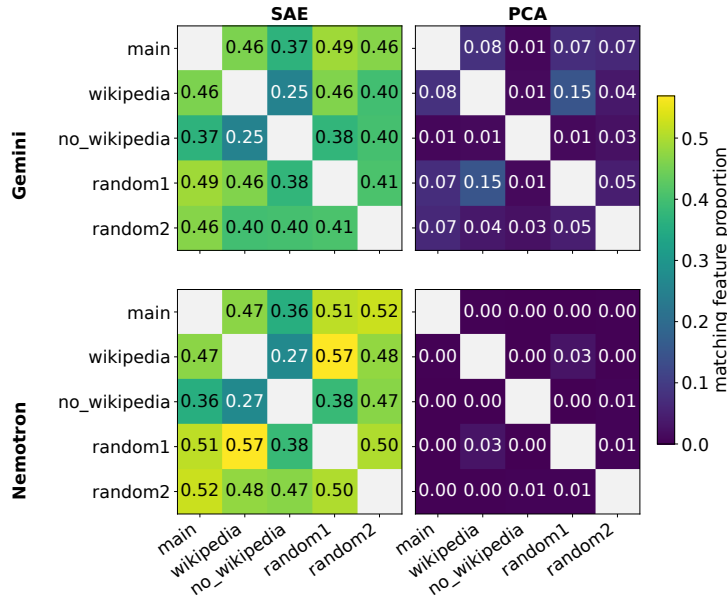


Figure 13: Cross-distribution SAE and PCA matches under the alternative exclusive criterion at $t = 0.7$, using signed cosine for SAEs and absolute cosine for PCA. Each counted pair exceeds the threshold and has no other above-threshold partner at either endpoint. Entries divide the pair count by the dictionary width. Compared with the maximum-cardinality matching in Figure 2b, this criterion discards all pairs involving competing above-threshold partners. It therefore gives a stricter measure of unambiguous agreement between dictionaries.

E.2 Cross-distribution stability by activation similarity

Figure 14 applies the activation metric, pairwise matching rule, and normalization in Section C.2 to the width-16,384 SAEs trained on `full`, `wiki`, `no_wiki`, `rand1`, and `rand2`. Within each embedding model, the cell at row D and column D' evaluates both SAEs on all examples in D . Reversing the row and column changes the evaluation distribution, so the heatmap need not be symmetric.

We construct sketches and candidate graphs for each row distribution using Section C.2.4, with moments computed over that distribution.

Matched proportions range from 22.0% to 48.9% for Gemini and from 29.3% to 60.7% for Nemotron.

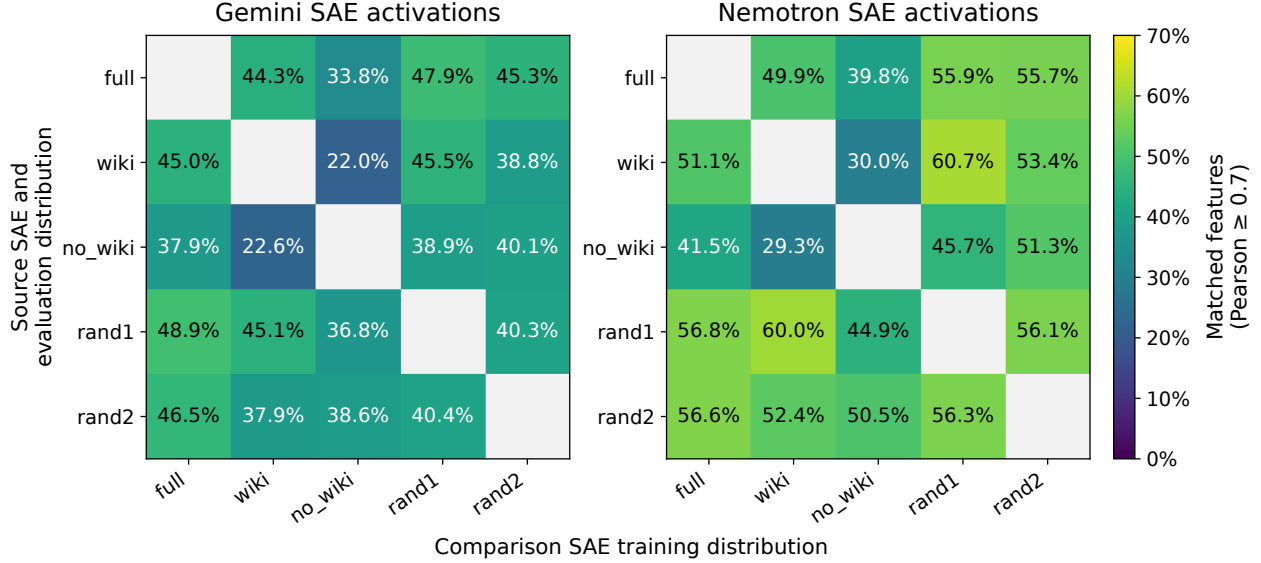


Figure 14: Cross-distribution SAE activation matching for Gemini (left) and Nemotron (right). Each cell reports the maximum number of one-to-one matches at signed Pearson correlation ≥ 0.7 , divided by the SAE width of 16,384. Both SAEs are evaluated on the row’s distribution; columns identify the comparison SAE’s training distribution.

E.3 Feature prevalence and cross-distribution matching

We test whether features matched between distribution-specific SAEs are more prevalent in both distributions, using the median-threshold definition in Equation (46). Corollary 2 suggests that the atomic features that are prevalent in both data distributions are recovered.

For each width-16,384 source SAE trained on D , we compute its pooled median positive activation on the embedding model’s full evaluation corpus, including MIRACL. We freeze that single threshold when evaluating the source SAE on both D and the target training distribution D' , and summarize prevalence by

$$b_i(D, D') = \min\{p_i(D), p_i(D')\}. \quad (47)$$

Both frequencies refer to the same feature in the same source SAE, including when it has no match in the target SAE. This avoids assigning a partner to unmatched features. Reversing source and target evaluates a different SAE, so the prevalence analysis is directional.

We use the disjoint `wiki/no_wiki` and `rand1/rand2` dataset partitions from Section C. These evaluation pairs exclude MIRACL, although threshold calibration uses the full corpus. They contain respectively 42,166,282/20,089,125 and 37,336,279/24,919,128 examples; they partition datasets rather than equal numbers of rows. The outcome is membership in one optimal SAE direction matching under the protocol in Section C.4.

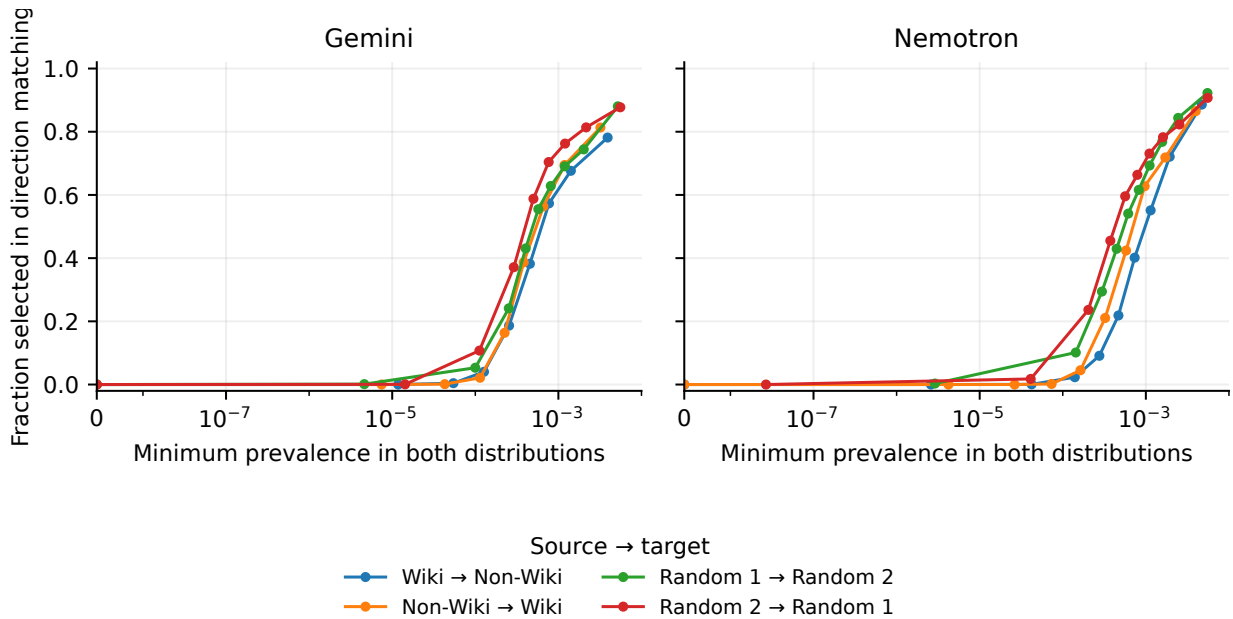


Figure 15: Median-threshold prevalence in both distributions and cross-distribution direction matching for Gemini (left) and Nemotron (right). Curves identify the source and target training distributions. The horizontal coordinate is the bin median of the minimum prevalence of a source feature across the two distributions; the vertical coordinate is the fraction selected in a one-to-one direction matching. Quantile binning preserves ties and includes all source features, with the same axis convention as Figure 12.

Higher prevalence in both distributions is associated with matching in all eight directional comparisons. AUCs, defined as in Section D.7, range from 0.864 to 0.923 for Gemini and from 0.839 to 0.927 for Nemotron. They remain large after restricting to features with nonzero prevalence in both distributions (ranges 0.844–0.905 and 0.828–0.921, respectively).

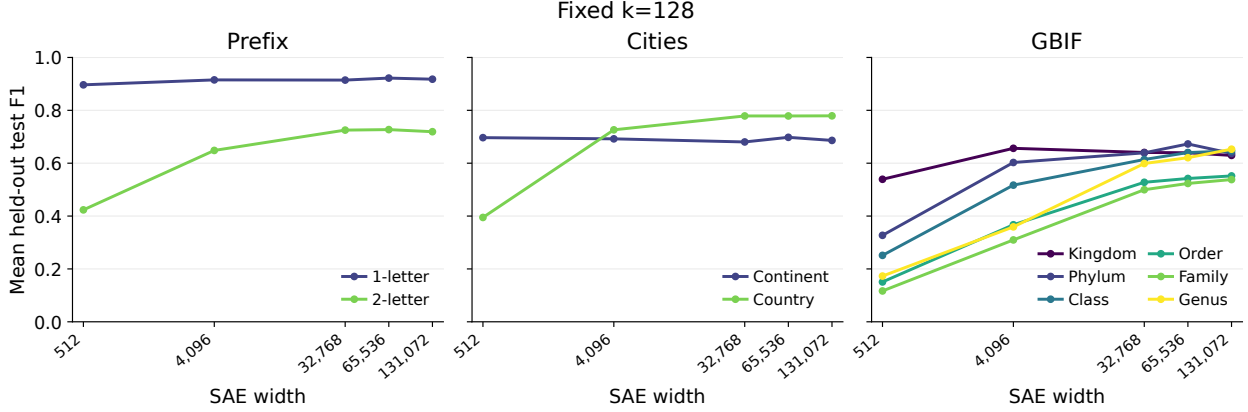


Figure 16: **Hierarchical feature recovery at fixed $k = 128$.** Mean held-out test F1 for Prefix, Cities, and GBIF at five SAE widths, pooling Gemini and Nemotron. Each curve averages the same category-model pairs as the corresponding curve in Figure 2c; each category’s feature and threshold are selected using training rows only.

F Extended Experiments for P3: Hierarchical Feature Recovery

We complement the aggregate hierarchy results in Figure 2c by controlling for sparsity, examining recovery of individual features and parent-child pairs, examining the prefix recovery examples, and discussing hierarchy extraction more broadly.

F.1 Hierarchical recovery at fixed sparsity

Figure 16 repeats the mean-F1 analysis of Figure 2c with $k = 128$ at widths 512, 4,096, 32,768, 65,536, and 131,072. We use the additional fixed- k checkpoints at the first three widths and the existing $k = 128$ checkpoints at the final two. The datasets, eligible categories, train/test splits, detector selection, and averaging over category-model pairs are the same as in the main analysis.

F.2 Counts and proportions of recovered categories and adjacent pairs

To complement mean F1, Figures 17 and 18 count categories and parent-child pairs (in gbif, this means adjacent levels of the hierarchy) whose held-out test F1 is strictly greater than 0.6.

F.3 Recovery of letter-prefix hierarchies

The upper panel of Figure 3a shows Gemini test F1 for the a parent and its eligible two-letter children, selecting a separate detector for each. The lower recall heatmap instead fixes the selection-chosen a detector at each width and measures its recall within each prefix group. That descriptive heatmap uses the full 200,000-word collection, including selection and test rows, rather than only the held-out split. Figure 19 extends the upper F1 panel to the 25 one-letter prefixes with eligible two-letter children, excluding x (since it does not have any qualifying two-letter children), using the same saved probes, nine widths, and category eligibility rules. These curves report continuous held-out test F1 without filtering by F1. We see more instances of mesoscale splitting (a U-shaped parent F1 curve). Figure 20 repeats this grid at fixed $k = 128$ for widths 512, 4K, 32K, 65K, and 131K, using the checkpoints described in Section F.1.

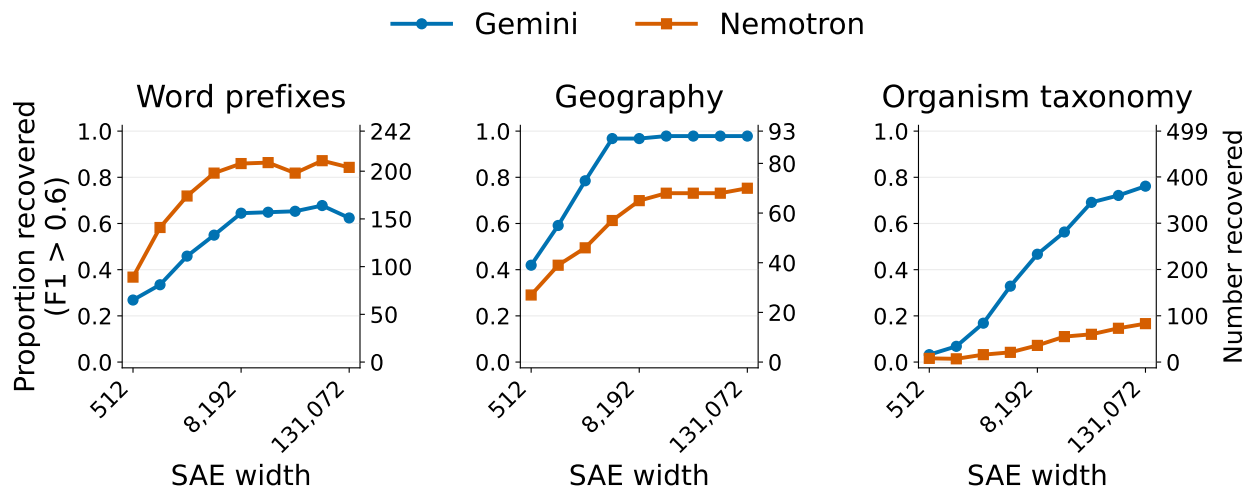


Figure 17: **Recovery of individual categories across SAE widths.** A category is recovered when its selection-chosen detector achieves held-out test F1 > 0.6. Left axes show proportions; right axes show raw counts among 242 word prefixes, 93 geographic categories, and 499 organism categories. Geography uses different continent and country prompts.

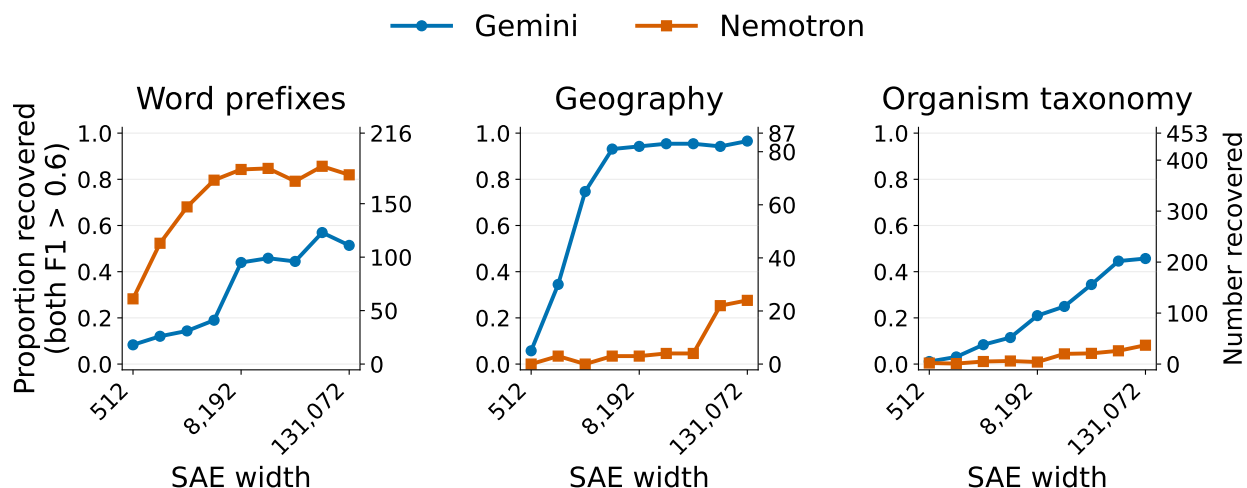


Figure 18: **Simultaneous recovery of adjacent parent-child pairs.** Both selection-chosen detectors must achieve held-out test F1 > 0.6 and use distinct SAE coordinates within the pair. Left axes show proportions; right axes show raw counts among 216 prefix pairs, 87 continent-country pairs, and 453 adjacent-rank organism pairs.

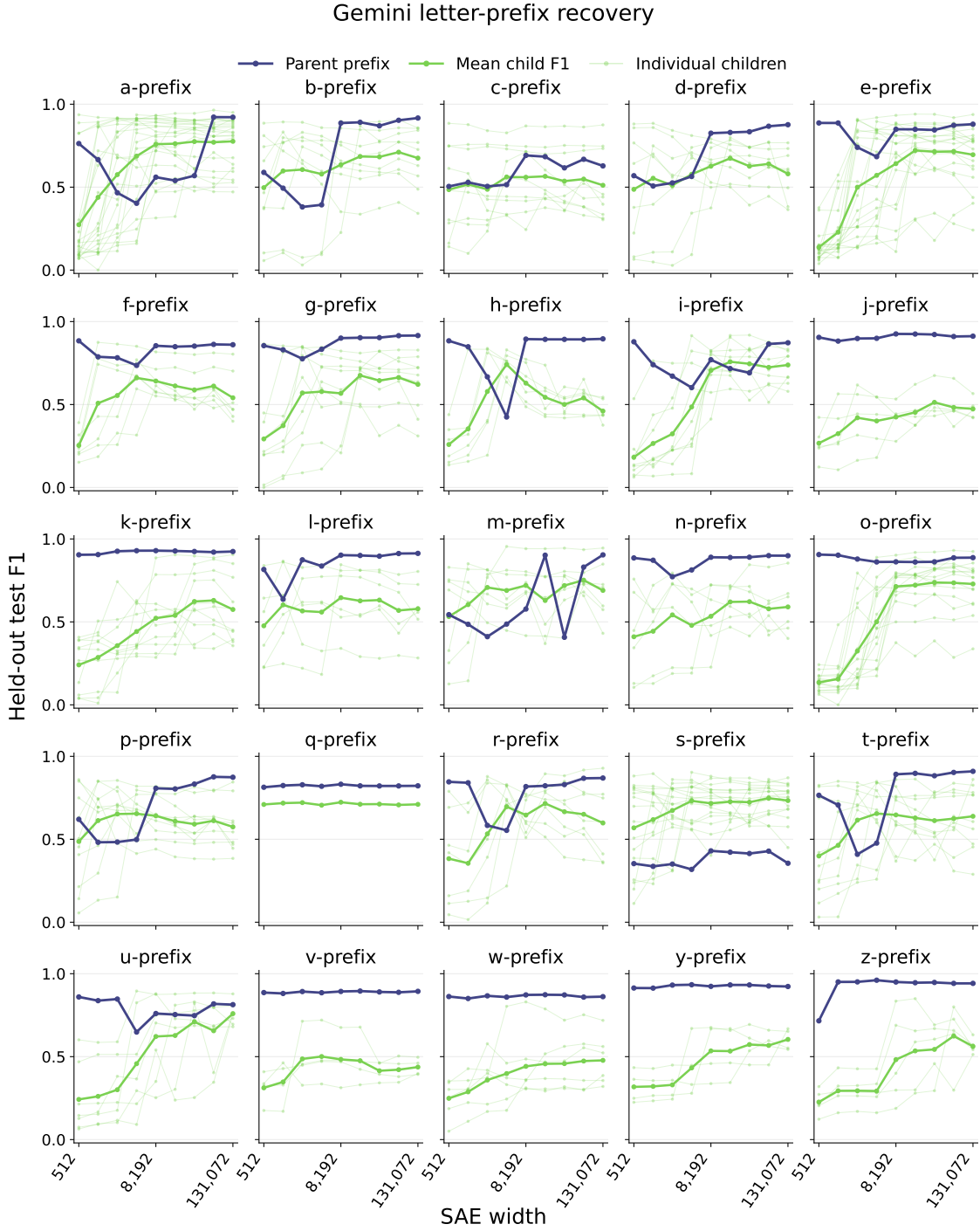


Figure 19: **Gemini F1 recovery for 25 letter prefixes.** Each panel is the analog of the upper panel of Figure 3a for one prefix from a to z, excluding x, across the original nine SAE widths and sparsity schedule. Blue curves show parent F1; bold green curves show the unweighted mean F1 of eligible two-letter children; faint green curves show individual children.

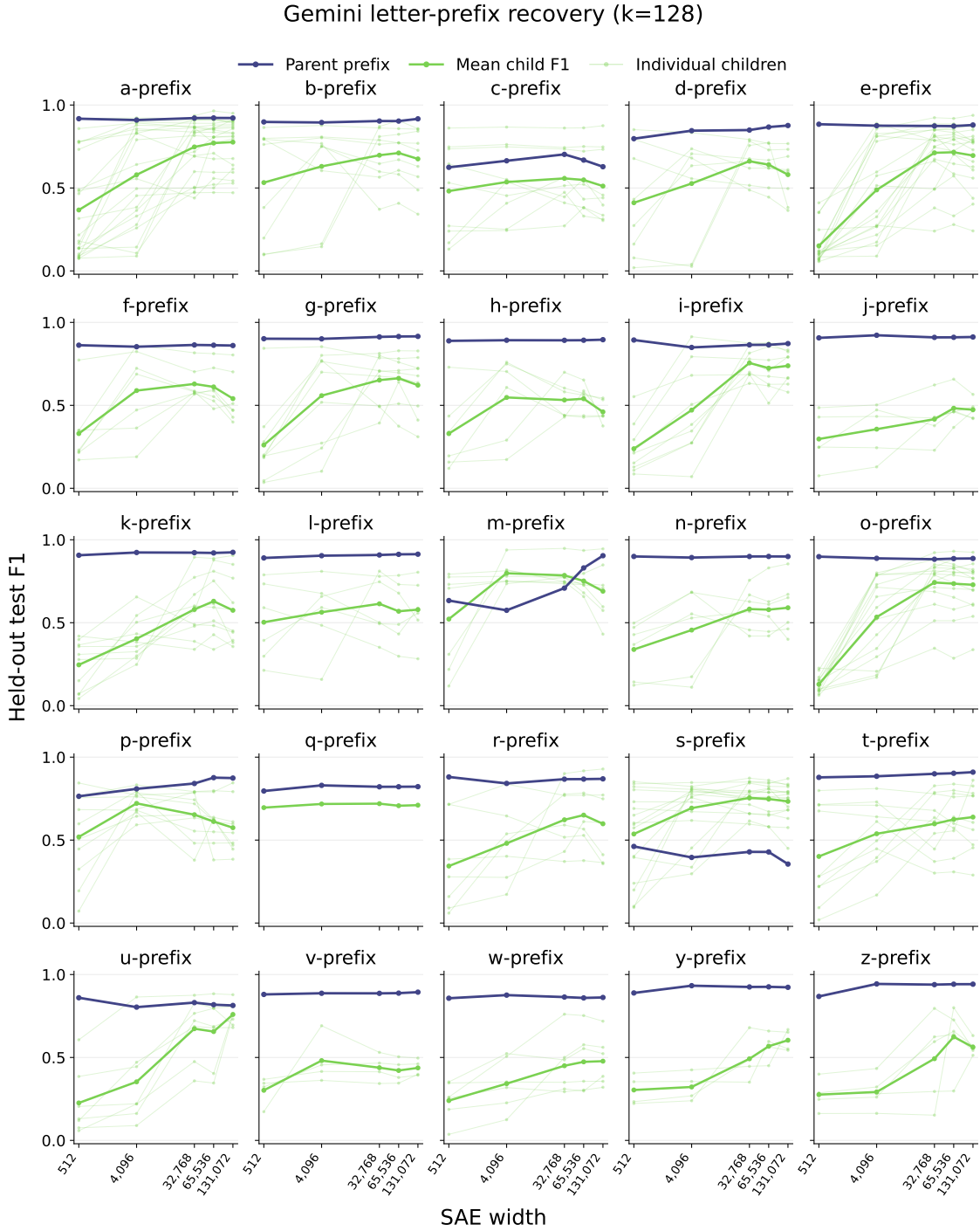


Figure 20: **Gemini F1 recovery for 25 letter prefixes at fixed $k = 128$.** The analog of Figure 19 at five SAE widths: 512, 4,096, 32,768, 65,536, and 131,072. Panels cover a–z, excluding x. Blue curves show parent F1; bold green curves show the unweighted mean F1 of eligible two-letter children; faint green curves show individual children. Each category’s detector is selected using selection rows and evaluated on held-out test rows, with no F1 filtering.

F.4 Empirical extraction of feature hierarchies

The food and music hierarchies in Figure 21 are curated subsets of a graph extracted from activation co-occurrences in the Gemini SAE with 131,072 features and $k = 128$. We use cached post-TopK activations on $N = 89,827,558$ corpus rows. A feature is active when its activation is strictly greater than 0.05; this threshold is the same for candidate parents and children. For a candidate parent P and child C , let n_P and n_C denote their activation counts and n_{PC} their co-activation count. We estimate forward and reverse containment as

$$\hat{p}_{P|C} = \frac{n_{PC}}{n_C}, \quad \hat{p}_{C|P} = \frac{n_{PC}}{n_P}. \quad (48)$$

High forward containment indicates that the parent usually activates when the child does; low reverse containment indicates that the parent also covers examples outside the child.

Candidate edges and directionality. We retain a candidate edge $P \rightarrow C$ only if $n_C \geq 100$, $n_{PC} \geq 100$, $n_P > n_C$, and

$$L_{P|C} \geq 0.70, \quad \hat{p}_{C|P} \leq 0.50, \quad \frac{\hat{p}_{P|C}}{n_P/N} \geq 5. \quad (49)$$

The last condition is a lift threshold: containment must be at least five times the parent’s marginal activation rate. The reverse-containment and support conditions orient edges from broader to narrower activation sets; the strict support ordering also rules out directed cycles.

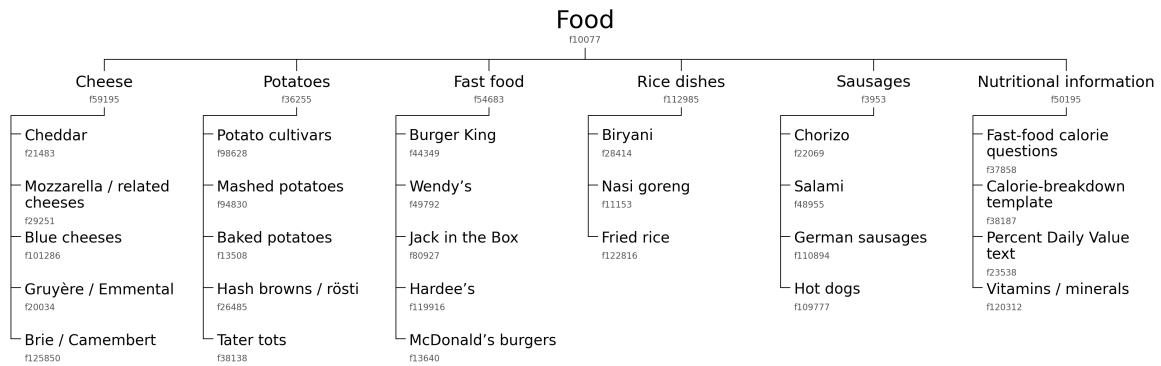
Here $L_{P|C}$ is the lower endpoint of the two-sided 95% Wilson interval for a binomial proportion. Writing $\hat{p} = \hat{p}_{P|C}$, $n = n_C$, and $z = 1.96$, it is

$$L_{P|C} = \frac{\hat{p} + z^2/(2n) - z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}. \quad (50)$$

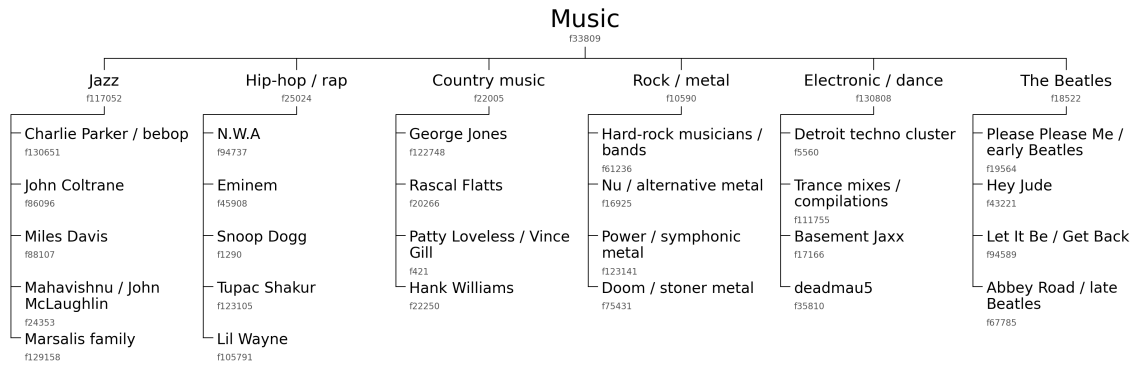
This penalizes estimates based on fewer observations: for example, 90/100 and 900/1000 both give 90% containment, but their lower bounds are approximately 82.6% and 88.0%. We use the bound as a screening heuristic, since corpus rows can be duplicated or dependent and the interval does not adjust for testing many feature pairs.

Removing ancestor shortcuts. After collecting all qualifying edges, we remove $P \rightarrow C$ whenever a feature M has both qualifying edges $P \rightarrow M$ and $M \rightarrow C$. Thus, Food $\rightarrow$ Cheddar is removed if Food $\rightarrow$ Cheese and Cheese $\rightarrow$ Cheddar both qualify. Multiple parents can remain, and a broader ancestor edge can remain if an intermediate edge fails the thresholds. The procedure yields 154,316 qualifying ancestor pairs and 114,762 retained parent–child pairs.

Selection and interpretation of the figures. Each hierarchy in Figure 21 displays 33 selected features and 32 retained edges. The compact main-text table in Figure 3b represents all 66 features and 64 parent–child relationships in three hierarchy-level columns, omitting only the feature IDs. We inspected cached activating texts to choose interpretable branches and assign descriptive labels; Tables 8 and 9 provide example texts. All edge statistics use the full corpus counts, rather than the sampled texts. The diagrams illustrate selected activation-containment relationships, not an exhaustive taxonomy or disjoint partition, and the labels are interpretations rather than independent ground truth.



Gemini 1.1K SAE - 33 selected features - Lines are retained parent-child relationships; labels summarize sampled texts.



Gemini 1.1K SAE - 33 selected features - Lines are retained parent-child relationships; labels summarize sampled texts.

Figure 21: Learned hierarchies for Music and Food. The labels here are automatically generated using Codex; activating examples for each feature in the hierarchy are given in Tables 8 and 9.

Table 8: Food hierarchy with five unique activating examples per feature.

Food

Food f10077	Food render • Food in England • Food Race • Food and Magic • Culinary diplomacy
Cheese f59195 Cheddar f21483 Mozzarella / related cheeses f29251 Blue cheeses f101286 Gruyère / Emmental f20034 Brie / Camembert f125850	チーズはいただき • Cheese (album) • Liste de mets à base de fromage • Premium Cheese! • Fromage à gogo Cheddar • CheddarU • Hombre de Cheddar • how cheddar cheese is made? • Cheddar (Käse) what is mozzarella • Mozzarella • 모차렐라 • موتزارالا • モッツアレッタ Bleue d'Auvergne • Roquefort • Bleu d'Auvergne • Gorgonzola • Fourme d'Ambert what is gruyere • Emmental de Savoie • what is gruyere cheese • 3. Jarlsberg. Jarlsberg is Norwegian variant of the cheese having holes and is a Swiss type cheese. This mellow and... • Emmental cheese Camembert • Coulommiers (Käse) • what does camembert cheese smell like • what's in brie cheese • what gives brie unique taste
Potatoes f36255 Potato cultivars f98628 Mashed potatoes f94830 Baked potatoes f13508 Hash browns / rösti f26485 Tater tots f38138	Potato Project • Potatoes & You • when is potato day • Похороны картофеля • Jus de pomme de terre Lapin Puikula • Golden Wonder potato • Umatilla Russet • Lenape potato • Golden Wonder (potato) how long will mashed potatoes stay good in the refrigerator? • do mashed potatoes go bad? • how long can mashed potatoes last? • how long are leftover mashed potatoes good for in the fridge? • do mashed potatoes go bad in the fridge? Check out this recipe for the Perfect Baked Potato: http://www.recipe4living.com/Recipes/Rec... It says to bake it for... • Jacket potato • how to check if baked potato done • Medium Plain Baked Potato: 150 Calories Size matters. A small baked potato (about 2 inches in diameter) has about 130... • how long is a baked potato good in the fridge how to oven cook hash browns • Hash Browns • what is rosti potatoes • هاش براونز • 쥬슈티 Tater Tots • With the lack of animal products not bearing any weight on making a decent batch of tots, tater tots are usually... • are tater tots bad for your diet? • are tater tots unhealthy for you? • are tater tots bad for your health?
Fast food f54683	which fast food chain launched a "value" campaign in 1988 that included drive-through • Wendy's • gillian clare mcdonald is the ceo of which fast food chain • which fast food chain did paramount partner with to release movies at participating restaurants • Burger King
Burger King f44349	Burger King • Burger King advertising • Burger King Specialty Sandwiches • who started burger king • burger king hours for breakfast
Wendy's f49792	The Wendy's Company • Wendy's • calories in wendys medium fries • ونديز • how many wendy's are in florida
Jack in the Box f80927	Jack in the Box • what are in jack in the box tacos? • are jack in the box tacos made with meat? • what are jack in the box tacos made with? • are jack in the box tacos made with real meat?
Hardee's f119916	Hardee's • Hardee's • The Carl's Jr near me has this also. Since Carl's Jr and Hardee's are the same company they are probably exactly the... • hardee's breakfast hours • are carl's jr and hardee's the same?
McDonald's burgers f13640	720 Cal. 720 Cal. Each Double Quarter Pounder with Cheese features two quarter pound* 100% fresh beef burger patties... • 510 Cal. 510 Cal. Each Quarter Pounder with Cheese burger features a ¼ lb. * of 100% fresh beef that's hot,... • 510 Cal. 510 Cal. Each Quarter Pounder with Cheese burger features a ¼ lb. • 1250 Cal. 1250 Cal. Get double the fresh beef** flavor with a Double Quarter Pounder® with Cheese made with fresh beef... • Double Quarter Pounder®* with Cheese Meal. 1260 Cal. 1260 Cal. Get double the fresh beef** flavor with a Double...
Rice dishes f112985 Biryani f28414 Nasi goreng f11153 Fried rice f122816	Locrio • دميختك • Galinhada • Pulao • Pelau what vessel is biriyani cooked in • What is the main ingredient in biriani? • where is the movie biriyani being performed • Each community and region of India has its own traditional style of making biriyani, some of the most famous styles of... • what is ulavacharu veg biryani? how long is the yangzhou fried rice • Nasi goreng pattaya • Yangzhou fried rice • 1 Mix and cook/fry the rice and egg together (in a wok is best). 2 Add the long green onion, chashaorou/bacon, carrots... • Yangzhou fried rice Yangzhou fried rice or Yeung Chow fried rice (Traditional Chinese: 揚州炒飯: Simplified Chinese :... Arroz frito Yangzhou • 揚州チャーハン • 揚州炒飯 • Nasi goreng pattaya • Nasi goreng tionghoa
Sausages f3953 Chorizo f22069 Salami f48955 German sausages f110894 Hot dogs f109777	Port sausage • Список колбас • sausage • Chinese sausage • Garlic sausage what is chorizo used for • what is chorizo • Spanish chorizo and Mexican chorizo are very different from each other in looks, texture and taste. Since the Spanish... • Mexican Chorizo. 1 How it's made and sold: Mexican chorizo is a spicy ground meat sausage that is most commonly sold... • Chorizo Salami Genoa • calories in hard salami lunch meat • how is salami processed • Yes, salami is very keto-friendly and low-carb. Salami has a very high-fat content and almost no carbs. Just be aware... • what is the difference between hard salami and dry salami? Wollwurst • Bratwurst • Regensburger Würstchen • what is bratwurst • Appenzeller Siedwurst Cooking Instructions. Grill: Place over medium heat and turn until browned. Boil: Bring water to a boil, reduce heat,... • For just two or three hot dogs, put them in a shallow bowl with about 1/3 cup of water. Cover with plastic wrap. Be... • To cook hot dogs on the stove top, add up to 10 hot dogs to saucepan of boiling water. Return to boil; cover. Remove... • Ingredients: Mechanically Separated Turkey, Water, Salt, Contains 2% or Less Modified Food Starch, Potassium Lactate,... • How long to cook hot dogs. OSCAR MAYER Franks or Wieners can be cooked in just minutes on the grill. Heat the grill to...
Nutritional information f50195 Fast-food calorie questions f37858 Calorie-breakdown template f38187 Percent Daily Value text f23538 Vitamins / minerals f120312	what is the nutritional value of flounder? • what is the nutritional value of eggplant? • Carbs: 45.6 grams. Fiber: 0.7 grams. Sodium: 399 mg. Folate: 24% of the Daily Value (DV) • The Basics. A 1-cup serving of fresh hearts of palm contains 4 grams of protein, 0.3 grams of total fat, 37 grams of... • what is the nutritional value of corn? how many calories in subway bacon egg and cheese flatbread? • how many calories in chicken noodle soup from panera? • how many calories in a burger king vanilla cone? • how many calories in a kfc donut chicken sandwich? • how many calories are in a mcdonald's egg white? There are 111 calories in 1 cup of Cheerios. Calorie breakdown: 13% fat, 75% carbs, 12% protein. • There are 74 calories in 1 serving of Nescafe 3 in 1 Coffee. Calorie breakdown: 26% fat, 72% carbs, 2% protein. • There are 270 calories in a 1 link serving of Johnsonville Italian Sausage. Calorie breakdown: 73% fat, 4% carbs, 22%... • There are 80 calories in a 1/3 cup serving of Carnation Nonfat Dry Milk Powder. Calorie breakdown: 0% fat, 60% carbs,... • There are 320 calories in 1 serving of Taco Bell Nacho Fries. Calorie breakdown: 51% fat, 44% carbs, 5% protein. * Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your... • Percent Daily Values are based on a 2000 calorie diet. Your daily values may be higher or lower depending on your... • Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your... • * Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your... • Percent Daily Values are based on a 2000 calorie diet. Your daily values may be higher or lower depending on your... what vitamins and minerals are in red peppers? • which vitamins and minerals are in bananas? • what are the vitamins and minerals in cucumbers? • what vitamins and minerals does cabbage have? • what are the vitamins of mango?

Table 9: Music hierarchy with five unique activating examples per feature.

Music

Music f33809	مجله موسیقی • موسیقی اسپانیا • Diamond Music • Unduhan musik • BlueBeat Music
Jazz f117052	Kids: Live at Dizzy's Club Coca-Cola • Sunset-Sunside • Sunset/Sunside • Robbins' Nest: Live at the Jazz Showcase • Love and Peace: A Tribute to Horace Silver
Charlie Parker / bebop f130651	Bird and Diz • Charlie Parker Memorial, Vol. 2 • Fats Navarro • Charlie Parker at Storyville • Groovin' High (Dizzy Gillespie album)
John Coltrane f86096	Coltrane for Lovers • コルトレーン • Coltrane (Album) • Coltrane (1962 album) • John Coltrane
Miles Davis f88107	Miles Davis: The Complete Columbia Album Collection • لیدل ایلز • میلز • 데이ویس • Дэвис, Майлз • Forever Miles Davis
Mahavishnu / John McLaughlin f24353	マハヴィシュヌ・オーケストラ • Mahavishnu Orchestra • Mahavishnu Orchestra: The Complete Columbia Albums Collection • mahavishnu orchestra vision of the emerald beyond album • John McLaughlin
Marsalis family f129158	Branford Marsalis • Марсалис, Уинтон • Wynton Marsalis • Branford Marsalis Quartet • 원튼 마살리스
Hip-hop / rap f25024	Hip-hop indonésien • Liste von Hip-Hop-Musikern • Hip-hop espagnol • Something from Nothing: The Art of Rap • Hip hop Tuga
N.W.A f94737	NWA: Straight Outta Compton • N.W.A • N.W.A. • NWA • Straight Outta Compton: N.W.A 10th Anniversary Tribute
Eminem f45908	Eminem • what is the name of the movie that features eminem • Eminem Is Back • Discographie d'Eminem • My Son Marshall, My Son Eminem
Snoop Dogg f1290	Snoop Dogg has worked with Death Row Records. • Snoop Dogg released an album on Death Row Records. • Snoop Dogg • Snoop Dogg comes from California. • Snoop Dogg had a soundtrack released under Death Row Records.
Tupac Shakur f123105	Tupac Shakur • Tupac Shakur Legacy • توباك شاكور • what movie is about tupac shakur • who wrote the book about tupac shakur
Lil Wayne f105791	what is lil wayne's record label called • Tha Carter III • what is lil wayne's label called • what is lil wayne's latest album called • lil wayne sorry 4 the wait album
Country music f22005	Now That's What I Call Country Volume 2 • This is a list of notable events in country music that took place in 2012. • 1988 in country music • My Life's Been a Country Song • List of number-one country singles of 1982 (U.S.)
George Jones f122748	Discographie de George Jones • George Jones Sings His Greatest Hits • 조지 존스 • how old was george jones when he died • George Jones
Rascal Flatts f20266	Discographie de Rascal Flatts • Rascal Flatts • who created rascal flatts music • what is the highest selling rascal flatts album • rascal flatts here comes goodbye album
Patty Loveless / Vince Gill f421	Pam Tillis • Patty Loveless • Suzy Bogguss • Kathy Mattea • pam tillis all the good ones are gone album
Hank Williams f22250	hank wiiliams where did bocephus come from • هانك وليامز • Hank Williams discography • what was hank williams first hit song • ハンク・ウィリアムズ
Rock / metal f10590	Phil Campbell and the Bastard Sons • Rhino Bucket • The Wildhearts • Taking Dawn • Ugly Kid Joe
Hard-rock musicians / bands f61236	Dokken • Dave Meniketti • Jeff Pilson • Bobby Blotzer • ドン・ドッケン
Nu / alternative metal f16925	Nonpoint • Spineshank • 40 Below Summer • Ill Niño • Stereomud
Power / symphonic metal f123141	Ecliptica – Revisited: 15th Anniversary Edition • Sonata Arctica • Liste der Lieder von Sonata Arctica • The Metal Opera Part II • A Tribute to Sonata Arctica
Doom / stoner metal f75431	Sleep's Holy Mountain • Spirit Caravan • Electric Wizard • The Atomic Bitchwax • Scott Weinrich
Electronic / dance f130808	Antoine Clamaran • Robbie Rivera • Sander Kleinenberg • Mauro Picotto • Ivan Gough
Detroit techno cluster f5560	The Belleville Three • Juan Atkins • Kevin Saunderson • where did the belleville three come from • where did the belleville three first perform
Trance mixes / compilations f111755	A State of Trance 2013 • A State of Trance • A State of Trance 2009 • A State of Trance 2012 • A State of Trance 2010
Basement Jaxx f17166	Basement Jaxx • Discographie de Basement Jaxx • basement jaxx never say never album • when did basement jaxx album come out • 베이스メント・ジャックス
deadmau5 f35810	what is the title of deadmau5's fourth album • Deadmau5 Circa 1998-2002 • Full Circle (Deadmau5 album) • when does deadmau5 upload stuff i used to do • Mau5trap
The Beatles f18522	The Beatles • Beatles Day • ビートルズ! • 비틀즈 • Beatles Stories
Please Please Me / early Beatles f19564	The Beatles' Please Please Me album was rush-released by Parlophone on 22nd March, 1963 to capitalise on the enormous... • Please Please Me (альбом) • Please Please Me • Please Please Me (album) • when was please please me released in uk
Hey Jude f43221	Hey Jude • where did the beatles film hey jude • what tv show featured the beatles singing hey jude • how many weeks did hey jude stay on the billboard hot 100 • Hey Jude (Wilson Pickett album)
Let It Be / Get Back f94589	Let It Be • Let It Be (song) "Let It Be" is a song by the English rock band the Beatles, released in March 1970 as a single, and... • Let It Be (песня) • Let It Be (song) • Let It Be (2004 film)
Abbey Road / late Beatles f67785	Abbey Road (album) • Abbey Road (album) • The Best of Abbey Road • what is the title of abbey road's last album • who sings lead vocals on best of abbey road

G Simulation experiments

G.1 Recovery under width and sparsity variation

We study recovery in finite synthetic distributions with known ground-truth atoms. These experiments assess whether trained TopK SAEs exhibit prefix recovery and how this behavior depends on the frequency distribution and the SAE sparsity. Figures 22 and 23 report approximate recovered prefix lengths, expressed in atoms and divided by SAE width.

Data-generating distributions. Inputs lie in $\mathbb{R}^{512}$ and are nonnegative linear combinations of exactly $K = 8$ atoms. We draw each ground-truth direction independently from a standard Gaussian and normalize it to unit length. In the flat setting, there are $M = 4096$ atoms. An eight-element support S is sampled according to

$$\Pr(S) = \frac{\prod_{i \in S} i^{-\alpha}}{\sum_{T: |T|=8} \prod_{j \in T} j^{-\alpha}}, \quad |S| = 8,$$

and its nonzero coefficients are independent $\text{Uniform}(0, 1)$ draws. This is a product-weight distribution over fixed-size subsets; the power law specifies the support weights, rather than the marginal activation probabilities directly. In the hierarchical setting, $M = 3072$ atoms form 1024 families, each containing a parent and two children. We sample four families using the same product-weight law over family ranks, then activate each selected parent and one of its two children, chosen uniformly. The eight active coefficients are again independent $\text{Uniform}(0, 1)$ draws. Thus a child can only occur with its parent, while either child can accompany the parent.

Training and sweeps. We train nonnegative TopK SAEs at widths $m \in \{128, 256, 512, 1024, 2048, M\}$. Generating sparsity $K = 8$ is fixed throughout; k denotes the SAE’s activation budget. The frequency sweep uses $\alpha \in \{0.8, 1.2, 1.6, 2.0, 2.4\}$ at $k = 8$. The sparsity sweep uses $k \in \{1, 2, 4, 6, 8, 10, 12, 16, 32\}$ at $\alpha = 1.6$. Theorem 2 does not give guarantees for $\alpha = 0.8$. We use three paired random seeds per setting, giving 468 distinct training trajectories after deduplicating the shared $\alpha = 1.6, k = 8$ setting.

Each trajectory trains on 1,024M examples using Adam with learning rate 10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.99$, and batch size 1024, with no learning-rate schedule. Decoder directions have unit norm and the decoder bias is fixed to zero. An auxiliary reconstruction loss with coefficient 0.25 targets up to 64 neurons inactive for 256 steps.

Recovery criteria. Geometric recovery requires absolute cosine similarity at least 0.9 with a learned decoder direction. For hierarchical data, child comparisons project both directions onto the orthogonal complement of the true parent; recovery of a complete family requires three distinct decoder directions meeting the parent and two projected-child criteria. Activation recovery uses 200,000 independent validation examples to select coordinate thresholds and a maximum-cardinality injective assignment among atom–coordinate pairs with validation F1 at least 0.8. A matched atom is recovered if its F1 is also at least 0.8 on a separate set of 400,000 test examples. An activation-recovered family must recover all three members.

For either criterion, let q_i indicate recovery of atom i in the flat setting or complete family i in the hierarchical setting, ordered by generating weight. We compute

$$r = \max \left(\{0\} \cup \left\{ s : \sum_{i=1}^s q_i \geq 0.9s \right\} \right).$$

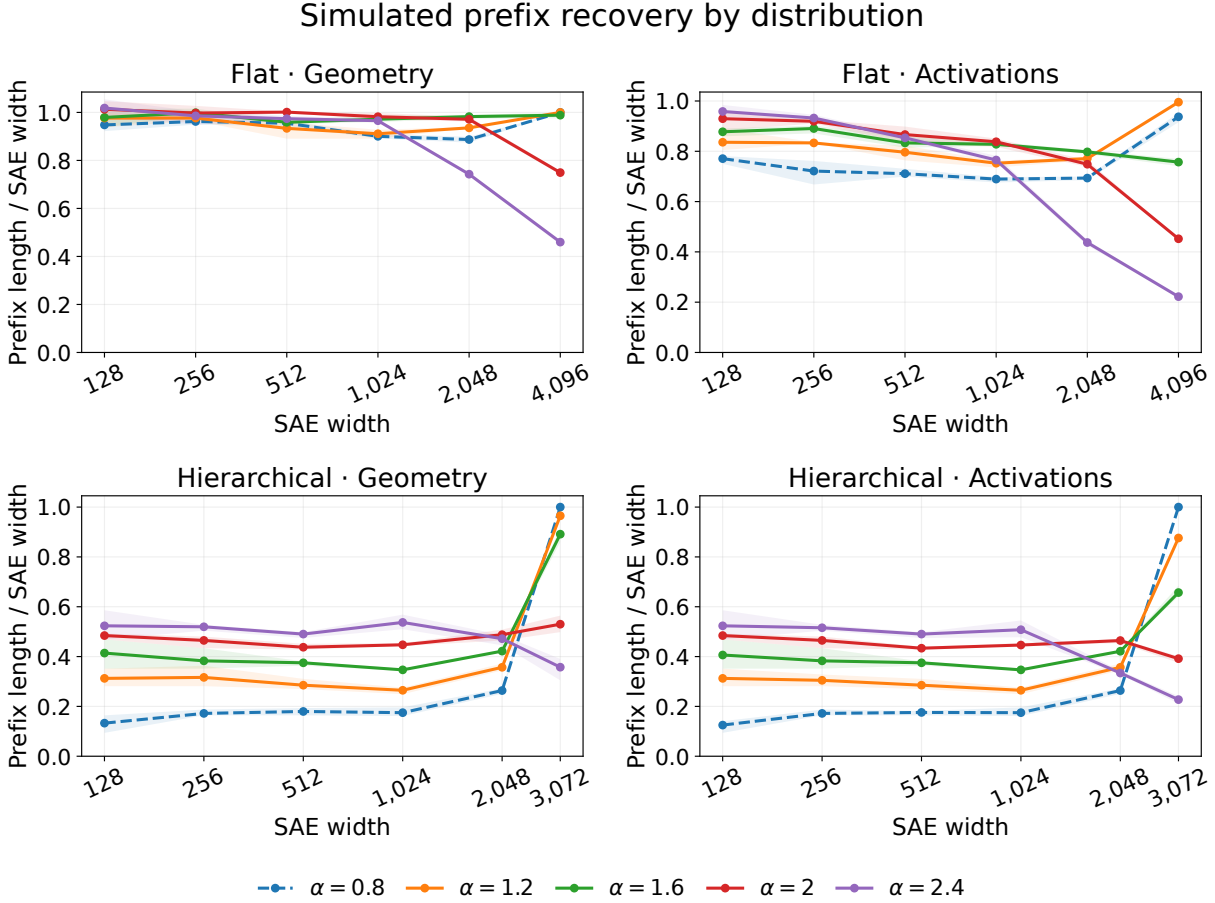


Figure 22: **Width-normalized prefix recovery across frequency distributions.** Generating sparsity $K = 8$ and SAE sparsity $k = 8$ are fixed; each curve corresponds to a different α . All models train on 1,024M examples. Rows show flat and hierarchical data; columns show geometric and activation recovery. The vertical axis is approximate prefix length in atoms divided by SAE width (three atoms per complete family for hierarchical data), with a 10% miss allowance. Lines show three-seed means and bands show seed ranges.

Simulated prefix recovery by sparsity parameter

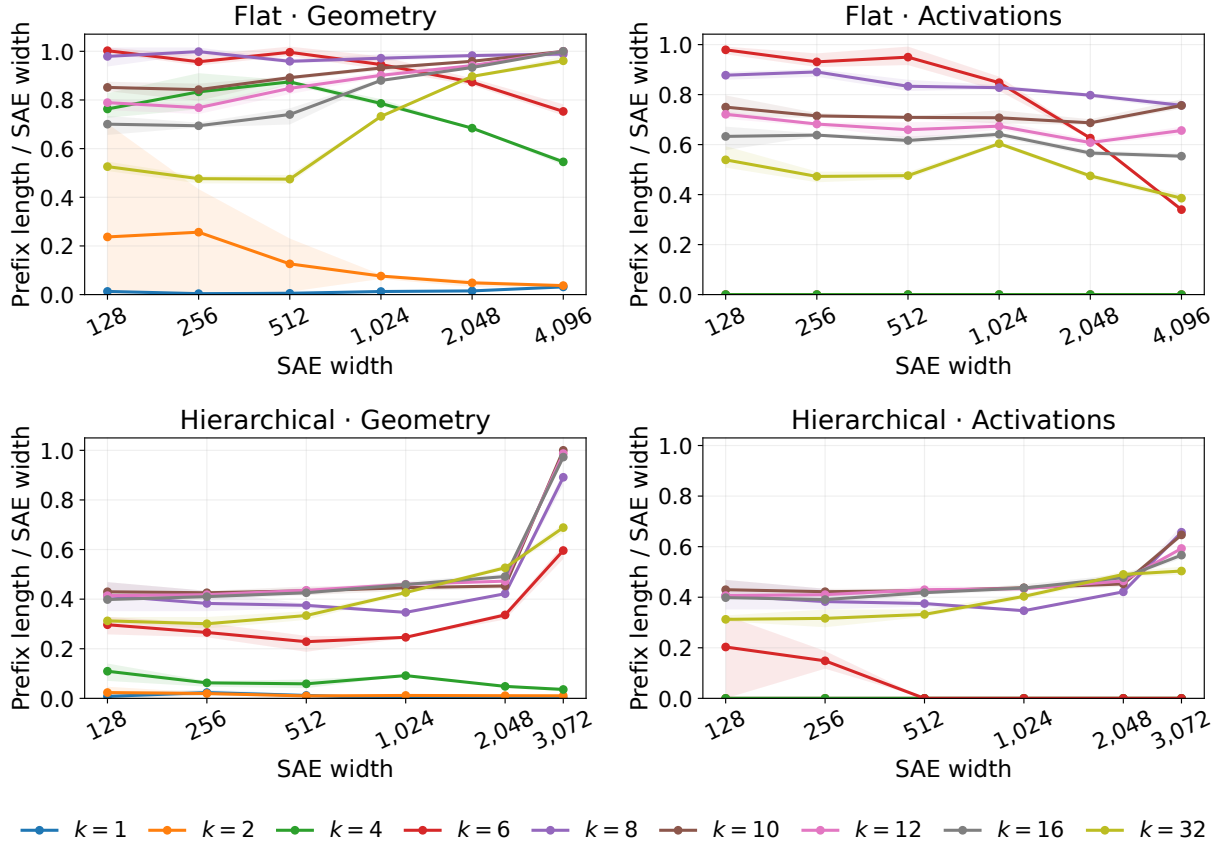


Figure 23: **Width-normalized prefix recovery across SAE sparsities.** Generating sparsity $K = 8$ and frequency exponent $\alpha = 1.6$ are fixed; curves vary SAE sparsity k . All models train on 1,024M examples. Panels, normalization, and the 10% prefix miss allowance follow Figure 22. Lines show three-seed means and bands show seed ranges.

G.2 Persistent features across simulated SAE widths

We apply the persistence metric in Section C.2.3. For each source width m , we independently compute a maximum matching into each larger trained SAE and intersect the matched source-feature sets, holding the data distribution, SAE sparsity, and paired seed fixed. Matches require signed decoder cosine similarity or signed Pearson correlation of continuous post-TopK activations at least $t = 0.8$. The largest width (4096 for flat data, 3072 for hierarchical data) is a matching target only. We report the persistent-feature proportion, dividing the size of the intersection by all m source features, including inactive features.

We evaluate each group of six widths on the same 400,000 held-out test examples and compute all pairwise similarities without sketching or candidate truncation. Constant activation columns cannot form Pearson edges. The procedure in Section C.2.3 certifies each pairwise maximum on these empirical threshold graphs and verifies their matched source-set intersection. This measures persistence between learned SAEs.

Figure 24 varies the frequency exponent; Figure 25 varies SAE sparsity. At $\alpha = 1.6, k = 8$, mean activation persistence across source widths ranges from 93.6–96.1% for flat data and 91.9–99.1% for hierarchical data. Hierarchical decoder persistence rises from 61.5% at width 128 to 91.8% at width 2048. Severe sparsity underspecification particularly reduces activation persistence, showing that persistent decoder directions need not imply persistent activation patterns.

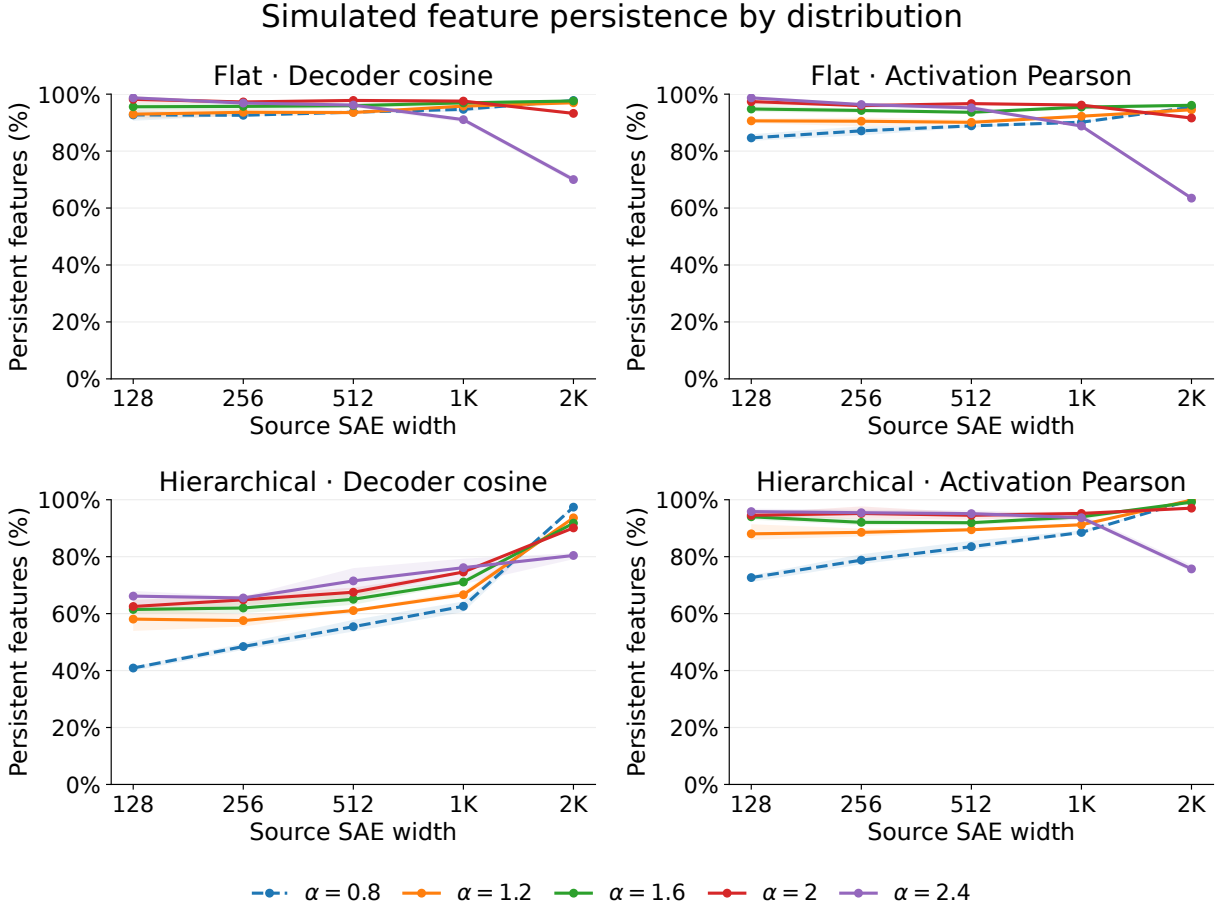


Figure 24: **Persistent-feature proportions across frequency distributions at $t = 0.8$.** The intersection of source sets matched by independent maximum matchings to every larger trained width, divided by source SAE width. Generating sparsity $K = 8$ and SAE sparsity $k = 8$ are fixed; colors vary α . Rows show flat and hierarchical data; columns show signed decoder cosine and activation Pearson similarity. All models train on 1,024M examples. Lines show three-seed means and bands show seed ranges. The dashed $\alpha = 0.8$ curve is the finite-distribution control outside the theorem’s guarantees.

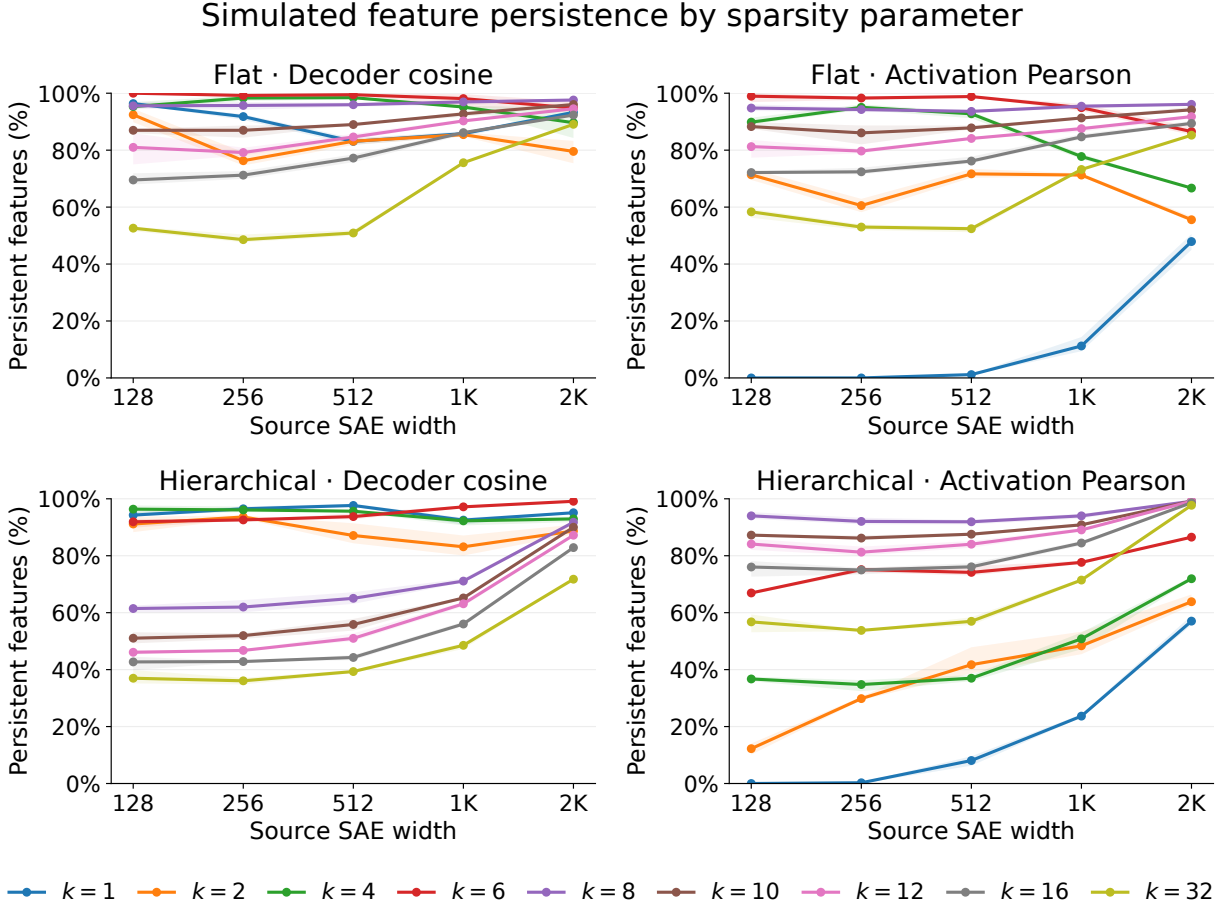


Figure 25: **Persistent-feature proportions across SAE sparsities at $t = 0.8$.** The intersection of source sets matched by independent maximum matchings to every larger trained width, divided by source SAE width. Generating sparsity $K = 8$ and frequency exponent $\alpha = 1.6$ are fixed; colors vary SAE sparsity k . Rows show flat and hierarchical data; columns show signed decoder cosine and activation Pearson similarity. All models train on 1,024M examples. Lines show three-seed means and bands show seed ranges.

dict. size	i	$\hat{a}_i$ with molecule illustration	$\hat{z}_i(\text{cat})$	$\hat{z}_i(\text{dog})$	$\hat{A}\hat{z}(\text{cat})$	$\hat{A}\hat{z}(\text{dog})$
1	1	$\frac{2}{\sqrt{6}}a_{\text{pet}} + \frac{1}{\sqrt{6}}a_{\text{cat}} + \frac{1}{\sqrt{6}}a_{\text{dog}}$ 	$\frac{3}{\sqrt{6}}$	$\frac{3}{\sqrt{6}}$	$\frac{2a_{\text{pet}} + a_{\text{cat}} + a_{\text{dog}}}{2}$	
2	1	$\frac{1}{\sqrt{2}}a_{\text{pet}} + \frac{1}{\sqrt{2}}a_{\text{cat}}$ 	$\sqrt{2}$	0	$a_{\text{pet}} + a_{\text{cat}} \quad a_{\text{pet}} + a_{\text{dog}}$	
	2	$\frac{1}{\sqrt{2}}a_{\text{pet}} + \frac{1}{\sqrt{2}}a_{\text{dog}}$ 	0	$\sqrt{2}$		
3	1	a_{pet} 	1	1	$a_{\text{pet}} + a_{\text{cat}} \quad a_{\text{pet}} + a_{\text{dog}}$	
	2	a_{cat} 	1	0		
	3	a_{dog} 	0	1		

Table 10: Learned directions illustrate molecular splitting. Corresponding learned features illustrate mesoscale feature splitting and absorption: at size 2 only, there is no learned feature that is active on both cat and dog.

H A Molecular Model of SAE Scaling

H.1 Computing the empirical molecules

The empirical molecules in Tables 1 and 2 approximate smaller-SAE decoder directions using decoder directions from the Gemini SAE of width 131,072 and sparsity $k = 128$ as proxy atoms. These are decompositions of learned directions in embedding space; the activation values and recalls are measured separately by evaluating the corresponding SAEs. The 131K features serve as empirical proxies, rather than certified atomic directions. Below, d denotes a target decoder row and b_j a proxy-atom decoder row, both at their stored checkpoint scale.

Prefix molecules: basis and least-squares fit. For Table 2, we use the word-prefix data, saved selection/test split, and single-feature probe procedure in Section C.5. For each of the 23 available one- and two-letter prefixes beginning with **a**, we take the 131K feature and positive activation threshold maximizing selection F1, without filtering by test F1. Their decoder rows form a matrix $B = [b_1, \dots, b_{23}]$. There are 22 distinct directions, since **ac** and **ak** select the same feature; the regression retains both columns. For each selected smaller-SAE decoder direction, we compute

$$c^* = \arg \min_{c \in \mathbb{R}^{23}} \|d - Bc\|_2^2,$$

using the minimum-norm solution, computed in double precision with `numpy.linalg.lstsq` and `rcond=None`. There is no intercept, regularization, or nonnegativity constraint, and no additional normalization of the decoder rows. The fit uses all 23 basis columns, even though the table displays only the five prefixes **a**, **ah**, **ab**, **ap**, and **as**. Their proxy-atom feature IDs are 38618, 121657, 100965, 120576, and 32170, respectively.

At each displayed width ($512/k = 32$, $2,048/k = 32$, and $131,072/k = 128$), we select the best probe for each of those five categories and deduplicate feature IDs. If a feature is selected for multiple categories, we use the threshold from the first category in the order **a**, **ah**, **ab**, **ap**, **as**. For a category q , its recall cell is

$$\frac{\#\{x \in T_q : \hat{z}_j(x) \geq \tau_j\}}{|T_q|},$$

where T_q is the set of held-out words beginning with q . The four displayed child categories contain 70, 339, 292, and 456 test words, respectively. We display only rows whose recall is at least 0.5 on at least one category for which that feature was selected, using the same threshold across all cells in a row. This is a display filter using held-out recall; the probes and thresholds themselves are selected on the selection split.

The displayed sums and circles retain only coefficients at least 0.015 among the five illustrated prefixes. Negative coefficients, smaller positive coefficients, and all other prefix components are omitted without refitting or renormalizing the retained coefficients. Circle radii are proportional to the cube root of the coefficient. Thus, these pictures summarize selected positive contributions rather than the complete signed least-squares reconstruction. At width 131,072, the illustrated directions are themselves basis columns, so their single-atom decompositions hold by construction; their held-out activation patterns are measured independently.

Polysemantic molecule: sparse decoder reconstruction. For feature 3290 in the Gemini width-4,096, $k = 32$ SAE (Table 1), the candidate dictionary contains all 131,072 proxy-atom decoder rows. We use orthogonal matching pursuit (OMP). Starting with residual $r = d$, each step selects the unselected atom maximizing $|\langle r, b_j / \|b_j\|_2 \rangle|$, refits unrestricted signed least-squares coefficients on the selected support, and updates the residual. Atom normalization is used for selection; reported coefficients are expressed against the stored decoder rows. We retain the first support along this greedy path satisfying

$$\left\| d - \sum_{j \in S} c_j b_j \right\|_2^2 \leq 0.05.$$

This is an absolute squared-error threshold on the stored target row, not a relative-error criterion or a guarantee of globally minimal support. The retained fit has 17 terms, squared error 0.049297, and cosine similarity 0.975040 between the target and reconstruction.

For readability, the diagram shows the five largest positive coefficients of this 17-term fit (also the first five atoms selected by OMP): 0.6763 on feature 8574, 0.1955 on 114331, 0.1665 on 68236, 0.1615 on 72509, and 0.0593 on 97729. These coefficients come from the full fit, not a refit restricted to five atoms, and the reported reconstruction error applies to all 17 terms. Circle areas approximately encode these positive coefficients; the smallest circle (97729) is enlarged for legibility. The other twelve signed terms are omitted from the illustration.

Activation examples. The numerical cells in Tables 1 and 11 are cached post-TopK SAE activations on the same corpus inputs, with absent features assigned zero. They are not inferred from the decoder-decomposition coefficients. The upper block of the main table is manually selected from activating texts to illustrate different senses of transformation, including examples containing and not containing the substring **transform**. The appendix gives twelve examples per proxy atom, curated across activation bins and semantic subtypes and sorted by decreasing activation. The lower block of the main table takes one example per atom from this appendix selection, with activation at least 0.09 on that atom and below 0.05 on each of the other four. These curated examples illustrate semantic selectivity rather than estimating its corpus frequency. Main-table shading uses a shared linear activation scale capped at 0.3.

Decoder-similarity heatmaps. For Figure 26, we select the best probes for all 23 available a-prefix categories separately at widths $32,768/k = 64$ and $131,072/k = 128$, again without a test-F1 filter. After deduplicating feature IDs, each panel contains 22 directions. Entries are signed decoder

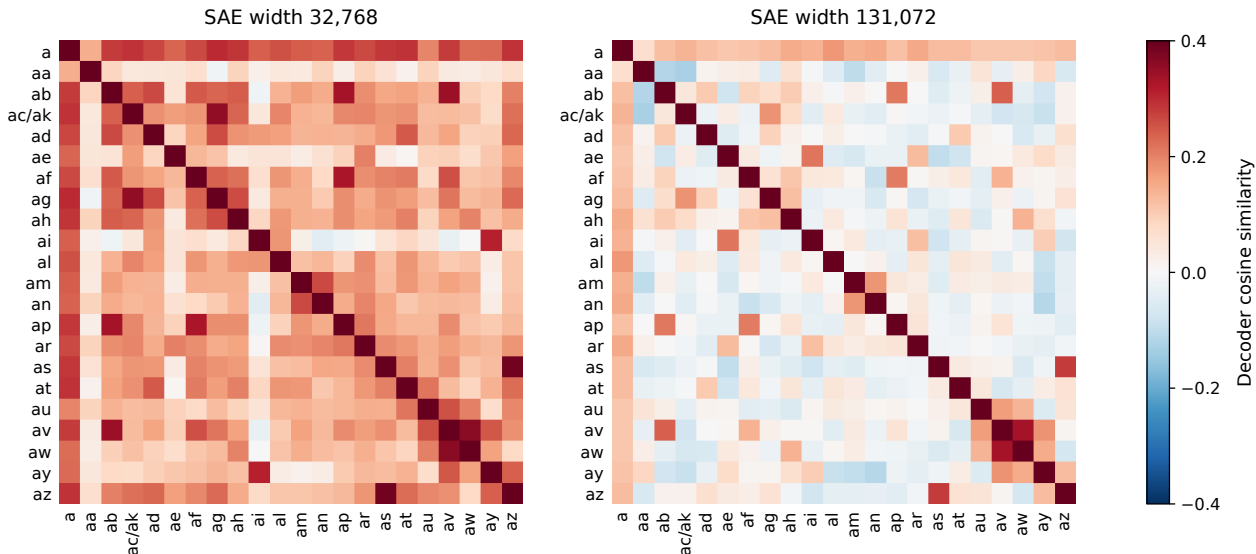


Figure 26: **Decoder direction similarities among a-prefix probes.** Signed pairwise cosine similarities for Gemini SAEs of widths 32,768 ($k = 64$, left) and 131,072 ($k = 128$, right). At each width, we take the best selected feature for each of the 23 available one- and two-letter a-prefix categories, without an F1 filter, and deduplicate feature IDs. This yields 22 unique decoder directions: **ac** and **ak** share a probe at both widths. Rows and columns use the same prefix order in both panels, with a common color scale from -0.4 to 0.4 . Diagonal self-similarities equal 1 and are shown at the maximum red. The median cosine over the 231 unique off-diagonal pairs is 0.1438 at width 32,768 and 0.0084 at width 131,072.

cosines $\langle d_i, d_j \rangle / (\|d_i\|_2 \|d_j\|_2)$, computed directly from the checkpoint rows, rather than from the molecule fits. The reported medians use the 231 unordered off-diagonal pairs in each panel.

Table 11 gives twelve activating examples for each of the five 131K features in Table 1.

Table 11: Activating examples for the five 131K atoms. Each column contains twelve examples, sorted by decreasing activation. These appear more monosemantic than the feature in Table 1.

8574	114331	68236	72509	97729
Transform (0.281)	Transformers: Cybertron (0.288)	is gewijzigd in of naar? (0.320)	Transformer (0.273)	* To multiply Row 1 by 3, use the elementary matrix: $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ [0 0 0 1]... (0.129)
meaning of transformed (0.273)	who does optimus prime fight in transformers autobots (0.270)	تغيير (0.313)	Transformer types (0.265)	The concept lies in the fact that every elementary row operation can be represented as a matrix m... (0.117)
Transformation (Politikwissenschaft) (0.265)	Transformers: Prime (0.250)	Cambios de imagen (0.265)	how does transformer work (0.265)	1. Reflect about the y-axis: The new endpoints become (1, 4) and (-2, 3). https://userupload... (0.117)
when was the transformation theory introduced in quantum mechanics (0.236)	what is the name of the god in transformers (0.245)	when was the roman liturgical celebration changed (0.247)	Why do transformers exhibit voltage ratios different from their specified values when connected i... (0.239)	Following the steps from the previous example, we first find the reflection matrix for the line y... (0.111)
To transform $\sqrt{2}$ into $1+\frac{1}{\sqrt{2}}$, we can follow these steps: 1. Start with ... (0.232)	Transformers: The Score (0.237)	how do i change my azure subscription? (0.239)	The buzzing noise from the transformer is caused by the rapid fluctuations of magnetic fields, ty... (0.238)	To rotate a vector around a point (2, 3) by 30 degrees, follow these steps: 1. Translate the vect... (0.108)
what are the steps in graph transformation called (0.197)	Transformers (film) voices of Ryan, Cullen, and Dunn, as well as Welker as Megatron. The Target c... (0.197)	when did the grammy award for best traditional pop vocal album change (0.231)	What are the consequences of mistakenly short-circuiting both the primary and secondary sides of... (0.228)	The Hadamard gate represents a 180-degree rotation around the axis connecting the states $ 0\rangle$ and... (0.104)
The transformations in the generating function are based on two key concepts: the derivative of a... (0.187)	a unit commanded by Beast Wars Megatron, who ordered them to attack following the deaths of Optim... (0.197)	If you want to change the unit vector $\vec{a}$ to $\vec{a}'$ with a distance d' from $\vec{b}$, n... (0.188)	How should I wire a center-tapped transformer with input voltages of 0V and 230V and output volta... (0.199)	To obtain the graph of $y=-2f(x)$ from the graph of $f(x)$, we can perform the following transfor... (0.102)
what is oncogenic transformation (0.186)	Transformers: Exodus Cybertron where each individual determining their own path in life highly ap... (0.194)	when did the west coast hockey league change to the east coast (0.177)	what type of transformer is needed to achieve the nominal impedance of valve outputs (0.186)	To represent the combined rotation of a 3D point (x, y, z), you can multiply the rotation matri... (0.102)
A Reflection is a transformation in which the figure is the mirror image of the other. 1 Reflecti... (0.142)	Greg "M Sipher" Sepelak is an American artist and writer for various Transformers comic ventu... (0.149)	The 2015 expenditure for Army research, development and acquisition changed from 32 billion proj... (0.141)	Commonly, transformers are used to increase or decrease the voltages of alternating current in el... (0.182)	To transform a vector from System (1) to System (2) using intrinsic rotations, follow these s... (0.101)
Is it correct to use Tietze transformations to transform a group into the group langle a, b, c... (0.121)	Sentinel Prime (0.148)	['Open your Google Account. You might need to sign in.', 'Under "Security," select Signing in to... (0.140)	Control transformers are used to provide single point wiring when the control voltage differs fro... (0.149)	In the context of rigid body transformations preserving the centroid at the origin, the transform... (0.052)
At transform plate boundaries plates grind past each other side by side. This type of boundary se... (0.095)	Barricade is the name of four fictional characters in the various Transformers universes. (0.139)	The integrals span one full period of the DTFT, which means that the $x[n]$ samples are also the co... (0.097)	Earthing transformers are used to create a neutral point in a three-phase system, which provides... (0.140)	Find the transformations applied to the function $f(x)=\frac{1}{x}$ to obtain the function $y=df...$ (0.040)
What is the correct interpretation of a Wick rotation, specifically regarding the transformation... (0.036)	에이수스 이패드 트랜스포머 (0.084)	Constant Change Constant Change is the sixth studio album by Filipino singer-songwriter Jose Mari... (0.086)	Will a step-down transformer work for using my US electronic devices (LED lights, string lights,... (0.137)	How can I effectively visualize Lorentz transformations in the context of Minkowski spacetime? Sp... (0.018)

I An Atomic View of Feature Manifolds

A growing body of work has observed that features do not necessarily exclusively have one-dimensional representations. For example, considering two-dimensional PCA projections of days of the year, one often recovers circular structure. The observation of such “manifolds” is sometimes used to argue against the linear representation hypothesis (which the atomic hypothesis strengthens).

Here, we point out that the atomic hypothesis is not inconsistent with the observation of higher-dimensional non-linear structure. However, the atomic hypothesis can be viewed as implying that it should be possible to reconstruct this non-linear structure using atomic features. In particular, if we believe that the information represented by f is captured entirely by z , we should see the same non-linear structure arise in z . Looking exclusively at z disentangles the information stored in the features themselves from information potentially stored by the directions corresponding to each feature.

We consider subsets of SAE features $S \subseteq [131072]$ and consider PCA projections of $z_S(x) \in \mathbb{R}^M$ for $x \in X^*$. We demonstrate that this procedure reveals geometric structure from atomic features. Notice this procedure only uses the learned features z and not the learned directions A . Importantly, this disentangles geometric structure arising from the embedding space (i.e., if different atoms are represented along correlated directions in $\mathbb{R}^d$) from geometric structure arising purely from correlation in feature activations in data.

We study two datasets: dates and colors. For each dataset, Figure 27 compares PCA of the raw embeddings with PCA of the continuous post-TopK activations restricted to a selected subset S of the 131K Gemini SAE’s coordinates. The feature subsets and color filtering were chosen through exploratory inspection.

Dates. The inputs are the 366 strings “[Month] [Day]” in calendar order, including “February 29”, with no year in the strings. We identify twelve month-related features in the gemini 131K SAE. We apply the full 131K encoder and TopK operation before extracting these twelve coordinates, assigning zero when a coordinate is not retained by TopK. PCA mean-centers the 366×3072 raw embedding matrix and the 366×12 activation matrix separately. The heatmap displays the same uncentered, continuous activation matrix in calendar order.

Colors. The original collection contains 865 color-name rows with hex-string embedding inputs. We deduplicate hex codes case-insensitively, retaining the first row for each of the 765 distinct colors. After scaling RGB channels to $[0, 1]$, we retain colors with HSV saturation $S \geq 0.60$, HSV value $V \geq 0.50$, and HLS lightness $L \leq 0.70$. This leaves 374 colors. We sort these by increasing hue, decreasing saturation, and then increasing value.

The eight selected 131K features were chosen from ranks 6, 7, 8, 10, 11, 12, 15, and 16, respectively, in the color collection’s feature-prevalence ranking at activation threshold 0.05. These corresponded to features that seemed color-related (rather than activating on hex patterns). That threshold is used only for selection: the plotted values are raw post-TopK activations. We refit three-component PCA separately to the 374×3072 raw embeddings and the 374×8 activation matrix using the full SVD solver, mean-centering without standardization, whitening, or additional normalization. These are fits on the retained colors, rather than subsets of PCA coordinates fitted to all 865 rows. The color heatmap uses the same raw activations, with rows ordered by activation-weighted circular mean position along the displayed hue sequence.

Text and artwork examples. Table 12 shows five texts and five WikiArt images for each of the eight color features: 40 texts and 40 images in total. Texts are ranked by raw post-TopK activation

over all 89,827,558 Gemini training-corpus rows, removing exact duplicate full texts separately for each feature. Images are ranked independently by the same activation definition over all 116,261 rows of the cached WikiArt corpus. We take the five highest positive image activations per feature, with ties broken by decreasing image-vector index; no hex-color, style, or enrichment filter is applied to these images.

WikiArt images are outside the SAE training corpus. These selected examples illustrate transfer beyond hex codes to multilingual text and images; they are not a held-out population performance estimate.

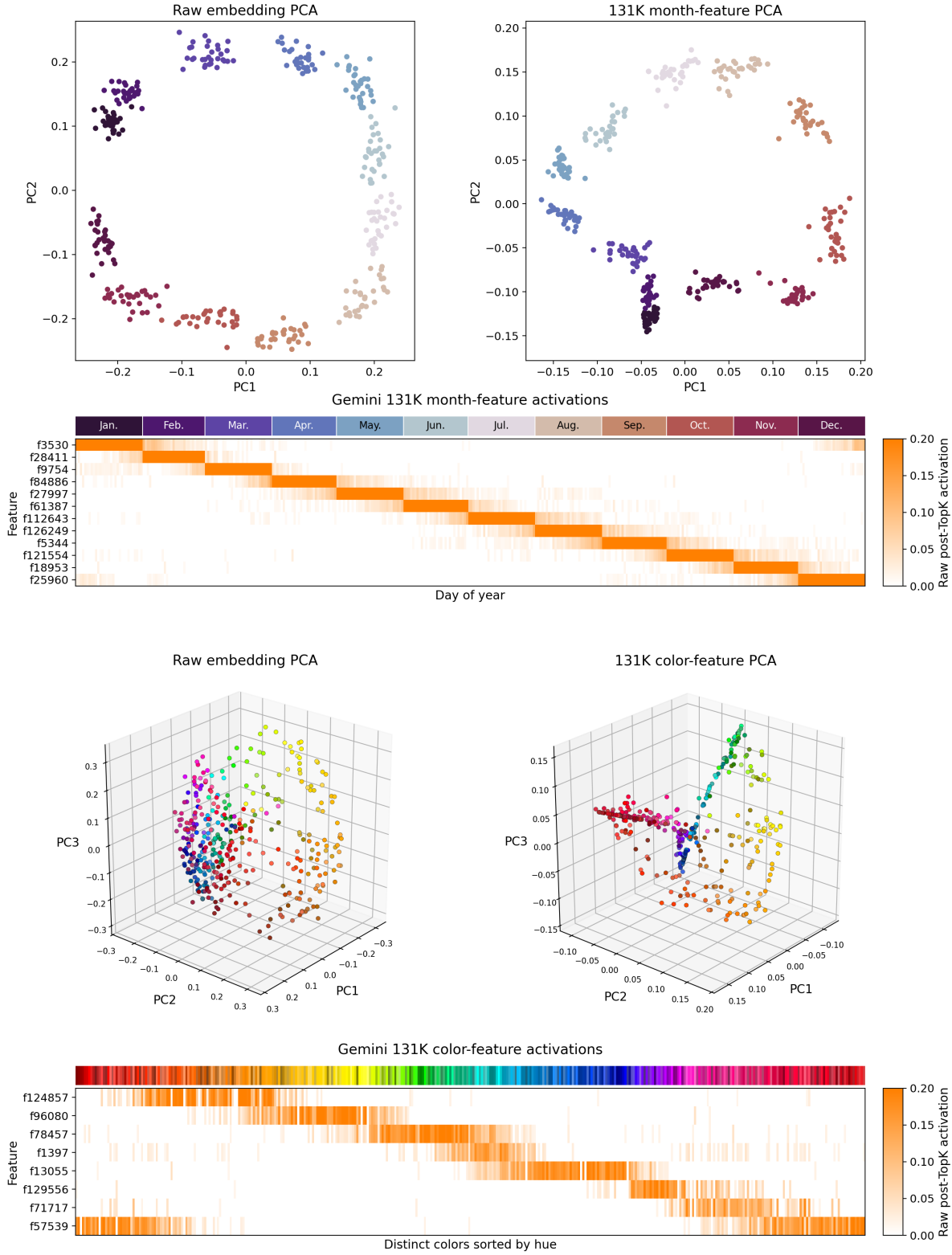


Figure 27: Manifolds arising from small sets of 131K gemini SAE activations. Top left shows PCA of raw embeddings, top right shows PCA of the SAE feature representation restricted to a subset of coordinates, and the bottom shows the activation pattern on these selected features. Importantly, the SAE-induced manifolds arise purely from correlation in activations, not correlation in feature directions.

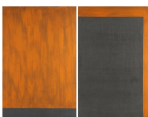
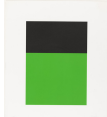
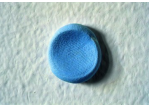
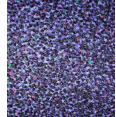
	top activating texts	top activating artwork				
f124857	Оранжевый — хит сезона (0.4119)					
	Orange Record Label (0.3469)					
	Orange Hat (0.3398)					
	Orange Link (0.3359)					
	Orange ribbon (0.3344)					
		(0.2292)	(0.2244)	(0.2110)	(0.2063)	(0.2052)
f96080	Yellow hat (0.3367)					
	Yellow Arrow (0.3251)					
	Yellow monitor (0.3236)					
	Yellow dog (0.3185)					
	Yellow Hair (0.3169)					
		(0.1967)	(0.1800)	(0.1798)	(0.1796)	(0.1787)
f78457	Green hat (0.3725)					
	Green and Silver (0.3619)					
	Green Rose (0.3583)					
	Green S (0.3556)					
	Green Light (0.3518)					
		(0.1837)	(0.1826)	(0.1816)	(0.1791)	(0.1762)
f1397	Turquoise is not a color choice you make for your car. It chooses you! I've been... (0.2814)					
	Turquoise ribbon (0.2587)					
	「ターコイズブルー」(C90 Y20) (0.2533)					
	turquoise number (0.2505)					
	Colors That Go With Turquoise. ... Different shades of turquoise don't have to match... (0.2317)					
		(0.1757)	(0.1655)	(0.1618)	(0.1565)	(0.1511)
f13055	Blue Rainbow (0.3702)					
	Blue hat (0.3697)					
	Blue Girl (0.3693)					
	Blue Snowman (0.3691)					
	Blue Monkey (0.3658)					
		(0.1800)	(0.1731)	(0.1716)	(0.1708)	(0.1637)
f129556	Purple Numbers (0.3519)					
	Purple crocodile (0.3409)					
	i feel like rich purple and gold are a match made in heaven and this reinforces that... (0.3329)					
	Purple Love (0.3328)					
	Purple Radio (0.3317)					
		(0.2225)	(0.2112)	(0.1996)	(0.1966)	(0.1950)
f71717	Pink Subaru (0.3593)					
	Pink contract (0.3525)					
	Pink Pineapple (0.3458)					
	Pink and Gray (0.3367)					
	Pink Man (0.3366)					
		(0.2032)	(0.1902)	(0.1841)	(0.1840)	(0.1839)
f57539	Красное на красном (0.4016)					
	Red Elephant (0.3942)					
	Red Bridge (0.3836)					
	Red Post (0.3779)					
	Red jersey (0.3765)					
		(0.1869)	(0.1832)	(0.1821)	(0.1800)	(0.1792)

Table 12: Top-activating text and image examples for eight color-selective features in the Gemini 131K sparse autoencoder. Each row shows the five highest-activating distinct general-corpus texts and five highest-activating WikiArt images for one feature, ordered by decreasing activation within each corpus. Parentheses give raw post-TopK activations. Texts are truncated for display.

J An Atomic View of the Platonic Representation Hypothesis

If we interpret the rigidity property at an even higher level, then we can arrive at an even stronger theory: that atomic features are not atomic only with respect to an individual model, but are actually atomic as descriptions of the world (or at least, of descriptions of human-generated text). We might describe these atomic features as also being “platonic” [48].

So while there is no reason for atomic directions to be model independent (one can have an equivalent representation that is a rotation), it seems possible that z could be model independent, in which case different representations f, f' are just different projections of the feature representation: that is, $f = Az$ and $f' = A'z$.

Same-width cross-model matching. We compare activations between models for all nine SAE widths from 512 to 131,072. At each width and threshold $t \in \{0.5, 0.6, 0.7, 0.8\}$, we apply the pairwise matching and candidate-graph procedures in Sections C.2.3 and C.2.4 to Gemini and Nemotron features.

Figure 28 shows that the number of matched pairs increases with SAE width, while the fraction of each dictionary that is matched generally decreases.

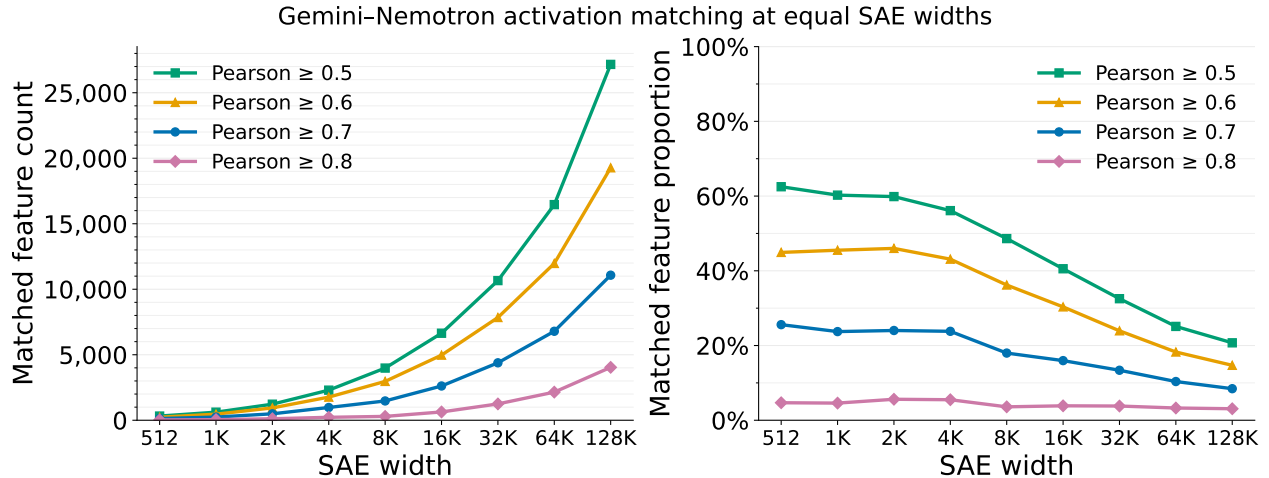


Figure 28: Feature-activation matching between Gemini and Nemotron SAEs of equal width. Curves give maximum-cardinality one-to-one matching at signed Pearson thresholds $t \in \{0.5, 0.6, 0.7, 0.8\}$ on 89,227,558 shared examples. Left: matched-pair count. Right: count divided by the common SAE width, including dead and unmatched features in the denominator. Counts are exact on the retained candidate graphs and lower bounds on exhaustive matching. The 131K comparison is included because matching across models does not require a larger SAE.

Example matches. Tables 13 to 24 show twelve illustrative correlated pairs of features: two main examples and ten additional diverse pairs, with Pearson similarity ranging from 0.3 to 0.9. They cover clarification of meaning, algebraic transformations, education, trigonometry, astronomy, trade, programming, mechanics, food, sign languages, music, and botany. The captions describe qualitative interpretations of the activating texts. Each table contains fifteen sampled texts from each model: eight with source-feature activation in $(0.05, 0.1]$ and seven above 0.1. Gemini feature 71850 is an exception: it has no cached activations above 0.1, so all fifteen of its examples come from $(0.05, 0.1]$.

Table 13: Clarifying the meaning or scope of a term. Pearson $r = 0.352$.

Text	Gemini 71850	Nemotron 119331
Secara khusus, Chang'an Avenue hanya terdiri dari Chang'an Avenue Barat dan Chang'an Avenue T...	0.073	0.067
The term "bile duct" is often used as a shorthand for the "common bile duct," but they are no...	0.064	0.109
In both Great Britain and Ireland, the Continent is widely and generally used to refer to the...	0.064	0.118
Wenn in diesem Text oder in der wissenschaftlichen Literatur "Mevalonsäure" ohne weiteren Nam...	0.064	0.037
While we often use the word "sun" to refer specifically to our star, the Sun, the term "sun"...	0.058	0.128
Typically, "abstraction" is used in the arts as a synonym for abstract art in general.	0.057	0.119
Kangaroo Island most commonly refers to Kangaroo Island in South Australia.	0.057	0.017
Stansted usually refers to London Stansted Airport.	0.056	0.000
The Entombment usually refers to the entombment, or burial, of Jesus.	0.056	0.026
what word is often used casually to refer to both a database and the dbms used	0.056	0.077
Generally yes. When people refer to the company as Coke or as the more formal "Coca-Cola" the...	0.052	0.135
By extension, Himalayas is also the name of the massive mountain system which includes the Hi...	0.052	0.078
North Africa The term "North Africa" has no single accepted definition. It is sometimes defin...	0.051	0.079
Some professional pool players still use the term billiards to describe what's more commonly...	0.051	0.130
Saat ini, istilah "gubernur jenderal" umumnya digunakan lebih ke arah bekas koloni Kerajaan B...	0.050	0.076
Although the term progeria applies strictly speaking to all diseases characterized by prematu...	0.025	0.141
In an older and more restricted sense (not used here), the term chytrids referred just to tho...	0.044	0.121
The term combat (French for fight) typically refers to armed conflict between military forces...	0.023	0.119
These terms are rarely used by the police themselves and instead are used by criminals, priso...	0.000	0.109
In U.S. usage, a "plant condensate" is the equivalent of products classified as "pentanes+" a...	0.021	0.104
While Mochica has been used in place of Moche by people describing this culture, the word Moc...	0.000	0.101
Definition Challenges Since the terms "risk appetite" and "risk tolerance" often are used int...	0.000	0.100
[This agency, the Brain Injury Network, had asked them to change their definition, which form...	0.000	0.085
Blindness is frequently used today to describe severe visual decline in one or both eyes with...	0.000	0.071
Hindi is an Indo-Aryan language or languages descended from Sanskrit , defined with various d...	0.023	0.071
I think the word is used in a electronics in a different way than what the word originally me...	0.000	0.067
The MENA acronym is often used in academia, military planning, disaster relief, and business...	0.000	0.059
As a broad generality, in the United States, the SI metric system of units is predominantly u...	0.021	0.058
My son is 3 and I still consider him a toddler, but I was talking to a friend of mine who has...	0.000	0.058
In computer science, this type of matrix representation is commonly known as a bit array, bit...	0.000	0.057

Table 14: Whether algebraic transformations preserve solutions. Pearson $r = 0.362$.

Text	Gemini 25799	Nemotron 102037
When we cancel out a common factor from both sides of an equation, we are essentially dividin...	0.124	0.128
When proving trigonometric identities, it is crucial to avoid manipulating them as if they we...	0.121	0.098
Manipulating each side of a trigonometric identity separately is advised because it ensures t...	0.116	0.064
Dividing by zero is undefined in mathematics. If we divide both sides of an equation by a var...	0.115	0.132
Dividing both sides of a trigonometric equation by another trigonometric expression, like c...	0.111	0.074
Explain why it is important to consider the possibility of losing solutions when dividing bot...	0.104	0.100
Which axioms or theorems in group theory support the justification of multiplying or adding a...	0.104	0.022
It is not accurate to say that there are solutions to a differential equation that cannot be...	0.091	0.098
When you have the equation AC = B and wish to isolate C by multiplying both sides by A^(-1)...	0.077	0.042
Explain why the method of factorizing a variable to solve a quadratic equation is incorrect,...	0.073	0.084
Unfortunately, removing the floor function from both sides of the equation is not valid becau...	0.070	0.043
If ab>0, then which one of the following must be true? (A) a/b>0 (B) a-b>0 (C)...	0.059	0.000
In the original question, it is mentioned that taking the natural logarithm of both sides of...	0.059	0.042
When we derive a new equation from a system of simultaneous equations, we are creating an equ...	0.056	0.119
Why does the third elementary row operation, adding a row to another one multiplied by a numb...	0.054	0.023
When is it incorrect to square both sides of an inequality to solve it?	0.082	0.149
By substituting the expression for y from the original equation into one of the derivatives a...	0.000	0.134
The zero-product property states that if the product of two factors is zero, then at least on...	0.000	0.122
Move all terms to one side of the inequality to simplify the expression.	0.041	0.108
When integrating a differential equation using separation of variables, why is it incorrect t...	0.034	0.107
Apply the product rule of differentiation to the potential parent term. If the result matches...	0.000	0.106
how to solve equation with completing the square?	0.000	0.101
['Isolate the x2u200b term on one side of the equal sign.', 'Take the square root of both si...	0.025	0.072
In the proof of the solution to a recurrence relation using the characteristic equation, how...	0.000	0.072
how to solve quadratic equations with variables?	0.000	0.067
To solve equations involving absolute values, follow these steps: 1. **Case Analysis**: Recog...	0.013	0.063
['Use the Rational Zero Theorem to list all possible rational zeros of the function.', 'Use s...	0.000	0.061
Suppose I have an equation f(x) = 0, and I know the solution exists. I perform a series of...	0.077	0.054
When calculating the Taylor expansion of a function that has different forms on either side o...	0.000	0.051
As for the second restriction, commas are interpreted as or, so desmos is graphing values tha...	0.000	0.051

Table 15: School activities and educational offerings. Pearson $r = 0.516$.

Text	Gemini 31660	Nemotron 42453
Roanoke Rapids High School's most prestigious club is the Student Council. Other clubs includ...	0.114	0.140
The Manhasset High School offers a variety of clubs and activities for our students.	0.110	0.129
Riverdale Country School Abraham Award, which is an award given annually to high schools that...	0.110	0.108
Untuk menyalurkan bakat para siswa di berbagai serta melatih jiwa kepemimpinan mereka, pihak se...	0.104	0.100
Popular extra curricular activities include National Cadet Corps (NCC), National Service Sche...	0.103	0.129
Die Musical-AG führt regelmäßig Musicals auf, unter anderem: Winnfried (2006), Anything Goes...	0.102	0.049
সাধারণ পাঠ্যপুস্তকের শিক্ষার বাইরে সাহিত্য ও সংস্কৃতি অঙ্গনে বিদ্যালয়ের শিক্ষার্থীরা বিভিন্ন...	0.100	0.076
softball, track and field (boys/girls), and cheer. Cave City students participate in competit...	0.089	0.100
what are some types of extracurricular activities?	0.083	0.064
productions, an annual talent show, and the annual ISAC Choir Festival. Schutz American Schoo...	0.068	0.042
why is extracurricular activities important in school?	0.063	0.094
History of Georgetown University or in opposition to it in later years, such as the short liv...	0.060	0.021
and events. It has more than 400 student organizations and 17 intercollegiate sports teams. A...	0.059	0.024
oldest still-running women's college radio station in the country. In the 80s, it earned the...	0.057	0.052
The top 10 countries represented are China, Greece, Hong Kong, India, Turkey, Singapore, Cana...	0.051	0.021
Early Childhood Each class in the Early Childhood section has a designated teacher, fully qua...	0.030	0.149
A few examples of common educational opportunities that may be considered co-curricular inclu...	0.099	0.145
Northland Preparatory Academy is a public charter high school and middle school in Flagstaff...	0.044	0.129
Wah Yan College, Hong Kong of Secondary Education Examination), Aestheic Education (Music and...	0.053	0.125
in the school assembly are: The National Pledge is also recited. Special classes are provided...	0.071	0.121
The Extended Student Services (ESS) Program extends Poway Unified School District's education...	0.027	0.108
Today (4/19) your student will bring home a permission slip for a wonderful opportunity espec...	0.030	0.105
Deer Valley High School offers all Arizona Interscholastic Association-sanctioned sports as o...	0.068	0.092
Outside the classroom, Iona students can join more than 75 campus clubs, including about 10 f...	0.045	0.092
Upon admittance to the school, students are given an academic level based on their proficienc...	0.024	0.091
Marianopolis is the only college that offers five special interest certificates: Environmenta...	0.073	0.083
2.) give first priority to EOF activities over other extra-curricular activities. 3.) make ev...	0.027	0.059
Nord Anglia Education International Baccalaureate. In some locations, A-levels, the French Ba...	0.000	0.058
organization on campus with over 280 members. The band program incorporates a competitive cor...	0.056	0.053
opportunity in which to engage youth and teachers in supporting English language-learning and...	0.019	0.053

Table 16: Trigonometric half-angle identities. Pearson $r = 0.578$.

Text	Gemini 23469	Nemotron 114122
The half-angle identities can be derived from the sum and difference formulas for sine and co...	0.154	0.104
To find the exact value of $\sin(\pi/6)$ using the half-angle identity, follow these steps: 1. St...	0.131	0.064
How do you use the half-angle formulas to simplify the expression $\sqrt{(1+\cos 4x)/2}$?	0.128	0.069
How do you use the half-angle identity to find the exact value of $\cos(\pi/8)$ degrees?	0.125	0.077
How do you use a half-angle formula to find the exact value of $\tan((9\pi)/8)$?	0.116	0.072
Find the exact value of $\cos(105^\circ)$ using the half-angle identity.	0.114	0.028
How do you use the half-angle identity to find the exact value of cosecant of $7\pi/8$?	0.113	0.076
How can we verify the identity $\# \cos 2x = (1 - \tan^2 x) / (1 + \tan^2 x)$ using the double angle identity?	0.098	0.000
Given that $\cos A = \frac{8}{17}$, find: a) $\sin 2A$ b) $\cos 2A$ Using the Pythagorean ide...	0.084	0.000
How do you find the value of $\sin(\theta/2)$ given $\tan(\theta) = 2$ and $0 < \theta < \pi/2$?	0.080	0.008
Starting with the double angle formula for tangent: $\tan(2x) = \frac{\tan(x)}{1 - \tan^2(x)}$...	0.076	0.024
We can use the double angle identity for cosine to write: $\# \cos(4x) = \cos(2(2x)) \# \cos(4x) = 2...$	0.073	0.013
The exact value of $\# \sec(75^\circ) \#$ can be found as follows: 1. Note that $\# \sec(75^\circ) = \frac{1}{\cos(75^\circ)}$...	0.066	0.029
To solve the equation $\# \cos 2x = -\frac{\sqrt{3}}{2} \#$, we can employ the double angle identity...	0.065	0.000
Using the double angle formula for sine, we have: $\# \sin(6\theta) = 2 \sin(3\theta) \cos(3\theta) \#$ Usi...	0.059	0.019
The half-angle identities are trigonometric identities that express the sine, cosine, and tan...	0.142	0.114
To use the half-angle identities to find the exact values of $\sin 165^\circ$, $\cos 165^\circ$, and $\tan 165^\circ$...	0.137	0.113
The half-angle identities can be derived from the sum and difference formulas for sine and co...	0.154	0.104
To derive the half-angle formulas for sine, cosine, and tangent, we can start with the identi...	0.155	0.104
Half-Angle Identities are trigonometric equations that relate the sine, cosine, and tangent o...	0.131	0.103
To find the exact value of trigonometric functions at specific angles using half-angle identi...	0.134	0.103
To derive the half-angle formulas, we can use the following steps: * Use the double-angle for...	0.123	0.102
How do you use the half-angle formulas to find the exact value of $\tan(3\pi/8)$?	0.136	0.077
How do you use the half angle identity to find the exact value of $\cos 101.25^\circ$?	0.121	0.074
The half-angle formula for sine is given by $\# \sin^2(\theta/2) = (1 - \cos(\theta))/2 \#$. To simplif...	0.094	0.062
To simplify $\# \csc((7\pi)/8) \#$ using the half-angle formula, follow these steps: 1. Recognize th...	0.098	0.062
To evaluate $\tan(-195)$ using the half-angle formula, we first need to find the value of $\tan(19...$	0.099	0.060
How do you use a half-angle formula to simplify $\tan[(9\pi)/8]$?	0.106	0.060
We can use the half-angle formula for cosine to find the value of $\cos(\pi/5)$ without using a c...	0.063	0.058
For the first question, we can use the half-angle formulas: $\sin(\frac{x}{2}) = \sqrt{\frac{1 - \cos(x)}{2}}$...	0.091	0.056

Table 17: Telescopes and astronomical optics. Pearson $r = 0.604$.

Text	Gemini 64628	Nemotron 31241
what are the two key properties of a telescope and why is each important?	0.222	0.127
when was the vega bray telescope built	0.170	0.040
Timeline of telescopes , observatories , and observing technology .	0.145	0.137
He therefore bought a telescope, which enabled him to watch as accurately as did the owner hi...	0.136	0.109
Can you see an American flag on the moon with a telescope? Even the powerful Hubble Space Tel...	0.128	0.127
the version of the track recorded on "Later... with Jools Holland" was included as a "bonus l...	0.110	0.033
sensitivity is about 10 mJy. Steering of the antenna main lobe is accomplished with phase shi...	0.108	0.061
طبق اعلام مجری رصدخانه ملی ایران ساخت بنای ثابت محفظه تلسکوپ ۳٫۴ متری، محوطه سازی و ساختمان...	0.100	0.069
Five hundred meter Aperture Spherical Telescope help change the focus to different areas of t...	0.099	0.097
The Square Kilometre Array (SKA) is a large multi radio telescope project aimed to be built...	0.094	0.033
from around 800 K (500 °C / 1000 °F) to 1500 K (1200 °C / 2200 °F) in just 6 hours. An observ...	0.079	0.041
Newsletter. Today's BAFact: The Sun is 400,000 times brighter than the full Moon in the sky...	0.074	0.063
Moon missions. Russian Federation spacecraft is planned to send cosmonauts to the moon orbit...	0.070	0.016
Holders of a bachelor's or a master's degree in astronomy often enter an unrelated field. How...	0.064	0.044
are swfa scopes any good?	0.057	0.057
As a rule you should avoid zoom binoculars, at least for astronomical use. Good zoom eyepiece...	0.023	0.203
I am planning to build a grism spectrometer based on F. Sigernes' work. Can I use single lens...	0.000	0.173
On the other hand, a compound microscope has two sets of lenses, an ocular or eyepiece lens a...	0.000	0.149
What online resources are available for amateur astronomy experiments, particularly those inv...	0.080	0.123
Coronagraph A coronagraph is a telescopic attachment designed to block out the direct light f...	0.025	0.122
I While looking at the objective lens and the stage from the side, turn the coarse focus knob...	0.000	0.116
از آینه‌های معروف جهان می‌توان به آینه‌ای که در تلسکوپ رصدخانه مانت پالومار کالیفرنیا کار گد...	0.112	0.102
what is the name of the spherical lens placed on top of text to magnify the	0.000	0.077
[This root (tel-) means reaching the end (aim).. It is well-illustrated with the old pirate's...	0.091	0.073
Bernard Ferdinand Lyot (27 February 1897 , in Paris – 2 April 1952 , in Cairo) was a Frenc...	0.000	0.067
This was the observation that Venus has phases, much like our Moon does. To the naked eye, Ve...	0.071	0.060
lens system was made obsolete by later developments, primarily advanced by Panavision, Cinema...	0.021	0.059
what year did the hubble space telescope not see neptune's great red spot	0.033	0.058
what does the parabolic reflector of holophones radio produce	0.000	0.054
Lens Creek	0.000	0.052

Table 18: Trade tariffs and protectionism. Pearson $r = 0.651$.

Text	Gemini 61225	Nemotron 124167
Undang-Undang Smoot-Hawley	0.153	0.030
Loi Hawley-Smoot	0.146	0.000
where were the first tariffs introduced in australia	0.145	0.088
what was the name of the tariff introduced by germany in 1925	0.136	0.137
what did the republicans impose to protect themselves from foreign trade	0.135	0.043
A commercial treaty is a formal agreement between states for the purpose of establishing mutu...	0.113	0.194
who was the president when steel tariffs were imposed	0.109	0.096
Customs duties in the United States	0.086	0.097
Canadian-American Reciprocity Treaty primarily to the United States. The treaty was abrogated...	0.083	0.098
define autarky	0.080	0.000
Rückzoll	0.072	0.000
the anti-cotton law league campaigned against the repeal of which laws	0.060	0.015
Aviation in Indonesia airlines by the end of 2015 or early 2016. Indonesia is considering ope...	0.057	0.039
what article did the jay treaty suspend in 1796	0.054	0.013
United Nations Conference on Trade and Development The organisation grew from the view that e...	0.053	0.162
Special 301 Report and enforcement of intellectual property rights". It also states that comp...	0.000	0.177
What are the main drawbacks of adopting a WTO Brexit (also known as a Clean Break)?	0.080	0.149
The so-called Really Good Friends (RGF) club is a sub-group of members of the World Trade Org...	0.027	0.148
Interpreting NAFTA Rules of Origin. A rule of origin might contain. 1) a change in tariff cla...	0.043	0.145
of Canada in the 1920s. Its platform was entitled the "New National Policy", and it advocated...	0.089	0.129
Pollution haven hypothesis estimate the indirect effect of trade liberalization on pollution...	0.036	0.129
Comprehensive and Progressive Agreement for Trans-Pacific Partnership favored that other coun...	0.023	0.118
Post-World War II economic expansion trade, GNP, and general quality of life. The high econom...	0.019	0.076
what act was passed in the kennedy round of trade negotiations	0.107	0.074
__ INDEX __ Arun M. Kumar (born 25 November 1952) has been the Chairman and Chief Executive...	0.000	0.072
Chile-United States Free Trade Agreement	0.067	0.057
Products may come from many different countries – one part of a product may come from China...	0.000	0.053
Both the United States and Japan belong to a variety of global organizations, including: the...	0.000	0.052
The International Jute Study Group (IJSG) is an intergovernmental organisation of states th...	0.027	0.051
which international agreement provides for the protection of sound recordings	0.000	0.050

Table 19: Comments in source code. Pearson $r = 0.667$.

Text	Gemini 96133	Nemotron 76359
In Emacs, the provided examples actually use 2-character comment delimiters, which are extend...	0.167	0.178
Modeling line comments in a CFG can be achieved as follows: 1. If you're not dealing with mul...	0.158	0.171
how to comment out multiple lines in jupyter?	0.151	0.165
are comments passed to the parser?	0.150	0.082
Comment programming	0.150	0.114
how to comment multiple lines in pycharm shortcut?	0.120	0.145
ulysses how to add comment?	0.102	0.022
How can I prevent syntax-propertize in Emacs from treating '#' within SPARQL URI's as the sta...	0.093	0.107
How can I make Yasnipet insert a comment sign on a new line when pressing Return? Q: When us...	0.089	0.108
জাভাস্ক্রিপ্ট এ ভেরিয়েবল ডিক্লেয়ার করা হয় var কি ওয়ার্ড দিয়ে : var x; // defines the var...	0.074	0.044
Important As of Internet Explorer 10, conditional comments are no longer supported by standar...	0.068	0.029
Here document	0.058	0.023
A feature that fits into the former category is the extended properties of databases. Introdu...	0.053	0.000
up vote 10 down vote favorite. 4. When I insert a comment or make a revision to some Microsof...	0.053	0.051
How to Query SQL Server 2000 for Stored Procedures Containing Text. For SQL Server 2000, the...	0.051	0.000
The recommended way to comment out multiple lines of code in Python is to use consecutive # s...	0.151	0.234
Multiline comment. A multiline comment starts with /* and can be found at any position in the...	0.160	0.201
Multiline Comments in Bash Unlike most of the programming languages, Bash doesn't support mul...	0.146	0.172
how to comment out multiple lines in mysql workbench?	0.145	0.161
While there isn't a direct way to insert line breaks in comments, they are designed to be con...	0.070	0.145
There is also another way to document Python code using comments that start with ##. These ty...	0.126	0.130
If the PyCharm IDE is used to write Python code - select multiple code rows to comment and pr...	0.090	0.113
Комментарий	0.015	0.094
Looks like you are back with WordPress comments. I have sites with WordPress, DNN Blog, and D...	0.000	0.073
En kommentoi	0.021	0.069
VBdocman allows commenting and the automatic generation of technical documentation from Visua...	0.077	0.065
There isn't a built-in command in c-mode that mimics LaTeX's latex-close-block for #ifdef blo...	0.053	0.064
This topic has been extensively discussed on various Meta sites within the Stack Exchange net...	0.051	0.053
Syntax highlighting for HTML code within a #+BEGIN_EXPORT HTML block in org-mode is affected...	0.020	0.053
Commentaries dealing with the zahir (outward aspects) of the text are called tafsir, and herm...	0.000	0.051

Table 20: Mechanical bearings. Pearson $r = 0.709$.

Text	Gemini 76787	Nemotron 40156
Bearing (mechanical) 1794. His was the first modern ball-bearing design, with the ball runnin...	0.203	0.282
Rolf Hoppler and Reinhold Errath evaluate the types of bearings used in the. cement industry...	0.191	0.184
A wheel bearing replacement can take anywhere from one to three hours depending on the type o...	0.157	0.195
Fluid bearing generators weighing hundreds of tons. They are also used in very heavy machiner...	0.146	0.218
Driving with 450+ answers. You didn't indicate front or rear, or whether it is a front-wheel...	0.137	0.153
SKF-89976A	0.108	0.097
The noise you're experiencing could be due to a faulty main bearing in the front of the trans...	0.107	0.073
Consider the following diagram: [Image of a diagram showing Jack and Jill's positions and the...	0.098	0.031
Pacific Bearing Corporation guides, linear slides, linear motors and nuts for lead screws. In...	0.096	0.198
ACER Racing is a large OEM manufacturing company specializing in precision high performance r...	0.091	0.067
taken from the shafts of VFD-driven motors throughout the commercial bakery to illustrate the...	0.075	0.085
1. Salvageability: The assembly can likely be salvaged with proper repairs. To verify the con...	0.073	0.128
Bearing an Hourglass is a fantasy novel by Piers Anthony. It is the second of eight books in...	0.064	0.092
A ship is anchored off a long straight shoreline that runs north and south. From two observat...	0.057	0.000
Nicolas Fatio de Duillier (also spelled Faccio or Facio ; 16 February 1664 – 12 May 1753)...	0.052	0.000
Bearing (mechanical) rolling element bearing is a wooden ball bearing supporting a rotating t...	0.214	0.274
608 STANDARD The most common bearing size is the "608." Characterized by an 8mm core, a 22mm...	0.167	0.247
tapered roller bearing in combination with modern lubricants is extremely durable and is used...	0.155	0.239
Slewing bearing. A slewing bearing or slew ring is a rotational rolling-element bearing or pl...	0.153	0.200
ABEC 5 bearings are the norm in skateboarding. You get a reasonable amount of speed, and at a...	0.177	0.174
Ball (bearing) the largest and smallest diameter measurement. For a given lot there is a "lot...	0.085	0.126
Oli pelumas harus selalu melumasi bagian-bagian poros engkol yang bergerak, hal ini tentunya...	0.104	0.117
when was the first roller bearing truck made	0.222	0.079
Tribology Online Read articles with impact on ResearchGate, the professional network for sc...	0.061	0.069
Cojnete magnético	0.199	0.060
• CHILDBEARING (noun). The noun CHILDBEARING has 1 sense: 1. the parturition process in human...	0.043	0.059
Im August 1993 übernahm die Minebea alle Anteile der PM DM und wurde somit alleiniger Eigentü...	0.039	0.058
Footnote on Crankshaft Rebuilders. Did another search on Crankshaft Rebuilders. They use a ne...	0.077	0.058
Tunnel crankcase there they were used in the more established diesel-electric designs. The fi...	0.074	0.053
The behavior of an object rolling on a surface is influenced by the coefficient of rolling re...	0.029	0.052

Table 21: Butter. Pearson $r = 0.788$.

Text	Gemini 59059	Nemotron 17701
which oil is used in sautéing to make butter	0.240	0.065
Grams in Butter Eight Tablespoon stick of unsalted butter weighs 110 grams. Therefore, 110 di...	0.171	0.235
The rock gunnel or Butterfish("Pholis gunnellus") is an eel-like fish found in the intertidal...	0.163	0.132
Basically, cookies made with butter spread more and are flatter and crisper if baked long eno...	0.158	0.213
If a recipe calls for butter in ounces, simply divide the number of ounces by 4 to arrive at...	0.154	0.244
is shea butter good for hair?	0.133	0.096
Hartkäse, Weichkäse und Butter sind relativ ergiebige Quellen für Vitamin K.	0.120	0.063
Fortunately for everyone who loves peanut butter, almond butter, and other nut butters, these...	0.076	0.102
...الجينة القابلة للدهن (الكاسات)، شرائح الجينة، الزبدة، السمن، كريمة الحفق، كريمة الطبخ، المورا...	0.070	0.038
otto mønsted aarhus was the founder of which famous margarine	0.069	0.014
There are a number of these cleaners available on the Internet, most of them free to try out...	0.064	0.025
Once melted, remove from heat and whisk in the hot sauce and salt. Coat the wings with the sa...	0.061	0.034
1 Bring a large pot of lightly salted water to a boil. 2 Add pasta and cook for 8 to 10 minut...	0.055	0.055
Directions. For the chocolate sauce: Combine the chocolate and 1/4 cup water in a heavy small...	0.054	0.018
Prize (marketing) prizes from German margarine were molded in a light cream color designed to...	0.051	0.071
what kind of butter is fiddlehead fern pickle	0.169	0.208
when did the buttery become part of connecticut hall	0.036	0.144
Don't use low fat spread-it might be better for you, but much of the fat is replaced by water...	0.032	0.136
Place the butter cubes on top of the carrots. They will melt all over and do yummy things to...	0.123	0.132
West Buttercreek Airport	0.063	0.121
After 2 hours begin mopping the brisket with a butter/cajun seasoning mix that I call "mop wa...	0.077	0.107
Pan Frying. 1 1. Shuck the clams. Remove the clams from their shells if they are still intact...	0.081	0.104
Calories in Kroger Peanut Butter based on the calories, fat, protein, carbs and other nutriti...	0.034	0.086
do you have to refrigerate earth balance butter?	0.152	0.085
Ingredients. 1 6-8 ounces salt pork, cut into 1/2 inch chunks. 2 4 Tbsp unsalted butter, divi...	0.070	0.080
A nervous stomach can give someone "butterflies" in their stomach, or even make a person feel...	0.000	0.077
A Franzbrötchen () is a small, sweet pastry, baked with butter and cinnamon. Sometimes othe...	0.042	0.076
how many grams is 1 tablespoon of margarine?	0.066	0.064
You're better off sticking with skim or low-fat dairy. Participants who ate whole-fat dairy (...)	0.046	0.060
Crème fraîche (English pronunciation : -LSB- ˌkʁɛmˈfʁɛʃ -RSB- , -LSB- kʁɛm fʁɛʃ -RSB- , lit...	0.042	0.055

Table 22: Sign languages. Pearson $r = 0.802$.

Text	Gemini 61020	Nemotron 89987
hawaii sign language is an independent language from what other language	0.157	0.117
A free sign language worksheet for kids (a great reference when learning the ASL alphabet) AS...	0.136	0.164
American Sign Language grammar signs: The hand shape is the initial of one of the English nam...	0.129	0.209
You could qualify by registering as a trainee sign language interpreter (TSLI). To register,...	0.118	0.185
Home sign is often the starting point for new deaf sign languages that emerge when deaf peopl...	0.116	0.203
Lithuanian Sign Language (LGK) (Lithuanian : Lietuvių gestų kalba) is the national sign l...	0.109	0.187
who was the founder of indian sign language school in madras	0.103	0.056
Cuban Sign Language is the deaf sign language of Cuba . Little information is available .	0.099	0.156
part. Thirteen of the most widely spoken indigenous languages are considered "national langua...	0.073	0.132
who did jean massieu's father play in sign gene	0.072	0.000
Fingerspelling Some writers have suggested that the body and hands were used to represent alp...	0.071	0.112
Naga Hills Sign Language was a village sign language of India. Early in the 20th century, a h...	0.068	0.099
Project Nim (film)	0.058	0.000
Starbucks Malaysia's Signing Store. Signing Store Workshop; Learn More . Policies; Looking fo...	0.053	0.000
Deaf with the capitalized D represents people that are members of the Deaf Community, a lingu...	0.050	0.048
Black American Sign Language 2004 by Melanie Metzger and Susan Mather found that Black male s...	0.130	0.213
hard to draw. A language 'continuum' is often described between signing with a strong sign-la...	0.117	0.205
ASLwrite () is a somacheiographic writing system that developed from si5s. It was created to...	0.099	0.180
Many Australian Aboriginal cultures have or traditionally had a manually coded language, a si...	0.082	0.136
Rennellese Sign Language is an extinct form of home sign documented from Rennell Island in 19...	0.081	0.117
estimates for the number of people who use ASL are relatively easy to find in research and pr...	0.108	0.109
can asl be used in other countries?	0.178	0.104
When the 26-year-old signed I'm proud of you, the president quickly signed back a Thank you....	0.081	0.099
Hanabiko "Koko" (July 4, 1971 - June 19, 2018) was a female western lowland gorilla known for...	0.081	0.086
She is now 13 months old. I started teaching her sign language as a tiny baby (signing as I t...	0.065	0.085
The World Trade Organization (WTO) is an intergovernmental organization that regulates inte...	0.000	0.084
Tom Clerc Tom Clerc is a fictional deaf superhero appearing in Sign Gene, the world's first f...	0.060	0.071
As in other states, a certified interpreter in Idaho will command a higher fee for legal inte...	0.018	0.059
Centre for Deaf Studies, Bristol The Centre for Deaf Studies was a department of the Universi...	0.046	0.054
The General Agreement on Trade in Services (GATS) is a treaty of the World Trade Organization...	0.000	0.052

Table 23: Eurovision Song Contest entries. Pearson $r = 0.865$.

Text	Gemini 123114	Nemotron 97094
Norway in the Eurovision Song Contest 1966	0.175	0.124
United Kingdom in the Eurovision Song Contest 1968	0.174	0.132
France in the Eurovision Song Contest 1956	0.173	0.139
Denmark in the Eurovision Song Contest 1989	0.169	0.144
Lithuania in the Eurovision Song Contest 2018	0.147	0.142
Greece in the Eurovision Song Contest 2014	0.138	0.157
Macedonia in the Junior Eurovision Song Contest 2013	0.124	0.126
Singapore in the ABU TV Song Festival	0.094	0.090
Bashkortostan in the Turkvision Song Contest 2013	0.087	0.094
Venezuela in the OTI Festival	0.082	0.074
Kazakhstan in the Bala Turkvision Song Contest 2015	0.078	0.093
France au Concours Eurovision de la chanson 1984	0.072	0.026
Ukraine in the Eurovision Young Musicians	0.071	0.108
Eurovision Song Contest 1971	0.056	0.051
Luxembourg au Concours Eurovision de la chanson 1961	0.052	0.036
Switzerland in the Eurovision Song Contest 2009	0.164	0.155
Denmark in the Eurovision Song Contest 2004	0.164	0.144
Belarus in the Eurovision Song Contest 2005	0.154	0.141
Netherlands in the Eurovision Song Contest 1983	0.179	0.141
United Kingdom in the Eurovision Song Contest 1981	0.177	0.135
Belgium in the Junior Eurovision Song Contest	0.116	0.119
Ireland in the Eurovision Young Musicians	0.109	0.118
Wales in Eurovision Choir	0.041	0.088
Switzerland in the Eurovision Song Contest 2019 2018 contest would be selected through a Die...	0.000	0.073
The Republic of Macedonia, participating under the name F.Y.R. Macedonia was represented at t...	0.000	0.065
Ukraine was represented in the Eurovision Song Contest 2005 by GreenJolly with the song "Razo...	0.000	0.060
Montenegro Eurovision laulukilpailussa	0.032	0.058
Spain in the Eurovision Song Contest 1965 Spain took part in the Eurovision Song Contest 1965...	0.000	0.054
Poland in the Early Middle Ages	0.000	0.054
Eurovision Song Contest 1975	0.060	0.052

Table 24: Astragalus and milkvetch species. Pearson $r = 0.871$.

Text	Gemini 104756	Nemotron 51458
Astragalus campylanthus	0.166	0.165
Astragalus brauntonii	0.165	0.151
Astragalus brachystachys	0.164	0.157
Astragalus annularis	0.163	0.160
Astragalus monanthemus	0.160	0.155
Astragalus sprucei is a species of legume in the Fabaceae family. It is found only in Ecuador...	0.120	0.160
Astragalus sabulorum is a species of milkvetch known by the common name gravel milkvetch. It...	0.120	0.226
Astracantha gummifer	0.091	0.075
Astragalus proimanthus is a species of flowering plant in the legume family known by the comm...	0.086	0.185
Astragalus cicer in regards to the inherent durability of the plant. In regards to obtaining...	0.081	0.143
The garage door gasket not only protects the bottom of the garage door from the floor each ti...	0.075	0.000
purtuka or nakhuna (Astragalus Tribuloids),	0.072	0.039
Mesorhizobium sangaii is a bacterium from the genus Mesorhizobium which was isolated from roo...	0.067	0.056
Циклоастрогенол	0.064	0.000
قناد سمسمي	0.063	0.000
Astragalus bernardinus , known by the common name San Bernardino milkvetch , is a species of...	0.138	0.200
astragalus (a·str·gae·gā l) , s N latin: Names astragalus, Membranaceus	0.111	0.182
astragalus; gummife...		
Astragalus cedreti (also called Cedar astragalus ;) is a species of flowering plant in the...	0.103	0.179
Astragalus crotalariae	0.163	0.157
Astragalus newberryi	0.162	0.152
Astragalus pelecinus pelecinus	0.162	0.138
Es una planta herbácea perennifolia originaria de Norteamérica donde se distribuye por Estado...	0.129	0.130
Astragalin is a chemical compound. It can be isolated from "Phytolacca americana" (the Americ...	0.109	0.088
what is the herbage of astragalus molybdenus coated in	0.140	0.086
Astragal	0.207	0.083
why is the seed bank of astragalus nitidiflorus so sensitive	0.104	0.071
Chinese Herbs In Western View - Huang (Bei) qi (Radix astragali) Health Benefits and Side eff...	0.080	0.067
where did the astragaloi perform divination in ancient greece	0.082	0.060
locoweed toxins, pinpointing levels of toxicity in animals once they have ingested locoweed,...	0.064	0.055
Astragalomancy pessomancy (also known as psephomancy) - another act of divination which uses...	0.053	0.054